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Thesis Title	Collaborative outbreak modelling for decision support: evaluating trade-offs from multi-model combination		
Primary Supervisor	Sebastian Funk		

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SECTION D – Multi-authored work

For multi-authored work, give full details of your role in the research included in the paper and in the preparation of the paper. (Attach a further sheet if necessary)	This work originated in real-time response work contributing Rt estimates to SPI-M from March 2020. Development of the Rt estimation and forecasting procedure using EpiNow and EpiNow2 was led by Sam Abbott and Sebastian Funk. I initially contributed data cleaning, processing, and analysis for UK and global surveillance data. I also supported running the end to end Rt estimation pipeline for regular SPI-M submission, and contributed to model development with testing, evaluation, and documentation. In May 2020, I worked together with Sam Abbott and Sebastian Funk on the initial hypothesis of exploring differences in Rt estimation by data source. I led work to develop this idea into this analysis, including developing code to run the Rt estimation pipeline using each data source and visualising the comparative results. I also led the write up of this into a short briefing note, presented to SPI-M first in early June and in three further iterations. I then led work to develop this into a paper. I developed the original work by using public data for estimating delay distributions, in order for all work to be presented in the public domain. I also designed and conducted further analysis to quantify the comparisons of Rt estimates, for example assessing peaks and wave durations. The specific further hypotheses for retrospectively understanding differences between Rt estimates over time were discussed jointly with Sam Abbott and Sebastian Funk, in turn drawing on wider discussions among SPI-M. I led work to source relevant data and formally add comparisons with these additional data sources, for example data on deaths in residential facilities. I led the initial and subsequent drafts of the paper. Sam Abbott and Sebastian Funk provided supervision and review before submission for peer review. All contributions are documented on Github: https://github.com/epiforecasts/rt-comparison-uk-public.

SECTION E

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Date	14 June 2024

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Date	14 June 2024

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Exploring surveillance data biases when estimating the reproduction number: with insights into subpopulation transmission of COVID-19 in England

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The time-varying reproduction number (R_t: the average number of secondary infections caused by each infected person) may be used to assess changes in transmission potential during an epidemic. While new infections are not usually observed directly, they can be estimated from data. However, data may be delayed and potentially biased. We investigated the sensitivity of R_t estimates to different data sources representing COVID-19 in England, and we explored how this sensitivity could track epidemic dynamics in population sub-groups. We sourced public data on test-positive cases, hospital admissions and deaths with confirmed COVID-19 in seven regions of England over March through August 2020. We estimated R_t using a model that mapped unobserved infections to each data source. We then compared differences in R_t with the demographic and social context of surveillance data over time. Our estimates of transmission potential varied for each data source, with the relative inconsistency of estimates varying across regions and over time. R_t estimates based on hospital admissions and deaths were more spatio-temporally synchronous than when compared to estimates from all test positives. We found these differences may be linked to biased representations of subpopulations in each data source. These included spatially clustered testing, and where outbreaks in hospitals, care homes, and young age groups reflected the link between age and severity of the disease. We highlight that policy makers could better target interventions by considering the source populations of R_t estimates. Further work should clarify the best way to combine and interpret R_t estimates from different data sources based on the desired use.

This article is part of the theme issue 'Modelling that shaped the early COVID-19 pandemic response in the UK'.

1. Background

Within six months of its emergence in late 2019, the novel coronavirus SARS-CoV-2 had caused over six million cases of disease (COVID-19) worldwide [1]. Its rapid initial spread and high death rate prompted global policy interventions to prevent continued transmission, with widespread temporary bans on social interaction outside the household [2]. Introducing and adjusting such policy measures depend on a judgement in balancing continued transmission potential with the multidimensional consequences of interventions. It is,

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therefore, critical to inform the implementation of policy measures with a clear and timely understanding of ongoing epidemic dynamics [3,4].

In principle, transmission could be tracked by directly recording all new infections. In practice, real-time monitoring of the COVID-19 epidemic relies on surveillance of indicators that are subject to different levels of bias and delay. In England, widely available surveillance data across the population include: (i) the number of positive tests, biased by changing test availability and practice, and delayed by the time from infection to symptom onset (if testing is symptom-based), from symptom onset to a decision to be tested and from test to test result; (ii) the number of new hospital admissions, biased by differential severity that triggers care seeking and hospitalization, and additionally delayed by the time to develop severe diseases; and (iii) the number of new deaths due to COVID-19, biased by the differential risk of death and the exact definition of a COVID-19 death, and further delayed by the time to death.

Each of these indicators provides a different view on the epidemic and therefore contains potentially useful information. However, any interpretation of their behaviour needs to reflect these biases and lags and is best done in combination with the other indicators. One approach that allows this in a principled manner is to use the different datasets to separately track the time-varying reproduction number, R_t, the average number of secondary infections generated by each new infected person [5]. Because R_t quantifies changes in infection levels, it is independent of the level of overall ascertainment as long as this does not change over time or is explicitly accounted for [6]. At the same time, the underlying observations in each data source may result from different lags from infection to observation. However, if these delays are correctly specified then transmission behaviour over time can be consistently compared via estimates of R_t.

Different methods exist to estimate the time-varying reproduction number, and in the UK a number of mathematical and statistical methods have been used to produce estimates used to inform policy [7–9]. Empirical estimates of R_t can be achieved by estimating time-varying patterns in transmission events from mapping to a directly observed time-series indicator of infection such as reported symptomatic cases. This can be based on the probabilistic assignment of transmission pairs [10], the exponential growth rate [11] or the renewal equation [12,13]. Alternatively, R_t can be estimated via mechanistic models that explicitly compartmentalize the disease transmission cycle into stages from susceptible through exposed, infectious and recovered [14,15]. This can include accounting for varying population structures and context-specific biases in observation processes, before fitting to a source of observed cases. Across all methods, key parameters include the time after an infection to the onset of symptoms in the infecting and infected, and the source of data used as a reference point for earlier transmission events [16,17].

In this study, we used a modelling framework based on the renewal equation, adjusting for delays in observation to estimate regional and national reproduction numbers of SARS-Cov-2 across England. The same method was repeated for each of three sources of data that are available in real time. After assessing differences in R_t estimates by data source, we explored why this variation may exist. We compared the divergence between R_t estimates with spatio-temporal variation in case detection, and the proportion at risk of severe disease, represented by the age distribution of test-positive cases and hospital admissions and the proportion of deaths in care homes.

2. Methods

(a) Data management

Three sources of data provided the basis for our R_t estimates. Time-series case data were available by specimen date of test. This was a de-duplicated dataset of COVID-19 positive tests notified from all National Health Service (NHS) settings (Pillar One of the UK Government's testing strategy) [18] and by commercial partners in community settings outside of healthcare (Pillar Two). Hospital admissions were also available by date of admission if a patient had tested positive prior to admission, or by the day preceding diagnosis if they were tested after admission. Death data were available by date of death and included only those that occurred within 28 days of a positive COVID-19 test in any setting. All data were publicly available and taken from the UK government source [19,20], and were aggregated to the seven English regions used by the NHS.

To provide context for R_t estimates, we sourced weekly data on regional and national test positivity (percentage positive tests of all tests conducted) from Public Health England [21], available as weekly average percentages from 10 May. From the same source, we also identified the age distributions of cases admitted to the hospital and all test-positive cases. Hospital admissions by age were available as age bands with rates per 100 000, so we used regional population data from 2019 [22] to approximate the raw count. We separately sourced daily data on the number of deaths in care homes by region from March 2020, available from 12 April [23]. Care homes are defined as supported living facilities (residential homes, nursing homes, rehabilitation units and assisted living units). Data were available by date of notification, which included an average 2–3 days' lag after the date of death. We also drew on a database that tracked COVID-19 UK policy updates by date and area [24].

(b) R_t estimation

We estimated R_t using EpiNow2 v. 1.2.0, an open-source package in R [13,25,26]. This package implements a Bayesian latent variable approach using the probabilistic programming language Stan [27]. To initialize the model, infections were imputed prior to the first observed case using a log-linear model with priors based on the first week of observed cases. This means that the initial observations both inform the initial parameters and are then also fit, which makes the initial R_t estimates less reliable than later estimates. This was a pragmatic choice to allow the model to be identifiable when only estimating part of the observed epidemic. We explored other parameterizations, but these suffered from poor model identification. For each subsequent time step with observed cases, new infections were imputed using the sum of previous modelled infections weighted by the generation time probability mass function, and combined with an estimate of R_t, to give the prevalence at time t [12]. The generation time was assumed to follow a gamma distribution that was fixed over time but varied between samples, with priors drawn from the literature for the mean and standard deviation [28].

These infection trajectories were mapped to reported case counts (D_t) by convolving over an incubation period distribution and report delay distribution (ξ). We assumed a negative binomial observation model for observed reported case counts (C_t),

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with overdispersion ϕ using an exponential prior with mean 1 and mean D_t. We combined this with a multiplicative day of the week effect (ω(tmod7)) with an independent effect for each day of the week. We controlled temporal variation using an approximate Gaussian process [29] with a squared exponential kernel (GP).

In mathematical notation:

$$R_t \sim R_{t-1} \times \mathrm{GP},$$

$$I_t = R_t \sum_\tau w_\tau I_{t-\tau},$$

$$D_t = \sum_\tau \xi_\tau I_{t-\tau},$$

$$C_t \sim \mathrm{NB}(D_{t^{\omega(t\mathrm{mod}7)}}, \phi).$$

The length scale and magnitude of the kernel were estimated during model fitting. We used an inverse gamma prior for the length scale, optimizing shape and scale values to give a distribution with 98% of the density between 2 and 21 days, and the prior on the magnitude was standard normal. Each region was fitted independently using Markov-chain Monte Carlo (MCMC). Eight chains were used with a warmup of 1000 samples and 2000 samples post warmup. Convergence was assessed using the R hat diagnostic.

We used a gamma-distributed generation time with mean 3.6 days (standard deviation (s.d.) 0.7), and s.d. of 3.1 days (s.d. 0.8), sourced from [28]. Instead of the incubation period used in the original study (which was based on fewer data points), we refitted using a lognormal incubation period with a mean of 5.2 days (s.d. 1.1) and s.d. of 1.52 days (s.d. 1.1) [30]. This incubation period was also used to convolve from unobserved infections to unobserved symptom onsets (or a corresponding viral load in asymptomatic cases) in the model. When fitting the model, the time interval distributions had independent priors placed on the mean and standard deviation of their respective lognormal distributions.

We estimated both the delay from symptom onset to positive test (either in the community or in hospital) and the delay from symptom onset to death as lognormal distributions using a subsampled Bayesian bootstrapping approach (with 100 subsamples each using 250 samples) from given data on these delays. Our delay from the date of onset to date of positive test (either in the community or in hospital) was taken from a publicly available linelist of international cases [31]. We removed countries with outlying delays (Mexico and the Philippines). The resulting delay data had a mean of 4.4 days and s.d. 5.6. Delays for hospital admissions and test positives were treated as having the same delay from infection to onset and observation. For the delay from onset to death we used data taken from a large observational UK study [32]. We re-extracted the delay from confidential raw data, with a mean delay of 14.3 days (s.d. 9.5). There were insufficient data available on the various reporting delays to estimate spatially or temporally varying delays, so they were considered to be static over the course of the epidemic, although we discuss the effects of this assumption. We have also discussed this approach more extensively in [25].

(c) Comparison of R_t estimates

We compared R_t estimates by data source, plotting each by region over time. To avoid the first epidemic wave obscuring visual differences, all plots were limited to the earliest date that any R_t estimate for England crossed below 1 after the peak. We also identified the time at which each R_t estimate fell below 1, the local minima and maxima of median R_t estimates and the number of times in the time-series that each R_t estimate crossed its own median, before comparing these across regions and against the total count of the raw data.

We investigated correlations between R_t estimates and the demographic and social context of transmission. We used linear regression to assess whether the level of raw data count influenced oscillations in R_t. We assessed the influence of local outbreaks using test positivity. We used a 5% threshold for positivity as the level at which testing is either insufficient to keep pace with widespread community transmission [33], or where outbreaks have already been detected and tests targeted to those more likely to be positive. We plotted this against raw data and R_t, and also used linear regression to test the association. We interpreted results in light of known outbreaks and policy changes. We plotted and qualitatively assessed variation in R_t estimates against the age distribution of cases over time, and similarly explored patterns in R_t estimates against the qualitative proportion of cases to all deaths. The latter was not assessed quantitatively due to differences in reference dates [23]. With the exception of fitting the delay from onset to death (held confidentially), code and data to reproduce this analysis are available [34].

3. Results

Across England, the COVID-19 epidemic peaked at 4798 reported test-positive cases (on 22 April 2020), 3099 admissions (1 April 2020) and 975 deaths (8 April 2020) per day (figure 1*a*). Following the peak, a declining trend continued for daily counts of admissions and deaths, while daily case counts from all reported test-positive cases increased from July and had more than tripled by August (from 571 on 30 June to 1929 on 1 September). Regions followed similar patterns over time to national trends. However, in the North East and Yorkshire, Midlands and North West, the incidence of test-positive cases did not decline to near the count of admissions as in other regions, and also saw a small temporary increase during the overall rise in case of counts in early August.

Following the initial epidemic peak in mid-March 2020, the date at which R_t estimates crossed below 1 varied by both data source and geography (figures 1*b* and 2). The first region to cross into a declining epidemic was London, on 26 March according to an R_t estimated from deaths (where the lower 90% credible interval (CrI) crossed below 1 on 24 March and the upper CrI on 28 March). However, the data source used to estimate R_t was as important as any regional variation in estimating the earliest date of epidemic decline. R_t estimated from hospital admissions gave the earliest estimate of a declining epidemic, while using all test-positive cases to estimate R_t took the longest time to reach a declining epidemic, in all but one region (East of England). This difference by data source varied by up to 21 days in the North East and Yorkshire, where hospital admissions gave a median R_t estimate under 1 on 1 April (90%CrIs 31 March, 2 April), but the median R_t estimate from test-positive cases crossed 1 on only the 22 April (90%CrIs 1 April, 25 April).

When not undergoing a clear state change, R_t estimates from all data sources oscillate, with oscillations damped when R_t estimates were transitioning to new levels. In England and all NHS regions, test-positive cases showed evidence of larger damped oscillations from July when a state change occurred to R_t over 1. In England, R_t estimates from test-positive cases increased from 0.99 (90%CrI 0.94–1.04) on 30 June to 1.37 (90%CrI 1.31–1.1.44) on 27 August. Meanwhile, the timing and duration of oscillations did not align

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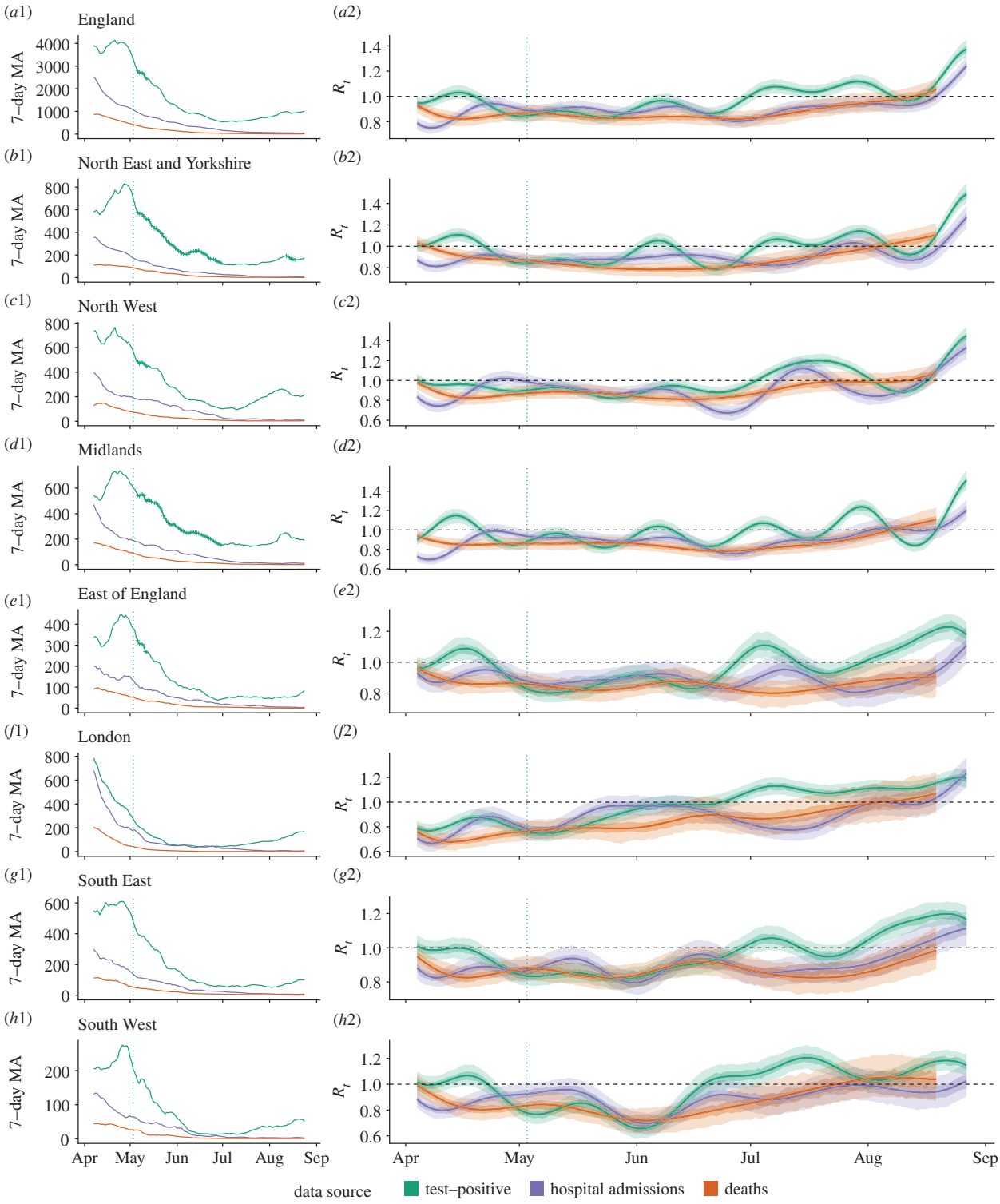

Figure 1. Epidemic dynamics across (*a*) England and (*b–h*) seven English National Health Service regions, 5 April through 27 August 2020. (*a*1–*h*1): Daily counts of confirmed cases by data source, as centred 7 days moving average. Counts marked with vertical dashes (on the green lines—see figure parts (*a*1,*b*1,*c*1,*d*1,*e*1)) indicate dates within weeks that averaged greater than 5% test-positivity (positive/all tests per week). Vertical dotted line indicates the start of national mass community testing on 3 May. (*a*2–*h*2): Estimates of R_t (median, with 50% (darker shade) and 90% (lightest shade) credible interval), derived from each data source. Data sources include all test-positive cases, hospital admissions and deaths with a positive test in the previous 28 days.

between R_t estimates (figure 1*b*). In some regions, the difference between R_t estimates was consistent over time, such as between R_t from admissions and deaths in the South East. In other regions such as the Midlands, this was not the case, with the divergence between the R_t estimates from test-positive cases, admissions, and deaths each varying over time. R_t estimates from test-positive cases were the most likely to differ from estimates derived from other data sources across all regions. Across all regions, R_t estimates from deaths had slower damped oscillations compared to estimates from test-positive cases or hospital admissions. However, oscillations in R_t estimates did not appear to be linked to the level of raw data counts in each source (electronic supplementary material, figure S2).

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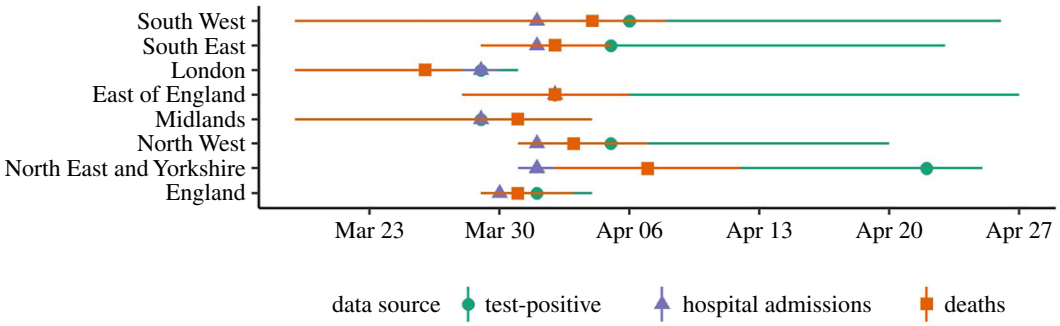

Figure 2. Dates in 2020 on which R_t estimate crossed 1 after first epidemic peak, median and 90% credible interval, by the data source for England and seven NHS regions.

More rapid oscillations in R_t estimates from test-positive cases appeared to be linked to targeted testing of case clusters, seen in high test positivity (electronic supplementary material, table SI2). Both the North East and Yorkshire and the Midlands saw more frequent oscillations in R_t estimates from test-positive cases than other regions. The R_t estimates from cases crossed its own median 10 times over the time-series in both regions, while in all other NHS regions this averaged 6 times, and oscillations in R_t estimates from cases also had a shorter duration in the North East and Yorkshire and the Midlands compared to other regions (electronic supplementary material, table SI1). Across all regions, 84% of weeks with over 5% positivity ($N = 19$) were in the North East and Yorkshire and the Midlands. In these regions, positivity peaked on the week of 9 May 2020 at 14% and 12%, respectively, and overall averaged 6% (95%CI 4.4–7.6%) and 5.9% (95%CI 4.6–7.2%, weeks of 10 May to 22 August), respectively. High test positivity is likely to have resulted from targeted testing among known local outbreaks in these regions. In the Midlands, these included local restrictions and increased testing across Leicester and in a Luton factory (restrictions between 4 and 25 July [35]). In Yorkshire case clusters were detected with local restrictions in Bradford, Calderdale and Kirklees (with restrictions from 5 August [36]).

In England, a divergence between R_t estimated from cases versus R_t estimated from deaths and admissions coincided with a decline in the age distribution among all test-positive cases in England to a younger population (electronic supplementary material, figure SI2A). From mid-April to June 2020, national estimates of R_t from test-positive cases remained around the same level as those from admissions or deaths, while after this, cases diverged to a higher steady state (figure 1*a*). On 23 May, the median R_t estimated from cases matched that of deaths at 0.83 (both with 90%CrIs 0.78–0.89), but this was followed by a 78 day period before the two estimates were again comparable, on 8 August. Over this period the median R_t estimate from cases was on average 14% higher (95%CI 12–15%). Meanwhile, the share of test-positive cases under age 50 increased from under one-quarter of cases in the week of 28 March (24%, $N = 16\,185$), to accounting for nearly three-quarters of cases by 22 August (77%, $N = 6733$). While the percentage of test-positive cases aged 20–49 increased consistently from April to August, the 0–19 age group experienced a rapid increase over mid-May through July, increasing by a mean 1% each week over 9 May through 1 August (from 4% of 18 774 cases to 14.8% of 5017 cases).

Similarly, R_t estimates from admissions in England oscillated over June through July 2020, potentially linked to the age distribution of hospital admissions. From 0.92 (90%CrI 0.87–0.98) on 11 June, R_t estimated from admissions fell to 0.8 (90%CI 0.75–0.85)) on 27 June. By contrast, this transition was not observed in the R_t estimate based on test-positive cases (figure 1*a*). Older age groups dominated COVID-19 hospital admissions, where 0–44 years never accounted for more than 12.8% of hospital-based cases (a maximum in the week of 22 August, $N = 690$; electronic supplementary material, figure SI2B). While the proportion of hospital admissions aged 75+ remained steady over May through mid-June, this proportion appeared to oscillate over July through August (standard deviation of weekly percentage at 6.1 over June–August, compared to 5.4 in months March–May). These variations were not seen in the proportion aged 70+ in the test-positive case data, which saw a continuous decline from 30% at the start of June to 7% by August.

R_t estimated from either admissions or deaths experienced near-synchronous local peaks across regions over April and May 2020. We compared this R_t estimated from deaths with its source data and a separate regional dataset of deaths in care homes. In the South East and South West, the R_t estimates from deaths rose over April, with a peak in early May. In the South West, the median R_t estimate from deaths increased by 0.04 from 22 April to 7 May (from 0.8 (90%CrI 0.72–0.88) to 0.84 (90%CrI 0.76–0.95)); and by 0.06 from 17 April to 4 May in the South East (from 0.82 (90% CrI 0.77–0.9) to 0.88 (90%CrI 0.72–0.88)). In both these regions, this early May peak in R_t estimates from deaths coincided with similarly rising R_t estimates from hospital admissions, while the reverse trend was seen in R_t estimates from cases. In all regions, care home deaths peaked over 22–29 April (by date of notification; electronic supplementary material, figure SI3). This was later than regional peaks in the raw count of all deaths in any setting (which peaked between 8 and 16 April, by date of death), even accounting for a 2–3 day reporting lag. This meant that the proportion of deaths from care homes varied over time, where in the South East and South West, deaths in care homes appeared to account for nearly all deaths for at least the period mid-May to July.

4. Discussion

We estimated the time-varying reproduction number for COVID-19 over March through August 2020 across England and English NHS regions, using test-positive cases, hospital admissions and deaths with confirmed COVID-19. Our estimates of transmission potential varied for each of these sources of infections, and the divergence between estimates from each data source was not consistent within or across

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regions over time, although estimates based on hospital admissions and deaths were more spatio-temporally synchronous than compared to estimates from cases. We compared differences in R_t estimates to the extent and context of transmission and found that the difference between R_t estimated from cases, admissions and deaths may be linked to uneven rates of testing, the changing age distribution of cases and outbreaks in care home populations.

R_t estimates varied by data source, and the extent of variation itself differed by region and over time. Following the initial epidemic peak in mid-March, the date at which R_t estimates crossed below 1 varied by both data source and geography, following which R_t estimates from all data sources varied when not undergoing a clear state change. The differences in these oscillations by data source may indicate different underlying causes. This implies that each data source was influenced differently by changes in subpopulations over time.

Increasingly rapid oscillations in R_t estimates from test-positive cases were associated with higher test-positivity rates. Increasing test-positivity rates could be an indication of inconsistent community testing, with the observation of an initial rise in transmission amplified by expanded testing and local interventions where a cluster of new, mild cases had been identified [18]. This targeted testing may have driven regionally localized instability in case detection and resulting R_t estimates but may not reflect changes in underlying transmission. This is a limitation of monitoring epidemic dynamics using test-positive surveillance data in areas where testing rates vary across the population and over time. This also suggests that R_t estimates from admissions may be more reliable than that from all test-positive cases for indicating the relative intensity of an epidemic over time [37].

We hypothesized that variations in R_t estimates were also related to changes in the age distribution of cases over time, because age is associated with severity [38,39]. If each data source represented a different sample of this age-severity gradient, and transmission also varied by age or severity, R_t estimates from each source would diverge. Early in the epidemic, tests were largely limited to hospital settings, and disproportionately represented healthcare workers compared to the general population. This sampling bias would be reflected in the R_t from test-positive cases. The early peak in R_t could then represent a substantial separate route of transmission in healthcare settings, in a wave of nosocomial infections [40]. If healthcare workers were less susceptible to severe disease than those older than working age, an early peak in R_t estimated from test-positive cases would not have been represented in R_t estimated from hospital admissions or deaths. Meanwhile, either hospital admissions or deaths data would be more representative of sampling a separate route of transmission among the general population. If infections spread through the general population later than nosocomial infections, then the timing of peaks in R_t estimates from each data source would not have matched.

From late spring, outbreaks in care homes may have contributed to a divergence between R_t estimates from test-positive cases and other data sources. All regions saw a near-synchronous local peak in R_t estimated from hospital admissions over spring, which was not seen in R_t estimated from test-positive cases. This may have reflected the known widespread regional outbreaks in care homes. The care home population is on average older and more clinically vulnerable than the general population, while also being less likely to appear for community testing [41,42]. Increased transmission in care homes would then be seen in an increased R_t from hospital admissions, but not observed in an R_t from test-positive cases.

Similarly, the age-severity gradient may have impacted transmission estimates later in the epidemic when community testing became more widely available. We found that from June 2020 onwards, R_t estimates from all test-positive cases appeared to increasingly diverge away from R_t estimates from admissions and deaths, transitioning into a separate, higher, steady state. This was followed by the observed age distribution of all test-positive cases becoming increasingly younger, while the age distribution of admissions remained approximately level. Because of the severity gradient, this suggested that the R_t estimates from all test-positive cases and admissions were more biased by the relative proportion of younger cases and older cases, respectively, than the R_t estimates from admissions or deaths.

Our analysis was limited where data or modelling assumptions did not reflect underlying differences in transmission. R_t estimates can become increasingly uncertain and unstable with lower case counts. Further, estimated unobserved infections were mapped to reported cases or deaths using two delay distributions: the time from infection to test in the community or hospital, and a longer delay from infection to death. Mis-specification of the priors would have created bias in the temporal distribution of all resulting R_t estimates, with estimated dates of infection and R_t incorrectly shifted too much or too little in time compared to the true infection curve, and decreased accuracy of R_t estimates [43].

We used the same distribution priors for both delays after symptom onset to positive test, and to hospital admission. This may be inaccurate where cases with mild symptoms take longer to present for testing than severe cases presenting for hospital admission, or vice versa. The difference between the two delays over time may also have varied, with a possible decrease in delay to reported tests when mass community testing became available over the summer of 2020. This would have had a differential impact on the accuracy of R_t estimates over time in either direction, which could explain some of the oscillations in R_t estimates from test-positive case data compared to hospital admissions. We had no data over time on delays from symptom onset to reporting in each data source with which to test this hypothesis. However, we have mitigated some of the impact of this by using a sub-sampled bootstrap of the available delay data when estimating the delay distribution priors. This inflated the uncertainty of these priors in line with the hypothesis that they varied over time. This adjustment may be conservative if the delay distributions are stable over time.

Spatial dependence in delay distributions may also have contributed to their mis-specification and increased uncertainty in R_t estimates. We observed that the variation in R_t estimates from admissions and deaths often showed comparable levels and patterns in oscillations over time but were out of phase with each other. This may have been due to using data sources from different populations for each delay estimate. To estimate the delay between symptom onset to either a positive test or hospitalization, we used a linelist of all patients publicly reported globally, which had a mean delay of 5.4 days (s.d. 5.6). This varied only slightly from an early estimate in the UK epidemic, where the delay from

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onset to hospitalization had a mean of 5.14 days (s.d. 4.2) in confidential Public Health England (FF100) data [44]. Meanwhile, the same global public linelist contained few records with delay from onset to death, with mean 11.4 (s.d. 16.5). We compared this to confidential UK data from an observational study that had mean delay 14.3 days (s.d. 9.5) [32].

Comparing each type and source of delay, we judged the benefits of using open data to outweigh the minor observed spatial variation of the delay from onset to test or admission, although at the expense of increased uncertainty. However, we judged that the difference in delay from onset to death in the UK compared to public (international) data was sufficiently meaningful to justify using confidential UK data in order to maintain the accuracy of the R_t estimate from deaths. The difference in the geographical source of delay distributions should not have substantially altered our conclusions about discrepancies between central estimates of R_t from either test-positives or admissions, compared to R_t estimated from deaths. However, using the international public linelist for the delay to test or admission may have introduced additional uncertainty around the respective R_t estimates, compared to greater accuracy (reduced uncertainty) in estimates of R_t from deaths based on a UK-specific delay distribution.

The data sources themselves may also have been inaccurate or biased, which would change the representation of the population we have assumed here. For example, we excluded data from other nations of the UK (Wales, Scotland and Northern Ireland) in our analysis, as these differed in both availability over time and in data collection and reporting practices [19,45]. English regional data may also contain bias where new parts of the population might be under focus for testing efforts, or the population characteristics of hospital admissions from COVID-19 may have changed over time with changes in clinical criteria or hospital capacity for admission. This would mean that an R_t estimate from these data sources would represent different source populations over time, limiting our ability to reliably compare against R_t estimates from other data sources. Where possible we highlighted this by comparing R_t estimates to known biases and changes in case detection and reporting.

Our approach is unable to make strong causal conclusions about varying transmission, and assumptions about sampling and the representation of subpopulations remain implicit. Alternatively, varying epidemics in subpopulations could have been addressed with mechanistic models that explicitly consider transmission in different settings and are fitted to multiple data sources. However, these require additional assumptions, detailed data to parameterize and may be time-consuming to develop. In the absence of data, the number of assumptions required for these models can introduce inherent structural biases. Our approach contains few structural assumptions and therefore may be more robust when data are sparse, or information is required in real-time.

We conclude that when estimating R_t, the choice of data source should be guided by the policy context in which the estimates will be used and interpreted. This work highlights that there is no clear superior choice of data source, while R_t estimates are sensitive to assumptions about the underlying population of each data source. This means that both producers and users of R_t estimates should understand relevant biases in the data source's population sampling strategy, such as by community case detection or patient severity, before drawing conclusions about transmission in the population as a whole.

We also recommend presenting concurrent R_t estimates jointly, rather than pooling estimates of R_t from different data sources. Pooling estimates would both suffer from unclear weighting and lose useful information about variation in subpopulation transmission. Although the reconstruction of the underlying transmission process from the reporting processes is robust, it is unclear how weights would be assigned based on likelihood to estimates from different data sources. Further, the variation in concurrent R_t estimates provides more information about population transmission than any single estimate, when considered in light of the sampling biases of each data source. This additional information can be useful to identify transmission intensity by subpopulation where access to high quality disaggregated data may not be available in real time. While this can be difficult to interpret without specific knowledge of population structure and dynamics, this information would be lost altogether in a single or pooled estimate of R_t. By contrast, if the policy were to be based on either a single or an averaged R_t estimate, it would be unclear what any recommendation should be and for whom.

Future work could explore systematic differences in the influence of data sources on R_t estimates by extending the comparison of R_t by data source to other countries or infectious diseases. Additionally, work should also clarify the potential for comparing R_t estimates in real-time tracking of outbreaks and explore the inconsistencies in case detection over time and space, where a cluster of cases leads to a highly localized expansion of community testing, creating an uneven spatial bias in transmission estimates. These findings may be used to improve R_t estimation and identify findings of use for epidemic control. Based on the work presented here we now provide R_t estimates, updated each day, for test positive cases, admissions, and deaths in each NHS region and in England. Our estimates are visualized on our website, are available for download, and are produced using publicly accessible code [46,47].

Tracking differences by data source can improve understanding of variation in testing bias in data collection, highlight outbreaks in new subpopulations, indicate differential rates of transmission among vulnerable populations and clarify the strengths and limitations of each data source. Our approach can quickly identify such patterns in developing epidemics that might require further investigation and early policy intervention. Our method is simple to deploy and scale over time and space using existing open-source tools, and all code and estimates used in this work are available to be used or re-purposed by others.

5. In context

In the UK, public policy and the media have prominently used the effective reproduction number (R_t) of COVID-19 to summarise ongoing pandemic transmission. Several teams in the UK have been contributing estimates of R_t that are aggregated into a consensus range, but the methods, approaches, and data sources for estimating transmission have varied among teams and over time. For example, data sources could, amongst others, include counts of test-positive cases, hospital admissions, or deaths due to COVID-19. In

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our team's submissions to the Scientific Pandemic Influenza Group on Modelling (SPI-M) from March onwards, we saw that even when using a consistent method, R_t estimates were not a single, clear-cut number, but varied depending on the source of data.

In late May, we started to explore whether these differences in transmission estimates from each data source could be a policy-relevant indicator of biased data sampling and subpopulation epidemics. We first presented a summary of the differences in our team's R_t estimates by data source to SPI-M as a short note in early June. From June onwards we used all three data sources to estimate R_t and contributed them separately to the weekly reproduction number estimates published by SPI-M and considered by the Scientific Advisory Group for Emergencies (SAGE). Over this time, we have adapted our work to support the changing UK policy context. This has meant there are several differences in available data, methods, and implications of this work between the time we first generated the SPI-M report and the time of this publication.

As COVID-19 data became more openly accessible, we started to publish a daily comparison of UK R_t estimates by data source (epiforecasts.io/covid/posts/national/united-kingdom). This had initially been impossible as there were very few sources of public subnational data. Thanks to the Public Health England dashboard (coronavirus.data.gov.uk), public data sources for England increased in both quantity and quality and from October we were able to produce subnational R_t estimates using a variety of public data sources. We felt that presenting these estimates publicly would be useful given the high level of interest in the government's claimed use of R_t as a policy decision tool.

Between generating the original SPI-M submission and this publication, we significantly developed and improved the software we have built to estimate R_t ("EpiNow2"). We continue to refine our methods for estimating R_t, although the improved methods did not substantially change the trend or direction of differences between estimates and our resulting conclusions.

Our interpretation of the differences in R_t estimates has changed over time as we saw new evidence for concentrated transmission in subpopulations. In the earliest paper presented to SPI-M, discussion centred on the likely effects of hospital-acquired infection and testing availability on differences between R_t from test-positives compared to admissions or deaths over March and May. However, increasing evidence for a widespread and severe epidemic in care homes provided an alternative explanation for such differences. We realised that, even without disaggregated data by age or residence, simply identifying the differences in R_t estimates could have been an early indicator of the epidemic in this vulnerable subpopulation. We therefore continued to track these differences, which once again became wider over the summer as transmission moved between age groups after restrictions were lifted and mass testing became available.

Most importantly, we continue to find new insights into the state of the UK pandemic from comparing R_t estimates. One of the clearest trends we have seen in varying R_t estimates by data source has followed from the National Health Service vaccination campaign. R_t estimates from deaths are now consistently below those from hospitalisations and cases. This is a strong indicator of the positive impact of vaccination, and an encouraging further use for this work.

Data accessibility. The code that supports the findings of this study is available on Github (https://github.com/epiforecasts/rt-comparison-uk-public, https://doi.org/10.5281/zenodo.4029075). Other sources of data were derived from the following resources available in the public domain: Office for National Statistics [23]. https://www.gov.uk/government/publications/national-covid-19-surveillance-reports [21]. See https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/datasets/populationestimatesforukenglandandwalesscotlandandnorthernireland [22]. Xu *et al.* [31]. See The Health Foundation: https://www.health.org.uk/news-and-comment/charts-and-infographics/covid-19-policy-tracker [24]. See UK Government Dashboard: https://coronavirus.data.gov.uk/about-data [19]. Data not yet available in the public domain included the delay from onset to death for a sample of COVID-19 positive cases in England. However, in the main code base for this paper we have included a table of summary statistics of the bootstrapped sample of these data (https://doi.org/10.5281/zenodo.4029075).

Authors' contributions. Conceptualization: S.A., K.S.; methodology: S.A., K.S.; data curation: K.S.; formal analysis: K.S.; writing—original draft: K.S.; writing—review and editing: S.A., S.F., M.J., S.R.M., J.H., J.D.M., N.B., CMMID COVID-19 working group; supervision: S.F.

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Supplementary Information:

Exploring surveillance data biases when estimating the reproduction number: with insights into subpopulation transmission of Covid-19 in England

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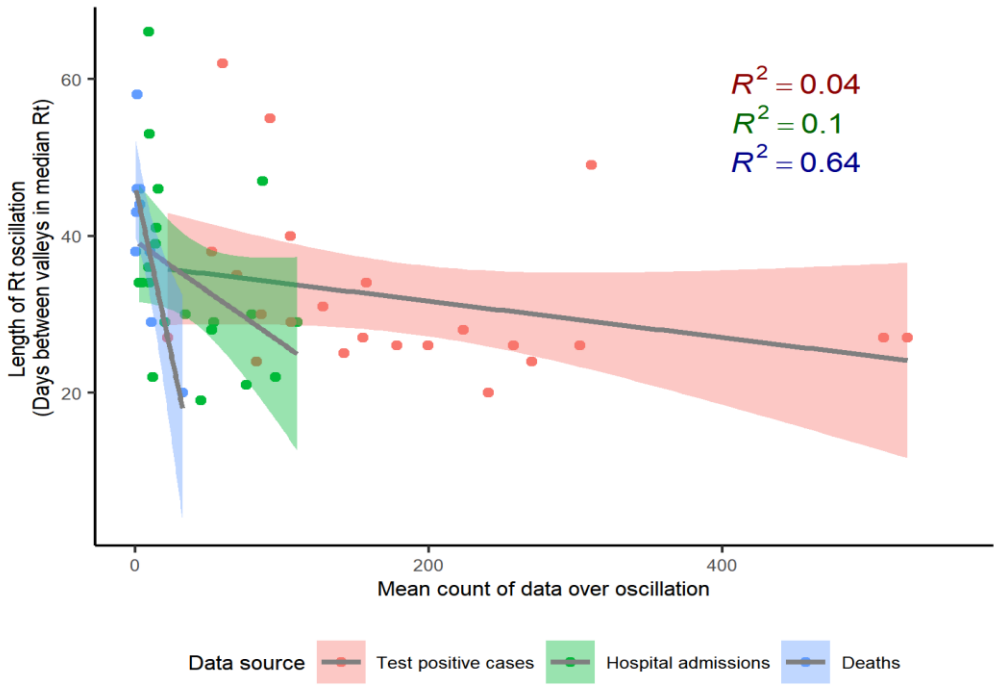

Figure SI1. Duration of each R_t oscillation against the mean of data over the oscillation. Shown with fitted linear models by data source. Scatter points represent each oscillation across 7 English NHS regions. For R_t derived from deaths, longer durations of oscillations appeared to be related to lower raw counts (coefficient: -0.88, R2 0.64), but this had a very small sample size (N oscillations = 9) and this relationship was not meaningful for cases or hospital admissions.

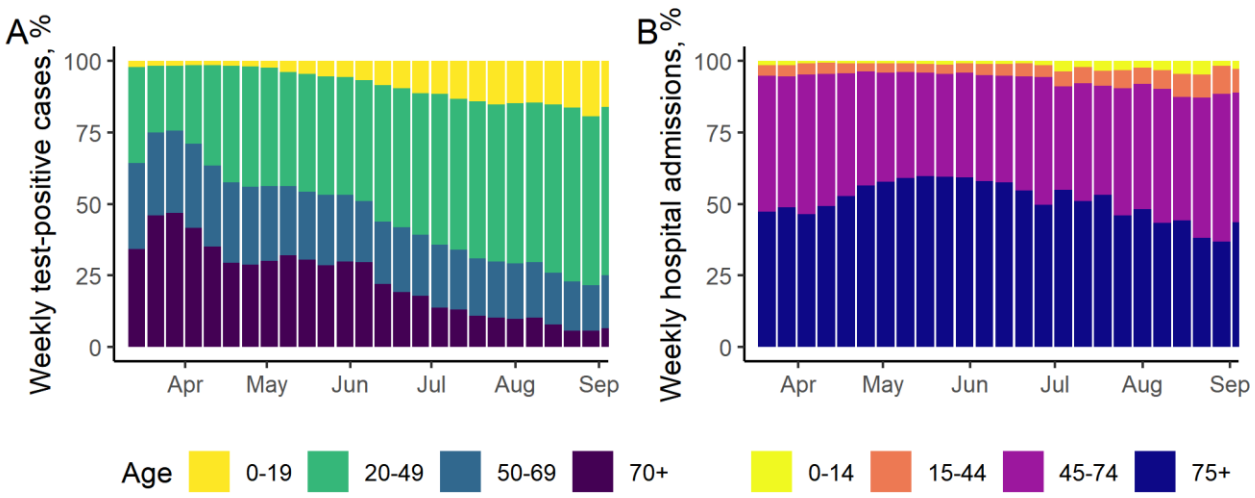

Figure SI2. Weekly percentage of cases in England by age, among all test-positive cases (A) and newly diagnosed hospital admissions (B).

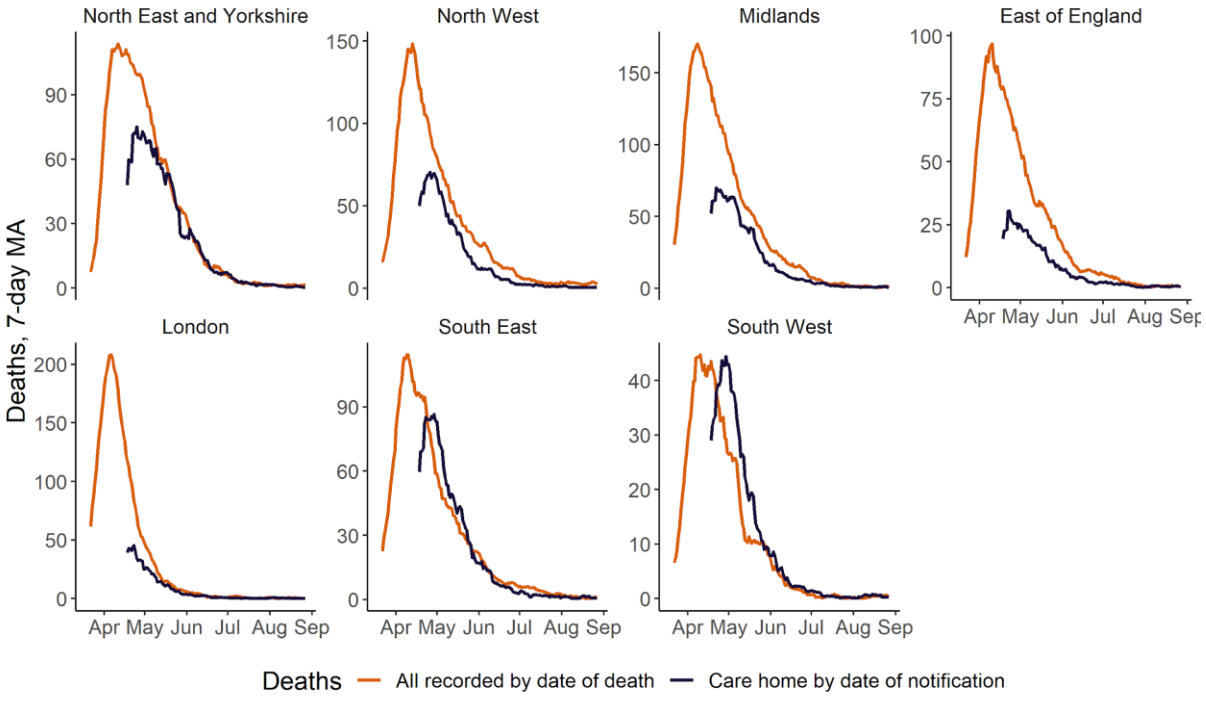

Figure SI3. Deaths in all settings by date of death, and deaths in care homes by date of notification to Clinical Quality Commission, shown by English NHS region, April to August 2020. Date of notification is typically 2-3 days after the date of death [22]. *Counts are shown as a seven-day moving average. Care homes data reported from April 2020.*

Region	Data source for Rt estimation	Earliest date median Rt <1	Median Rt after first wave* (90% CrI)	Minimum median Rt after first wave* (date, median [90%CrI])	Maximum median Rt after first wave* (date, median [90%CrI])	Number of days where median Rt crossed series median	Mean number of days between troughs** (mean, 95% CI)
England	*Test-positive cases*	01 Apr	0.97 (0.85-1.11)	23 May (0.83 [0.78-0.89])	27 Aug (1.37 [1.31-1.44])	6	32 (21-44, n=4)
	Hospital admissions	30 Mar	0.9 (0.81-0.98)	08 Apr (0.75 [0.7-0.79])	27 Aug (1.24 [1.17-1.31])	8	27 (24-29, n=3)
	Deaths	31 Mar	0.86 (0.8-0.98)	27 Jun (0.82 [0.76-0.89])	19 Aug (1.06 [0.97-1.14])	4	40 (n=1)
North East and Yorkshire	*Test-positive cases*	22 Apr	0.96 (0.82-1.13)	22 Jun (0.79 [0.73-0.85])	27 Aug (1.49 [1.4-1.58])	10	26 (23-29, n=5)
	Hospital admissions	01 Apr	0.89 (0.82-1.01)	09 Apr (0.81 [0.76-0.87])	27 Aug (1.27 [1.16-1.38])	8	31 (23-39, n=4)
	Deaths	07 Apr	0.86 (0.77-1.02)	04 Jun (0.78 [0.72-0.85])	19 Aug (1.11 [1-1.21])	2	NA
North West	*Test-positive cases*	05 Apr	0.94 (0.85-1.19)	26 May (0.82 [0.76-0.87])	27 Aug (1.45 [1.37-1.54])	4	34 (22-46, n=3)
	Hospital admissions	01 Apr	0.91 (0.74-1.09)	25 Jun (0.67 [0.6-0.75])	27 Aug (1.33 [1.22-1.44])	8	38 (30-47, n=3)
	Deaths	03 Apr	0.88 (0.8-1.02)	15 Jun (0.81 [0.73-0.88])	19 Aug (1.08 [0.97-1.17])	4	NA
Midlands	*Test-positive cases*	29 Mar	0.96 (0.84-1.17)	23 May (0.82 [0.75-0.88])	27 Aug (1.51 [1.41-1.61])	12	27 (25-28, n=5)
	Hospital admissions	29 Mar	0.91 (0.76-1.03)	07 Apr (0.7 [0.64-0.75])	27 Aug (1.2 [1.11-1.31])	6	26 (22-30, n=4)
	Deaths	31 Mar	0.86 (0.78-0.99)	24 Jun (0.78 [0.71-0.85])	19 Aug (1.11 [1.01-1.23])	8	33 (12-54, n=2)
East of England	*Test-positive cases*	02 Apr	0.96 (0.82-1.15)	09 May (0.8 [0.74-0.86])	22 Aug (1.23 [1.15-1.31])	6	36 (34-38, n=3)
	Hospital admissions	02 Apr	0.89 (0.81-0.99)	28 Jul (0.8 [0.71-0.9])	27 Aug (1.11 [0.99-1.24])	8	36 (28-45, n=3)
	Deaths	02 Apr	0.85 (0.78-0.94)	06 Jul (0.8 [0.71-0.88])	02 Apr (0.99 [0.93-1.05])	6	38 (24-51, n=2)
London	*Test-positive cases*	29 Mar	0.98 (0.77-1.15)	07 May (0.74 [0.68-0.8])	25 Aug (1.2 [1.13-1.27])	4	35 (27-43, n=2)
	Hospital admissions	29 Mar	0.89 (0.73-1.04)	08 Apr (0.67 [0.6-0.73])	27 Aug (1.23 [1.12-1.36])	4	47 (16-78, n=2)
	Deaths	26 Mar	0.84 (0.7-1.02)	13 Apr (0.68 [0.61-0.75])	19 Aug (1.07 [0.91-1.24])	4	41 (36-46, n=2)
South East	*Test-positive cases*	05 Apr	0.96 (0.82-1.15)	28 May (0.82 [0.75-0.87])	22 Aug (1.2 [1.14-1.27])	4	35 (19-51, n=3)
	Hospital admissions	01 Apr	0.89 (0.81-1.02)	31 May (0.8 [0.73-0.87])	27 Aug (1.11 [1-1.21])	8	29 (23-34, n=3)
	Deaths	02 Apr	0.86 (0.79-0.95)	25 May (0.82 [0.76-0.9])	19 Aug (0.99 [0.87-1.12])	6	48 (32-64, n=2)
South West	*Test-positive cases*	06 Apr	1.02 (0.73-1.18)	02 Jun (0.66 [0.58-0.74])	16 Jul (1.21 [1.12-1.3])	6	39 (21-58, n=3)
	Hospital admissions	01 Apr	0.9 (0.76-1.02)	03 Jun (0.7 [0.61-0.78])	27 Aug (1.02 [0.91-1.18])	6	44 (28-59, n=2)
	Deaths	04 Apr	0.85 (0.72-1.06)	03 Jun (0.72 [0.62-0.81])	07 Aug (1.05 [0.9-1.24])	2	43 (n=1)

Table SI1. Key summary statistics for R_t from when R_t crossed 1, indicating the end of the first wave of the epidemic, to the most recent estimate. 90%CrI = credible interval (quantiles around median); 95%CI = confidence interval (standard error around mean)

**End of first wave = earliest date Rt <1. Last estimate of Rt from cases and admissions: 28th August, from deaths: 19th August 2020*

***Troughs defined as local minima within a sequential fall and rise in the median Rt estimate*

Effect on regional average duration between troughs in Rt	**Influence of % test positive**		
Source of Rt	Intercept (95% CI)	Coefficient (95% CI)	Adjusted R^2
Test positive cases	39 (36.7 to 42)	-2.2 (-2.8 to -1.4)	0.87 (*p* <0.001)
Hospital admissions	41 (29.5 to 52)	-2.1 (-5.5 to 1.2)	0.18 (*p* 0.2)
Deaths	45.4 (38 to 52.5)	-3.7 (-4.7 to 0.37)	0.48 (*p* 0.07)

Table SI2. Results of individual linear regressions of test positive on each Rt estimate from test-positive cases, hospital admissions, and deaths separately. Test positivity is the average % positive of all tests conducted. Data for seven English NHS regions and England (N=8).

Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations

Sherratt K, Gruson H, Grah R, Johnson H, Niehus R, Prasse B, Sandmann F, Deuschel J, Wolffram D, Abbott S, Ullrich A, Gibson G, Ray EL, Reich NG, Sheldon D, Wang Y, Wattanachit N, Wang L, Trnka J, Obozinski G, Sun T, Thanou D, Pottier L, Krymova E, Meinke JH, Barbarossa MV, Leithauser N, Mohring J, Schneider J, Wlazlo J, Fuhrmann J, Lange B, Rodiah I, Baccam P, Gurung H, Stage S, Suchoski B, Budzinski J, Walraven R, Villanueva I, Tucek V, Smid M, Zajicek M, Perez Alvarez C, Reina B, Bosse NI, Meakin SR, Castro L, Fairchild G, Michaud I, Osthus D, Alaimo Di Loro P, Maruotti A, Eclerova V, Kraus A, Kraus D, Pribylova L, Dimitris B, Li ML, Saksham S, Dehning J, Mohr S, Priesemann V, Redlarski G, Bejar B, Ardenghi G, Parolini N, Ziarelli G, Bock W, Heyder S, Hotz T, Singh DE, Guzman-Merino M, Aznarte JL, Morina D, Alonso S, Alvarez E, Lopez D, Prats C, Burgard JP, Rodloff A, Zimmermann T, Kuhlmann A, Zibert J, Pennoni F, Divino F, Catala M, Lovison G, Giudici P, Tarantino B, Bartolucci F, Jona Lasinio G, Mingione M, Farcomeni A, Srivastava A, Montero-Manso P, Adiga A, Hurt B, Lewis B, Marathe M, Porebski P, Venkatramanan S, Bartczuk RP, Dreger F, Gambin A, Gogolewski K, Gruziel-Slomka M, Krupa B, Moszyński A, Niedzielewski K, Nowosielski J, Radwan M, Rakowski F, Semeniuk M, Szczurek E, Zielinski J, Kisielewski J, Pabjan B, Holger K, Kheifetz Y, Scholz M, Przemyslaw B, Bodych M, Filinski M, Idzikowski R, Krueger T, Ozanski T, Bracher J, Funk S. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. Elife. 2023 Apr 21;12:e81916. doi: 10.7554/eLife.81916.

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SECTION A – Student Details

Student ID Number	1701639	**Title**	Ms
First Name(s)	Katharine		
Surname/Family Name	Sherratt		
Thesis Title	Collaborative outbreak modelling for decision support: evaluating trade-offs from multi-model combination		
Primary Supervisor	Sebastian Funk		

If the Research Paper has previously been published please complete Section B, if not please move to Section C.

SECTION B – Paper already published

Where was the work published?	eLife		
When was the work published?	June 2023		
If the work was published prior to registration for your research degree, give a brief rationale for its inclusion	PhD by Publication		
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Stage of publication	Choose an item.

SECTION D – Multi-authored work

For multi-authored work, give full details of your role in the research included in the paper and in the preparation of the paper. (Attach a further sheet if necessary)	My contribution to this work is twofold. Firstly, from January 2021 I jointly developed, built, and maintained the infrastructure for the European COVID-19 Forecast Hub, together with Sebastian Funk, Johannes Bracher, Hugo Gruson, and many collaborators. I led the adaptation of code for forecast processing and validation; data sourcing, processing, and validation; all documentation; and the initial ensemble. I also contributed code for the forecast evaluation procedures, and website design. This drew on existing software from both the US and Germany/Poland COVID-19 Forecast Hubs, and was planned and reviewed jointly with Sebastian Funk. From March 2021, I led work supporting forecasters to contribute to the Hub, and continuously maintained the weekly updating of data sources, forecasts, ensemble, and evaluation. Second, I led work developing this paper focussing on the performance of the resulting Hub ensemble. I led this work from conceptualisation to analysis and drafting, with supervision by Sebastian Funk. I designed the plan for this work and developed all code to conduct analysis, including accessing forecasts, observed data, and evaluation scores; descriptive summaries of individual model and ensemble performance; and visualisations. Sebastian Funk contributed code for weighted ensembles and their evaluation. I wrote the first and subsequent drafts of the paper. We invited initial feedback from close collaborators at the ECDC and in Germany (Rene Niehus, Rok Grah, Frank Sandmann, Bastian Prasse, and Johannes Bracher). We invited a further round of review from all individuals who were named in metadata of the forecasts contributed to the Hub, who had therefore contributed component forecasts to the ensemble. I led work on all subsequent drafts, submission for peer review, and revisions. All contributions are documented on Github: https://github.com/epiforecasts/euro-hub-ensemble.

SECTION E

Student Signature	Katharine Sherratt
Date	14 June 2024

Supervisor Signature	Sebastian Funk
Date	14 June 2024

RESEARCH ARTICLE

Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations

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Abstract

Background: Short-term forecasts of infectious disease burden can contribute to situational awareness and aid capacity planning. Based on best practice in other fields and recent insights in infectious disease epidemiology, one can maximise the predictive performance of such forecasts if multiple models are combined into an ensemble. Here, we report on the performance of ensembles in predicting COVID-19 cases and deaths across Europe between 08 March 2021 and 07 March 2022.

Methods: We used open-source tools to develop a public European COVID-19 Forecast Hub. We invited groups globally to contribute weekly forecasts for COVID-19 cases and deaths reported by a standardised source for 32 countries over the next 1–4 weeks. Teams submitted forecasts from March 2021 using standardised quantiles of the predictive distribution. Each week we created an ensemble forecast, where each predictive quantile was calculated as the equally-weighted average (initially the mean and then from 26th July the median) of all individual models' predictive quantiles. We measured the performance of each model using the relative Weighted Interval Score (WIS), comparing models' forecast accuracy relative to all other models. We retrospectively explored alternative methods for ensemble forecasts, including weighted averages based on models' past predictive performance.

Results: Over 52 weeks, we collected forecasts from 48 unique models. We evaluated 29 models' forecast scores in comparison to the ensemble model. We found a weekly ensemble had a consistently strong performance across countries over time. Across all horizons and locations, the

ensemble performed better on relative WIS than 83% of participating models' forecasts of incident cases (with a total N=886 predictions from 23 unique models), and 91% of participating models' forecasts of deaths (N=763 predictions from 20 models). Across a 1–4 week time horizon, ensemble performance declined with longer forecast periods when forecasting cases, but remained stable over 4 weeks for incident death forecasts. In every forecast across 32 countries, the ensemble outperformed most contributing models when forecasting either cases or deaths, frequently outperforming all of its individual component models. Among several choices of ensemble methods we found that the most influential and best choice was to use a median average of models instead of using the mean, regardless of methods of weighting component forecast models.

Conclusions: Our results support the use of combining forecasts from individual models into an ensemble in order to improve predictive performance across epidemiological targets and populations during infectious disease epidemics. Our findings further suggest that median ensemble methods yield better predictive performance more than ones based on means. Our findings also highlight that forecast consumers should place more weight on incident death forecasts than incident case forecasts at forecast horizons greater than 2 weeks.

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Editor's evaluation

This large-scale collaborative study is a timely contribution that will be of interest to researchers working in the fields of infectious disease forecasting and epidemic control. This paper provides a comprehensive evaluation of the predictive skills of real-time COVID-19 forecasting models in

Europe. The conclusions of the paper are well supported by the data and are consistent with findings from studies in other countries.

Introduction

Epidemiological forecasts make quantitative statements about a disease outcome in the near future. Forecasting targets can include measures of prevalent or incident disease and its severity, for some population over a specified time horizon. Researchers, policy makers, and the general public have used such forecasts to understand and respond to the global outbreaks of COVID-19 (***Van Basshuysen et al., 2021***; ***CDC, 2020***; ***European Centre for Disease Prevention and Control, 2021c***). At the same time, forecasters use a variety of methods and models for creating and publishing forecasts, varying in both defining the forecast outcome and in reporting the probability distribution of outcomes (***Zelner et al., 2021***; ***James et al., 2021***).

Within Europe, comparing forecasts across both models and countries can support a range of national policy needs simultaneously. European public health professionals operate across national, regional, and continental scales, with strong existing policy networks in addition to rich patterns of cross-border migration influencing epidemic dynamics. A majority of European countries also cooperate in setting policy with inter-governmental European bodies such as the European Centre for Disease Prevention and Control (ECDC). In this case, a consistent approach to forecasting across the continent as a whole can support accurately informing cross-European monitoring, analysis, and guidance (***European Centre for Disease Prevention and Control, 2021c***). At a regional level, multi-country forecasts can support a better understanding of the impact of regional migration networks. Meanwhile, where there is limited capacity for infectious disease forecasting at a national level, forecasters generating multi-country results can provide an otherwise-unavailable opportunity for forecasts to inform national situational awareness. Some independent forecasting models have sought to address this by producing multi-country results (***Aguas et al., 2020***; ***Adib et al., 2021***; ***Agosto and Giudici, 2020***; ***Agosto et al., 2021***).

Variation in forecast methods and presentation makes it difficult to compare predictive performance between forecast models, and from there to derive objective arguments for using one forecast over another. This confounds the selection of a single representative forecast and reduces the reliability of the evidence base for decisions based on forecasts. A 'forecast hub' is a centralised effort to improve the transparency and usefulness of forecasts, by standardising and collating the work of many independent teams producing forecasts (***Reich et al., 2019a***). A hub sets a commonly agreed-upon structure for forecast targets, such as type of disease event, spatio-temporal units, or the set of quantiles of the probability distribution to include from probabilistic forecasts. For instance, a hub may collect predictions of the total number of cases reported in a given country for each day in the next 2 weeks. Forecasters can adopt this format and contribute forecasts for centralised storage in the public domain.

This shared infrastructure allows forecasts produced from diverse teams and methods to be visualised and quantitatively compared on a like-for-like basis, which can strengthen public and policy use of disease forecasts. The underlying approach to creating a forecast hub was pioneered in climate modelling and adapted for collaborative epidemiological forecasts of dengue (***Johansson et al., 2019***) and influenza in the USA (***Reich et al., 2019a***; ***Reich et al., 2019b***). This infrastructure was adapted for forecasts of short-term COVID-19 cases and deaths in the US (***Cramer et al., 2021a***; ***Ray et al., 2020***), prompting similar efforts in some European countries (***Bracher et al., 2021c***; ***Funk et al., 2020***; ***Bicher et al., 2020***).

Standardising forecasts allows for combining multiple forecasts into a single ensemble with the potential for an improved predictive performance. Evidence from previous efforts in multi-model infectious disease forecasting suggests that forecasts from an ensemble of models can be consistently high performing compared to any one of the component models (***Johansson et al., 2019***; ***Reich et al., 2019b***; ***Viboud et al., 2018***). Elsewhere, weather forecasting has a long-standing use of building ensembles of models using diverse methods with standardised data and formatting in order to improve performance (***Buizza, 2019***; ***Moran et al., 2016***).

The European COVID-19 Forecast Hub (***European Covid-19 Forecast Hub, 2023d***) is a project to collate short-term forecasts of COVID-19 across 32 countries in the European region. The Hub is

funded and supported by the ECDC, with the primary aim to provide reliable information about the near-term epidemiology of the COVID-19 pandemic to the research and policy communities and the general public (***European Centre for Disease Prevention and Control, 2021c***). Second, the Hub aims to create infrastructure for storing and analysing epidemiological forecasts made in real time by diverse research teams and methods across Europe. Third, the Hub aims to maintain a community of infectious disease modellers underpinned by open science principles.

We started formally collating and combining contributions to the European Forecast Hub in March 2021. Here, we investigate the predictive performance of an ensemble of all forecasts contributed to the Hub in real time each week, as well as the performance of variations of ensemble methods created retrospectively.

Materials and methods

We developed infrastructure to host and analyse prospective forecasts of COVID-19 cases and deaths. The infrastructure is compatible with equivalent research software from the US (***Cramer et al., 2021c***; ***Wang et al., 2021***) and German and Polish COVID-19 (***Bracher et al., 2020***) Forecast Hubs, and easy to replicate for new forecasting collaborations.

Forecast targets and models

We sought forecasts for the incidence of COVID-19 as the total reported number of cases and deaths per week. We considered forecasts for 32 countries in Europe, including all countries of the European Union, European Free Trade Area, and the United Kingdom. We compared forecasts against observed data reported for each country by Johns Hopkins University (JHU, ***Dong et al., 2020***). JHU data sources included a mix of national and aggregated subnational data. We aggregated incidence over the Morbidity and Mortality Weekly Report (MMWR) epidemiological week definition of Sunday through Saturday.

Teams could express their uncertainty around any single forecast target by submitting predictions for up to 23 quantiles (from 0.01 to 0.99) of the predictive probability distribution. Teams could also submit a single point forecast. At the first submission, we asked teams to add a pre-specified set of metadata briefly describing the forecasting team and methods (provided online and in supplementary information). No restrictions were placed on who could submit forecasts. To increase participation, we actively contacted known forecasting teams across Europe and the US and advertised among the ECDC network. Teams submitted a broad spectrum of model types, ranging from mechanistic to empirical models, agent-based and statistical models, and ensembles of multiple quantitative or qualitative models (described at ***European Covid-19 Forecast Hub, 2023a***). We maintain a full project specification with a detailed submissions protocol (***European Covid-19 Forecast Hub, 2023c***).

We collected forecasts submitted weekly in real time over the 52-week period from 08 March 2021 to 07 March 2022. Teams submitted at latest 2 days after the complete dataset for the latest forecasting week became available each Sunday. We implemented an automated validation programme to check that each new forecast conformed to standardised formatting. Forecast validation ensured a monotonic increase of predictions with each increasing quantile, integer-valued non-negative counts of predicted cases, as well as consistent date and location definitions.

Each week we used all available valid forecasts to create a weekly real-time ensemble model (referred to as 'the ensemble' from here on), for each of the 256 possible forecast targets: incident cases and deaths in 32 locations over the following one through 4 weeks. The ensemble method was an unweighted average of all models' forecast values, at each predictive quantile for a given location, target, and horizon. From 08 March 2021, we used the arithmetic mean. However we noticed that including highly anomalous forecasts in a mean ensemble produced extremely wide uncertainty. To mitigate this, from 26th July 2021 onwards the ensemble instead used a median of all predictive quantiles.

We created an open and publicly accessible interface to the forecasts and ensemble, including an online visualisation tool allowing viewers to see past data and interact with one or multiple forecasts for each country and target for up to 4 weeks' horizon (***European Covid-19 Forecast Hub, 2023b***). All forecasts, metadata, and evaluations are freely available and held on Github (***European Covid-19 Forecast Hub, 2023d***) (archived in real-time at ***Sherratt, 2022***), and Zoltar, a platform for hosting

epidemiological forecasts (***EpiForecasts, 2021***; ***Reich et al., 2021***). In the codebase for this study (***covid19-forecast-hub-europe, 2022***) we provide a simple method and instructions for downloading and preparing these data for analysis using R. We encourage other researchers to freely use and adapt this to support their own analyses.

Forecast evaluation

In this study, we focused only on the comparative performance of forecasting models relative to each other. Performance in absolute terms is available on the Hub website (***European Covid-19 Forecast Hub, 2023b***). For each model, we assessed calibration and overall predictive performance. We evaluated all previous forecasts against actual observed values for each model, stratified by the forecast horizon, location, and target. We calculated scores using the *scoringutils* R package (***Bosse et al., 2023***). We removed any forecast surrounding (both the week of, and the first week after) a strongly anomalous data point. We defined anomalous as where any subsequent data release revised that data point by over 5%.

To investigate calibration, we assessed coverage as the correspondence between the forecast probability of an event and the observed frequency of that event. This usage follows previous work in epidemic forecasting (***Bracher et al., 2021a***), and is related to the concept of reliability for binary forecasts. We established the accuracy of each model's prediction boundaries as the coverage of the predictive intervals. We calculated coverage at a given interval level k, where $k \in [0, 1]$, as the proportion p of observations that fell within the corresponding central predictive intervals across locations and forecast dates. A perfectly calibrated model would have $p = k$ at all 11 levels (corresponding to 22 quantiles excluding the median). An underconfident model at level k would have $p > k$, i.e. more observations fall within a given interval than expected. In contrast, an overconfident model at level k would have $p < k$, i.e. fewer observations fall within a given interval than expected. We here focus on coverage at the $k = 0.5$ and $k = 0.95$ levels.

We also assessed the overall predictive performance of weekly forecasts using the Weighted Interval Score~(WIS) across all available quantiles. The WIS represents a parsimonious approach to scoring forecasts based on uncertainty represented as forecast values across a set of quantiles (***Bracher et al., 2021a***), and is a strictly proper scoring rule, that is, it is optimal for predictions that come from the data-generating model. As a consequence, the WIS encourages forecasters to report predictions representing their true belief about the future (***Gneiting and Raftery, 2007***). Each forecast for a given location and date is scored based on an observed count of weekly incidence, the median of the predictive distribution and the predictive upper and lower quantiles corresponding to the central predictive interval level.

Not all models provided forecasts for all locations and dates, and we needed to compare predictive performance in the face of various levels of missingness across each forecast target. Therefore we calculated a relative WIS. This is a measure of forecast performance which takes into account that different teams may not cover the same set of forecast targets (i.e. weeks and locations). The relative WIS is computed using a *pairwise comparison tournament* where for each pair of models a mean score ratio is computed based on the set of shared targets. The relative WIS of a model with respect to another model is then the ratio of their respective geometric mean of the mean score ratios, such that smaller values indicate better performance.

We scaled the relative WIS of each model with the relative WIS of a baseline model, for each forecast target, location, date, and horizon. The baseline model assumes case or death counts stay the same as the latest data point over all future horizons, with expanding uncertainty, described previously in ***Cramer et al., 2021b***. In this study, we report the relative WIS of each model with respect to the baseline model.

Retrospective ensemble methods

We retrospectively explored alternative methods for combining forecasts for each target at each week. A natural way to combine probability distributions available in the quantile format ***Genest, 1992*** used here is

$$F^{-1}(\alpha) = \sum_{i=1}^{n} w_i F_i^{-1}(\alpha),$$

Where $F_1 \ldots F_n$ are the cumulative distribution functions of the individual probability distributions (in our case, the predictive distributions of each forecast model i contributed to the hub), w_i are a set of weights in $[0, 1]$; and α are the quantile levels, such that following notation introduced in ***Genest, 1992***,

$$F^{-1}(\alpha) = \inf\{t : F_i(t) \geq \alpha\}.$$

Different ensemble choices then mainly translate to the choice of weights w_i. An arithmetic mean ensemble uses weights at $w_i = 1/n$, where all weights are equal and sum up to 1.

Alternatively, we can choose a set of weights to apply to forecasts before they are combined. Numerous options exist for choosing these weights with the aim to maximise predictive performance, including choosing weights to reflect each forecast's past performance (thereby moving from an untrained to a trained ensemble). A straightforward choice is so-called inverse score weighting. In this case, the weights are calculated as

$$w_i = \frac{1}{S_i},$$

where S_i reflects the forecasting skill calculated as the relative WIS of forecaster i, calculated over all available model data, and normalised so that weights sum to 1. This method of weighting was found in the US to outperform unweighted scores during some time periods (***Taylor and Taylor, 2023***) but this was not confirmed in a similar study in Germany and Poland (***Bracher et al., 2021c***).

When constructing ensembles from quantile means, a single outlier can have an oversized effect on the ensemble forecast. Previous research has found that a median ensemble, replacing the arithmetic mean of each quantile with a median of the same values, yields competitive performance while

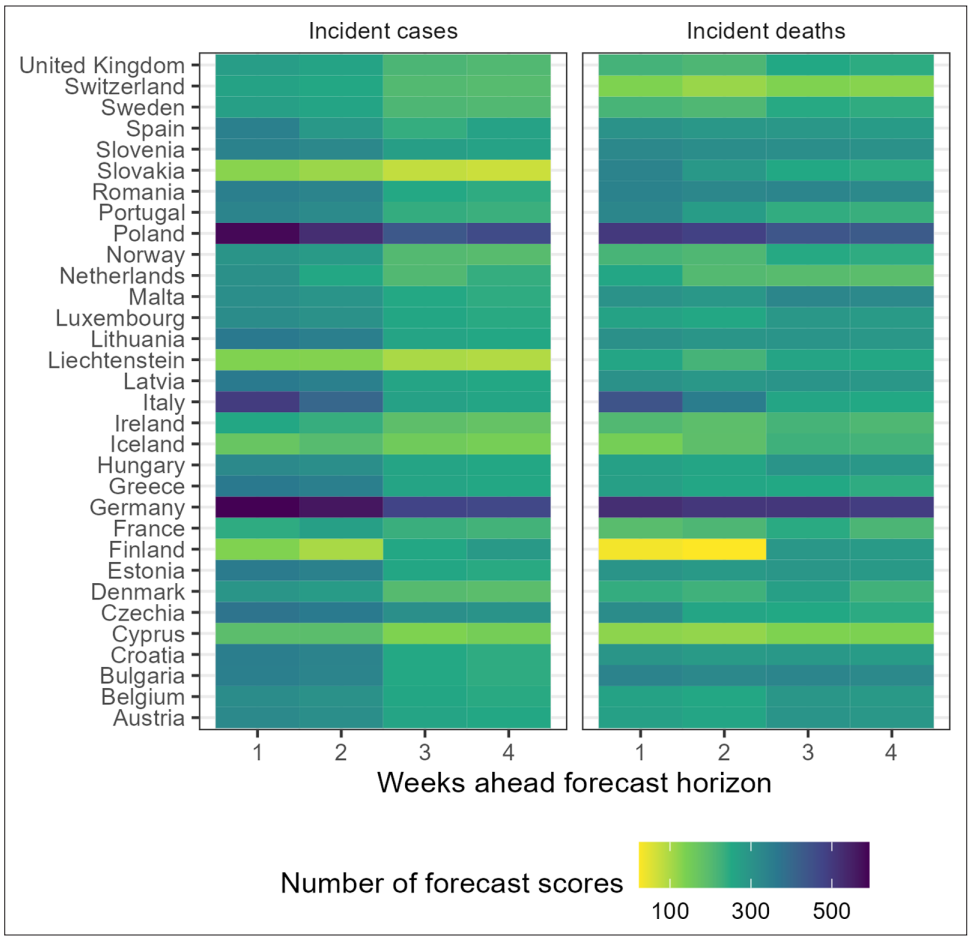

Figure 1. Total number of forecasts included in evaluation, by target location, week ahead horizon, and variable.

maintaining robustness to outlying forecasts (***Ray et al., 2022***). Building on this, we also created weighted median ensembles using the weights described above and a Harrel-Davis quantile estimator with a beta function to approximate the weighted percentiles (***Harrell and Davis, 1982***). We then compared the performance of unweighted and inverse relative WIS weighted mean and median ensembles, comparing the ratio of interval scores between each ensemble model relative to the baseline model.

Results

For 32 European countries, we collected, visualised, and made available online weekly COVID-19 forecasts and observed data (***Sherratt, 2022***). Over the whole study period, we collected forecasts from 48 unique models. Modellers created forecasts choosing from a set of 32 possible locations, four time horizons, and two variables, and modellers variously joined and left the Hub over time. This meant the number of models contributing to the Hub varied over time and by forecasting target. Using all models and the ensemble, we created 2139 forecasting scores, where each score summarises a unique combination of forecasting model, variable, country, and week ahead horizon (***Figure 1***).

Of the total 48 models, we received the most forecasts for Germany, with 29 unique models submitting 1-week case forecasts, while only 12 models ever submitted 4-week case or death forecasts for Liechtenstein. Modelling teams also differed in how they expressed uncertainty. Only three models provided point forecasts with no estimate of uncertainty around their predictions, while 41 models provided the full set of 23 probabilistic quantiles across the predictive distribution for each target.

In this evaluation we included 29 models in comparison to the ensemble forecast (***Figure 1***). We have included metadata provided by modellers in the supplement and online (***Sherratt, 2022***). In this evaluation, at most 15 models contributed forecasts for cases in Germany at the 1 week horizon, with an accumulated 592 forecast scores for that single target over the study period. In contrast, deaths in Finland at the 2 week horizon saw the smallest number of forecasts, with only 6 independent models contributing 24 forecast scores at any time over the 52-week period. Of the 29 models included in this evaluation, 5 models provided less than the full set of 23 quantiles, and were excluded when creating the ensemble. No ensemble forecast was composed of less than 3 independent models.

We visually compared the absolute performance of forecasts in predicting numbers of incident cases and deaths. We observed that forecasts predicted well in times of stable epidemic behaviour,

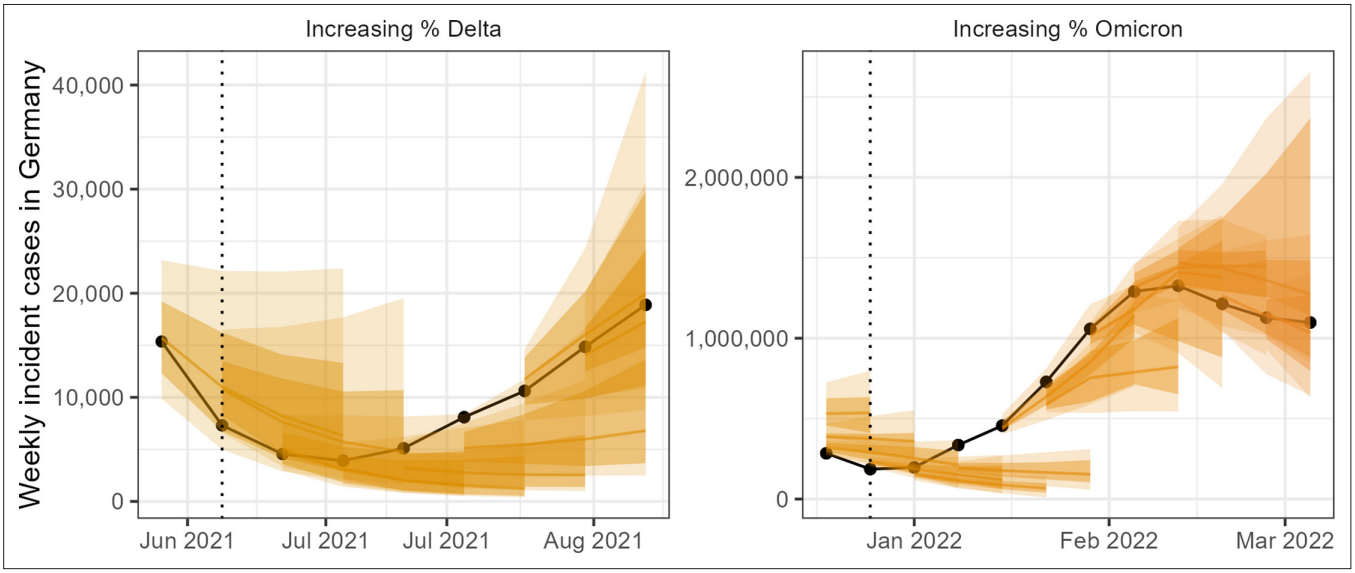

Figure 2. Ensemble forecasts of weekly incident cases in Germany over periods of increasing SARS-CoV-2 variants Delta (B.1.617.2, left) and Omicron (B.1.1.529, right). Black indicates observed data. Coloured ribbons represent each weekly forecast of 1–4 weeks ahead (showing median, 50%, and 90% probability). For each variant, forecasts are shown over an x-axis bounded by the earliest dates at which 5% and 99% of sequenced cases were identified as the respective variant of concern, while vertical dotted lines indicate the approximate date that the variant reached dominance (>50% sequenced cases).

while struggling to accurately predict at longer horizons around inflection points, for example during rapid changes in population-level behaviour or surveillance. Forecast models varied widely in their ability to predict and account for the introduction of new variants, giving the ensemble forecast over these periods a high level of uncertainty. An example of weekly forecasts from the ensemble model is shown in ***Figure 2***.

In relative terms, the ensemble of all models performed well compared to both its component models and the baseline. By relative WIS scaled against a baseline of 1 (where a score <1 indicates outperforming the baseline), the median score of forecasts from the Hub ensemble model was 0.71, within an interquartile range of 0.61 at 25% probability to 0.88 at 75% probability. Meanwhile the median score of forecasts across all participating models (excluding the Hub ensemble) was 1.04 (IQR 0.82–1.36).

Across all horizons and locations, the ensemble performed better on scaled relative WIS than 83% of forecast scores when forecasting cases (with a total N=886 from 23 unique models), and 91% of scores for forecasts of incident deaths (N=763 scores from 20 models). We also saw high performance from the ensemble when evaluating against all models including those who did not submit the full set of probabilistic quantile predictions (80% for cases with N=1006 scores from 28 models, and 88% for deaths, N=877 scores from 24 models).

The performance of individual and ensemble forecasts varied by length of the forecast horizon (***Figure 3***). At each horizon, the typical performance of the ensemble outperformed both the baseline model and the aggregated scores of all its component models, although we saw wide variation between individual models in performance across horizons. Both individual models and the ensemble saw a trend of worsening performance at longer horizons when forecasting cases with the median scaled relative WIS of the ensemble across locations worsened from 0.62 for 1-week ahead forecasts to 0.9 when forecasting 4 weeks ahead. Performance for forecasts of deaths was more stable over one through 4 weeks, with median ensemble performance moving from 0.69 to 0.76 across the 4-week horizons.

We observed similar trends in performance across horizon when considering how well the ensemble was calibrated with respect to the observed data. At 1 week ahead the case ensemble was well calibrated (ca. 50% and 95% nominal coverage at the 50% and 95% levels, respectively). This did not hold at longer forecast horizons as the case forecasts became increasingly over-confident. Meanwhile, the ensemble of death forecasts was well calibrated at the 95% level across all horizons, and the calibration of death forecasts at the 50% level improved with lengthening horizons compared to being underconfident at shorter horizons.

The ensemble also performed consistently well in comparison to individual models when forecasting across countries (***Figure 4***). In total, across 32 countries forecasting for 1 through 4 weeks, when forecasting cases the ensemble outperformed 75% of component models in 22 countries, and outperformed all available models in 3 countries. When forecasting deaths, the ensemble outperformed 75% and 100% of models in 30 and 8 countries, respectively. Considering only the the 2-week horizon shown in ***Figure 4***, the ensemble of case forecasts outperformed 75% models in 25 countries and all models in only 12 countries. At the 2-week horizon for forecasts of deaths, the ensemble outperformed 75% and 100% of its component models in 30 and 26 countries, respectively.

We considered alternative methods for creating ensembles from the participating forecasts, using either a mean or median to combine either weighted or unweighted forecasts. We evaluated each alternative ensemble model against the baseline model, taking the mean score ratio across all targets (***Table 1***). Across locations we observed that the median outperformed the mean across all one through 4 week horizons and both cases and death targets, for all but cases at the 1 week horizon. This held regardless of whether the component forecasts were weighted or unweighted by their individual past performance. Between methods of combination, weighting made little difference to the performance of the median ensemble, but appeared to improve performance of a mean ensemble in forecasting deaths.

Discussion

We collated 12 months of forecasts of COVID-19 cases and deaths across 32 countries in Europe, collecting from multiple independent teams and using a principled approach to standardising both forecast targets and the predictive distribution of forecasts. We combined these into an ensemble

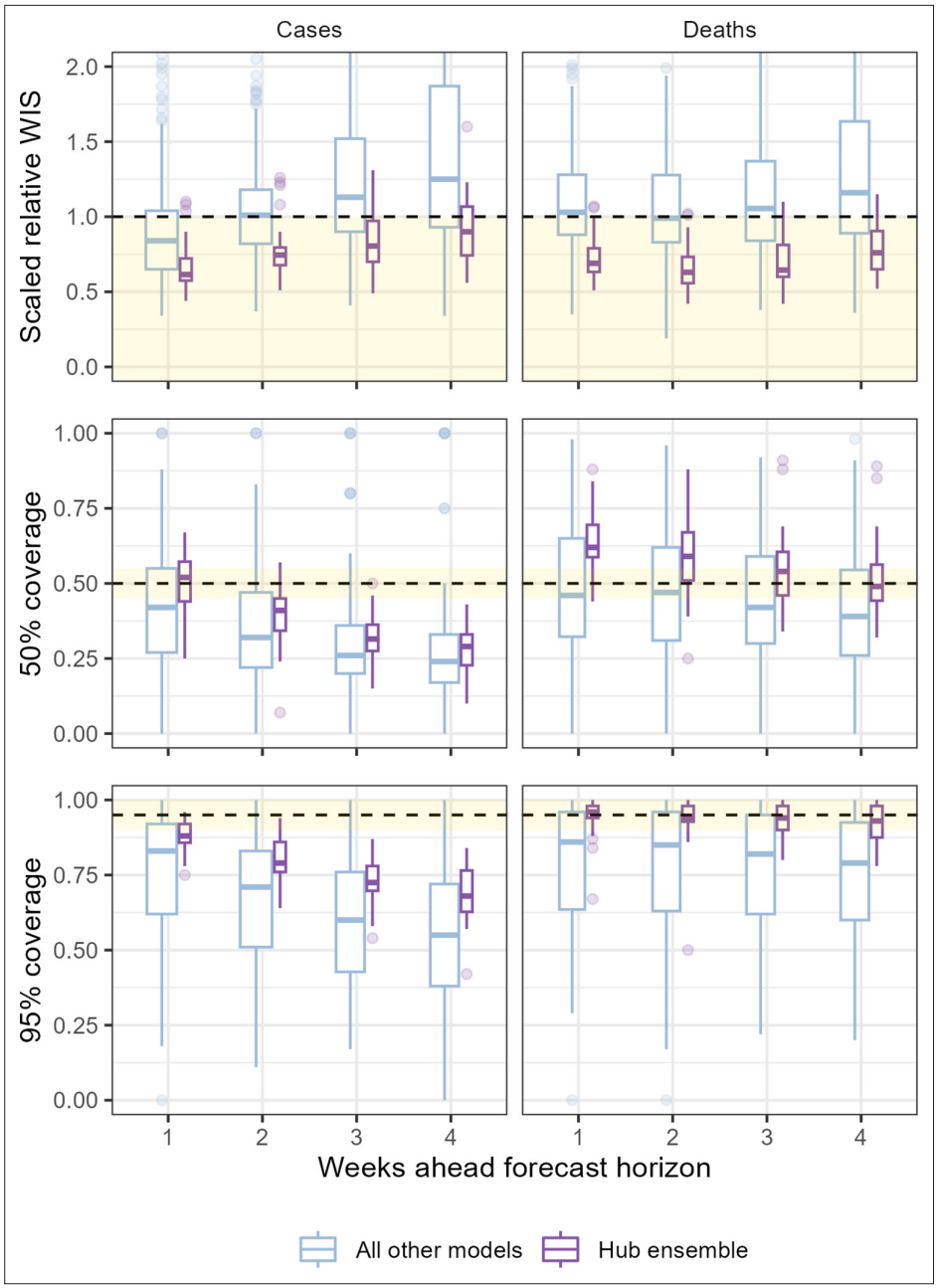

Figure 3. Performance of short-term forecasts aggregated across all individually submitted models and the Hub ensemble, by horizon, forecasting cases (left) and deaths (right). Performance measured by relative weighted interval score scaled against a baseline (dotted line, 1), and coverage of uncertainty at the 50% and 95% levels. Boxplot, with width proportional to number of observations, show interquartile ranges with outlying scores as faded points. The target range for each set of scores is shaded in yellow.

forecast and compared the relative performance of forecasts between models, finding that the ensemble forecasts outperformed most individual models across all countries and horizons over time.

Across all models we observed that forecasting changes in trend in real time was particularly challenging. Our study period included multiple fundamental changes in viral-, individual-, and population-level factors driving the transmission of COVID-19 across Europe. In early 2021, the introduction of vaccination started to change population-level associations between infections, cases, and deaths (***European Centre for Disease Prevention and Control, 2021b***), while the Delta variant emerged and became dominant (***European Centre for Disease Prevention and Control, 2021a***). Similarly from

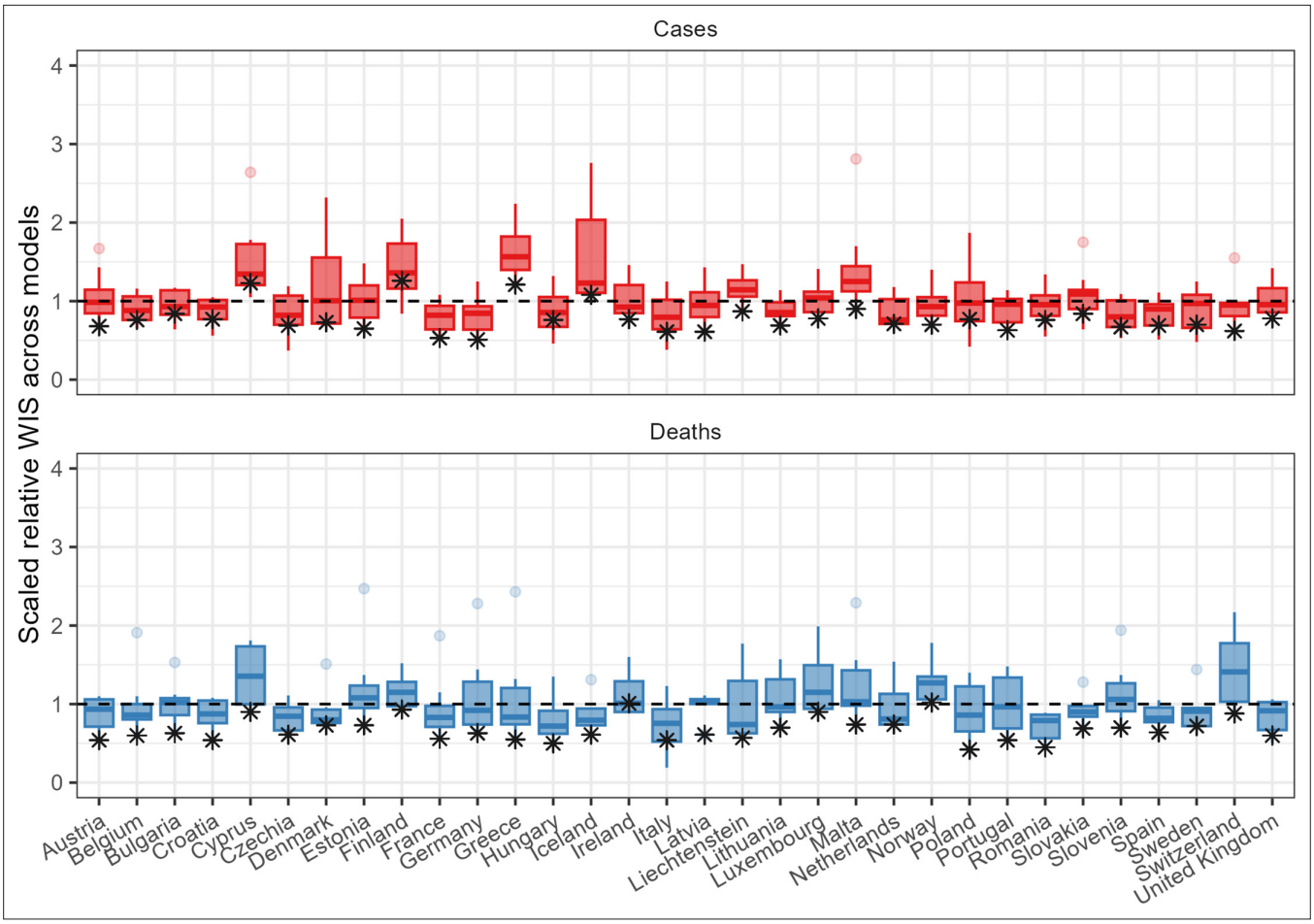

Figure 4. Performance of short-term forecasts across models and median ensemble (asterisk), by country, forecasting cases (top) and deaths (bottom) for 2-week ahead forecasts, according to the relative weighted interval score. Boxplots show interquartile ranges, with outliers as faded points, and the ensemble model performance is marked by an asterisk. y-axis is cut-off to an upper bound of 4 for readability.

Table 1. Predictive performance of main ensembles, as measured by the mean ratio of interval scores against the baseline ensemble.

Horizon	Weighted mean	Weighted median	Unweighted mean	Unweighted median
Cases				
1 week	0.63	0.64	0.61	0.64
2 weeks	0.72	0.71	0.69	0.69
3 weeks	0.82	0.76	0.82	0.72
4 weeks	1.07	0.86	1.12	0.78
Deaths				
1 week	0.65	0.61	1.81	0.61
2 weeks	0.58	0.54	1.29	0.54
3 weeks	0.64	0.57	1.17	0.53
4 weeks	0.82	0.67	0.84	0.62

late 2021 we saw the interaction of individually waning immunity during the emergence and global spread of the Omicron variant (***European Centre for Disease Prevention and Control, 2022b***). Neither the extent nor timing of these factors were uniform across European countries covered by the Forecast Hub (***European Centre for Disease Prevention and Control, 2023***). This meant that the performance of any single forecasting model depended partly on the ability, speed, and precision with which it could adapt to new conditions for each forecast target.

We observed a contrast between a more stable performance of forecasting deaths further into the future compared to forecasts of cases. Previous work has found rapidly declining performance for case forecasts with increasing horizon (***Cramer et al., 2021b***; ***Castro et al., 2020***), while death forecasts can perform well with up to 6 weeks lead time (***Friedman et al., 2021***). We can link this to the specific epidemic dynamics in this study.

First, COVID-19 has a typical serial interval of less than a week (***Alene et al., 2021***). This implies that case forecasts of more than 2 weeks only remain valid if rates of both transmission and detection remain stable over the entire forecast horizon. In contrast, we saw rapid changes in epidemic dynamics across many countries in Europe over our study period, impacting the longer term case forecasts.

Second, we can interpret the higher reliability of death forecasts as due to the different lengths and distributions of time lags from infection to case and death reporting (***Jin, 2021***). For example, a spike in infections may be matched by a consistently sharp increase in case reporting, but a longer tailed distribution of the subsequent increase in death reports. This creates a lower magnitude of fluctuation in the time-series of deaths compared to that of cases. Similarly, surveillance data for death reporting is substantially more consistent, with fewer errors and retrospective corrections, than case reporting (***Català et al., 2021***).

Third, we also note that the performance of trend-based forecasts may have benefited from the slower changes to trends in incident deaths caused by gradually increasing vaccination rates. These features allow forecasters to incorporate the effect of changes in transmission more easily when forecasting deaths, compared to cases.

We found the ensemble in this study continued to outperform both other models and the baseline at up to 4 weeks ahead. Our results support previous findings that ensemble forecasts are the best or nearly the best performing models with respect to absolute predictive performance and appropriate coverage of uncertainty (***Funk et al., 2020***; ***Viboud et al., 2018***; ***Cramer et al., 2021b***). While the ensemble was consistently high performing, it was not strictly dominant across all forecast targets, reflecting findings from previous comparable studies of COVID-19 forecasts (***Bracher et al., 2021c***; ***Brooks, 2020***). Our finding suggests the usefulness of an ensemble as a robust summary when forecasting across many spatio-temporal targets, without replacing the importance of communicating the full range of model predictions.

When exploring variations in ensemble methods, we found that the choice of median over means yielded the most consistent improvement in predictive performance, regardless of the method of weighting. Other work has supported the importance of the median in providing a stable forecast that better accounts for outlier forecasts than the mean (***Brooks, 2020***), although this finding may be dependent on the quality of the individual forecast submissions. In contrast, weighing models by past performance did not result in any consistent improvement in performance. This is in line with existing mixed evidence for any optimal ensemble method for combining short term probabilistic infectious disease forecasts. Many methods of combination have performed competitively in analyses of forecasts for COVID-19 in the US, including the simple mean and weighted approaches outperforming unweighted or median methods (***Taylor and Taylor, 2023***). This contrasts with later analyses finding weighted methods to give similar performance to a median average (***Ray et al., 2020***; ***Brooks, 2020***). We can partly explain this inconsistency if performance of each method depends on the outcome being predicted (cases, deaths), its count (incident, cumulative) and absolute level, the changing disease dynamics, and the varying quality and quantity of forecasting teams over time.

We note several limitations in our approach to assessing the relative performance of an ensemble among forecast models. While we have described differences in model scores, we have not used any formal statistical test for comparing forecast scores, such as the Diebold-Mariano test (***Diebold and Mariano, 1995***), recognising that it is unclear how this is best achieved across many models. Our results are the outcome of evaluating forecasts against a specific performance metric and baseline, where multiple options for evaluation exist and the choice reflects the aim of the evaluation process.

Further, our choice of baseline model affects the given performance scores in absolute terms, and more generally the choice of appropriate baseline for epidemic forecast models is not obvious when assessing infectious disease forecasts. The model used here is supported by previous work (***Cramer et al., 2021b***), yet previous evaluation in a similar context has suggested that choice of baseline affects relative performance in general (***Bracher et al., 2021b***), and future research should be done on the best choices of baseline models in the context of infectious disease epidemics.

Our assessment of forecast performance may further have been inaccurate due to limitations in the observed data against which we evaluated forecasts. We sourced data from a globally aggregated database to maintain compatibility across 32 countries (***Dong et al., 2020***). However, this made it difficult to identify the origin of lags and inconsistencies between national data streams, and to what extent these could bias forecasts for different targets. In particular, we saw some real time data revised retrospectively, introducing bias in either direction where the data used to create forecasts was not the same as that used to evaluate it. We attempted to mitigate this by using an automated process for determining data revisions, and excluding forecasts made at a time of missing, unreliable, or heavily revised data. We also recognise that evaluating forecasts against updated data is a valid alternative approach used elsewhere (***Cramer et al., 2021b***). More generally it is unclear if the expectation of observation revisions should be a feature built into forecasts. Further research is needed to understand the perspective of end-users of forecasts in order to assess this.

The focus of this study was describing and summarising an ensemble of many models. We note that we have little insight into the individual methods and wide variety of assumptions that modellers used. While we asked modellers to provide a short description of their methods, we did not create a rigorous framework for this, and we did not document whether modellers changed the methods for a particular submitted model over time. Both the content of and variation in modelling methods and assumptions are likely to be critical to explaining performance, rather than describing or summarising it. Exploring modellers' methods and relating this to forecast performance will be an important area of future work.

In an emergency setting, access to visualised forecasts and underlying data is useful for researchers, policymakers, and the public (***CDC, 2020***). Previous European multi-country efforts to forecast COVID-19 have included only single models adapted to country-specific parameters (***Aguas et al., 2020***; ***Adib et al., 2021***; ***Agosto et al., 2021***).

The European Forecasting Hub acted as a unique tool for creating an open-access, cross-country modelling network, and connecting this to public health policy across Europe. By opening participation to many modelling teams and with international high participation, we were able to create robust ensemble forecasts across Europe. This also allows comparison across forecasts built with different interpretations of current data, on a like for like scale in real time. The European Hub has supported policy outputs at an international, regional, and national level, including Hub forecasts cited weekly in ECDC Communicable Disease Threats Reports (***European Centre for Disease Prevention and Control, 2022a***).

For forecast producers, an easily accessible comparison between results from different methods can highlight individual strengths and weaknesses and help prioritise new areas of work. Collating time-stamped predictions ensures that we can test true out-of-sample performance of models and avoid retrospective claims of performance. Testing the limits of forecasting ability with these comparisons forms an important part of communicating any model-based prediction to decision makers. For example, the weekly ECDC Communicable Disease Threats reports include the specific results of this work by qualitatively highlighting the greater uncertainty around case forecasts compared to death forecasts.

This study raises many further questions which could inform epidemic forecast modellers and users. The dataset created by the European Forecast Hub is an openly accessible, standardised, and extensively documented catalogue of real time forecasting work from a range of teams and models across Europe (***European Covid-19 Forecast Hub, 2023b***), and we recommend its use for further research on forecast performance. In the code developed for this study, we provide a worked example of downloading and using both the forecasts and their evaluation scores (***covid19-forecast-hub-europe, 2022***).

Future work could explore the impact on forecast models of changing epidemiology at a broad spatial scale by combining analyses of trends and turning points in cases and deaths with forecast

performance, or extending to include data on vaccination, variant, or policy changes over time. There is also much scope for future research into methods for combining forecasts to improve performance of an ensemble. This includes altering the inclusion criteria of forecast models based on different thresholds of past performance, excluding or including only forecasts that predict the lowest and highest values (trimming) (***Taylor and Taylor, 2023***), or using alternative weighting methods such as quantile regression averaging (***Funk et al., 2020***). Exploring these questions would add to our understanding of real time performance, supporting and improving future forecasting efforts.

We see additional scope to adapt the Hub format to the changing COVID-19 situation across Europe. We have extended the Forecast Hub infrastructure to include short term forecasts for hospitalisations with COVID-19, which is a challenging task due to limited data across the locations covered by the hub. As the policy focus shifts from immediate response to anticipating changes brought by vaccinations or the geographic spread of new variants (***European Centre for Disease Prevention and Control, 2023***), we are also separately investigating models for longer term scenarios in addition to the short term forecasts in a similar framework to existing scenario modelling work in the US (***Borchering et al., 2021***).

In conclusion, we have shown that during a rapidly evolving epidemic spreading through multiple populations, an ensemble forecast performed highly consistently across a large matrix of forecast targets, typically outperforming the majority of its separate component models and a naive baseline model. In addition, we have linked issues with the predictability of short-term case forecasts to underlying COVID-19 epidemiology, and shown that ensemble methods based on past model performance were unable to reliably improve forecast performance. Our work constitutes a step towards both unifying COVID-19 forecasts and improving our understanding of them.

Additional information

Competing interests

Prasith Baccam, Heidi Gurung, Steven Stage, Bradley Suchoski: Affiliated with IEM, Inc. The author has no financial interests to declare. The other authors declare that no competing interests exist.

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Additional files

Supplementary files

- Supplementary file 1. EPIFORGE reporting guidelines Completed checklist following reporting guidelines on epidemic forecasting research.
- Supplementary file 2. Participating team metadata Team metadata for teams participating in the European Forecast Hub and evaluated in this study.
- MDAR checklist

Data availability

All source data were openly available before the study, originally available at: https://github.com/covid19-forecast-hub-europe/covid19-forecast-hub-europe (copy archived at swh:1:rev:b4d66c495e07c12d88384506154cf58f08592365). All data and code for this study are openly available on Github: covid19-forecast-hub-europe/euro-hub-ensemble.

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Supplement: EPIFORGE reporting checklist

Section of manuscript	Item	Checklist item	Reported on page
Title/Abstract	1	Describe the study as forecast or prediction research in at least the title or abstract	1
Introduction	2	Define the purpose of study and forecasting targets	4
Methods	3	Fully document the methods	4,5,6,7,8
Methods	4	Identify whether the forecast was performed prospectively, in real time, and/or retrospectively	5
Methods	5	Explicitly describe the origin of input source data, with references	5
Methods	6	Provide source data with publication, or document reasons as to why this was not possible	see Github epiforecasts/euro-hub-ensemble
Methods	7	Describe input data processing procedures in detail	5,6
Methods	8	State and describe the model type, and document model assumptions, including references	5,6, Supplement Table 1
Methods	9	Make the model code available, or document the reasons why this was not possible	see Github epiforecasts/euro-hub-ensemble
Methods	10	Describe the model validation, and justify the approach	5,6
Methods	11	Describe the forecast accuracy evaluation method used, with justification	6,7
Methods	12	Where possible, compare model results to a benchmark or other comparator model, with justification of comparator choice	6,7
Methods	13	Describe the forecast horizon, with justification of its length	5
Results	14	Present and explain uncertainty of forecasting results	8,9,10,11,12
Results	15	Briefly summarize the results in nontechnical terms, including a nontechnical interpretation of forecast uncertainty	12,13,14
Results	16	If results are published as a data object, encourage a time-stamped version number	see Github epiforecasts/euro-hub-ensemble
Discussion	17	Describe the weaknesses of the forecast, including weaknesses specific to data quality and methods	12,13,14
Discussion	18	If the research is applicable to a specific epidemic, comment on its potential implications and impact for public health action and decision-making	14,15
Discussion	19	If the research is applicable to a specific epidemic, comment on how generalizable it may be across populations	15

Supplement: Team metadata

Team	Model	Authors	Methods	Website	Metadata
BIOCOMSC	BIOCOMSC-Gompertz	Martí Català, Enric Álvarez, Sergio Alonso, Daniel López, Clara Prats	Empirical model based on cases and deaths dynamics.	https://biocomsc.upc.edu/en/covid-19	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/BIOCOMSC-Gompertz.yml
University of Cologne Covid Metrics	CovidMetrics-epiBATS	Tom Zimmermann, Arne Rodloff	Forecasts are based on TBATS - models (DeLivera, Hyndman and Snyder (2011)) and are updated daily for each German state.	https://tomz.shinyapps.io/coronaLandkreise/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/CovidMetrics-epiBATS.yml
Epiforecasts / London School of Hygiene and Tropical Medicine	epiforecasts-EpiNow2	Nikos Bosse, Sam Abbott, Sebastian Funk	Semi-mechanistic estimation of the time-varying reproduction number for latent infections mapped to reported cases/deaths.	https://epiforecasts.io/EpiNow2	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/epiforecasts-EpiNow2.yml
epiforecasts	epiforecasts-weeklygrowth	Sam Abbott	A Bayesian autoregressive model using weekly incidence data, application of the forecast.vocs R package.	https://samabbott.co.uk	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/epiforecasts-weeklygrowth.yml
epiMOX	epiMOX-SUIHTER	Giovanni Ardenghi, Giovanni Ziarelli, Luca Dede', Nicola Parolini, Alfio Quarteroni	Compartmental model SUIHTER	https://www.epimox.polimi.it	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/epiMOX-SUIHTER.yml
European COVID-19 Forecast Hub	EuroCOVIDhub-ensemble	Katharine Sherratt, Nikos Bosse, Sebastian Funk	An ensemble, or model average, of submitted forecasts to the European COVID-19 Forecast Hub.	https://covid19forecasthub.eu/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/EuroCOVIDhub-ensemble.yml
Frankfurt Institute for Advanced Studies & Forschungszentrum Jülich	FIAS_FZJ-Epi1Ger	Maria V. Barbarossa, Jan Fuhrmann, Stefan Krieg, Jan H. Meinke	An extended SEIR model with additional compartments for undetected cases	https://www.fz-juelich.de/SharedDocs/Meldungen/IAS/JSC/DE/2021/2021-01-covid-19.html;jsessionid=F4D5FB4027E871A6F4C2FCAF0F08FC35	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/FIAS_FZJ-Epi1Ger.yml

Supplement: Team metadata

Helmholtz Zentrum fuer Infektionsforschung	HZI-AgeExtendedSEIR	Isti Rodiah, Berit Lange, Pratizio Vanella, Alexander Kuhlmann, Wolfgang Bock	Deterministic SEIR type model	https://www.helmholtz-hzi.de/en/nc/research/research-topics/bacterial-and-viral-pathogens/epidemiology/team/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/HZI-AgeExtendedSEIR.yml
ICM / University of Warsaw	ICM-agentModel	Rafał Bartczuk, Łukasz Górski, Magdalena Gruziel-Słomka, Artur Kaczorek, Jan Kisielewski, Antoni Moszyński, Karol Niedzielewski, Jędrzej Nowosielski, Maciej Radwan, Franciszek Rakowski, Marcin Semeniuk, Jakub Zieliński	Agent-based model	https://covid-19.icm.edu.pl/en/model-description/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ICM-agentModel.yml
IEM Health	IEM_Health-CovidProject	Brad Suchoski, Steve Stage, Heidi Gurung, Sid Baccam	SEIR model projections for daily incident confirmed COVID cases and deaths by using AI to fit actual cases observed.	https://iem.com/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/IEM_Health-CovidProject.yml
ILM	ILM-EKF	Stefan Heyder, Thomas Hotz	Extended Kalman filter based on reproduction equation	https://github.com/Stochastik-TU-Ilmenau	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ILM-EKF.yml
Fraunhofer Institute for Industrial Mathematics ITWM	itwm-dSEIR	Jan Mohring, Neele Leithäuser, Michael Helmling	Integral equation model based on age cohorts taking into account vaccination and testing. The parameters are adjusted to the counted cases and deaths.	https://www.itwm.fraunhofer.de/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/itwm-dSEIR.yml
ITWW	ITWW-county_repro	Przemyslaw Biecek, Viktor Bezborodov, Marcin Bodych, Jan Pablo Burgard, Stefan Heyder, Thomas Hotz, Tyll Krüger	Forecasts of county level incidence based on regional reproduction numbers.	https://github.com/Stochastik-TU-Ilmenau	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ITWW-county_repro.yml
JBUD	JBUD-HMXK	Jozef Budzinski	Heavily modified infection-age SIR-X model with waning immunity, vaccinations, seasonality and undetected cases.	https://joebud.github.io/covid-19-analyses/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/JBUD-HMXK.yml

Supplement: Team metadata

MOCOS group	MOCOS-agent1	Marek Bawiec, Marcin Bodych, Tyll Krueger, Tomasz Ozanski, Barbara Pabjan, Agata Migalska, Przemyslaw Biecek, Viktor Bezborodov, Ewa Szczurek, Ewaryst Rafajłowicz, Ewa Rafajłowicz, Wojciech Rafajłowicz	Agent-based microsimulation model	https://mocos.pl/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/MOCOS-agent1.yml
Masaryk University	MUNI-ARIMA	Andrea Kraus, David Kraus	ARIMA model with outlier detection fitted to transformed weekly aggregated series.	https://krausstat.shinyapps.io/covid19global/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/MUNI-ARIMA.yml
Department of Mathematics and Statistics Masaryk University Team	MUNI_DMS-SEIAR	Veronika Eclerova, Lenka Pribylova	SEIAR model with A compartment of absent unobserved infected estimated from hospital data with incorporated mobility data dependence; optimized to the compartment of all exposed (unobserved included)	https://webstudio.shinyapps.io/MAMES/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/MUNI_DMS-SEIAR.yml
Grzegorz Redlarski	PL_GRedlarski-DistrictsSum	Grzegorz Redlarski	Modified SIR method, applied to all districts. Forecasts for districts are summed up.	https://docs.google.com/spreadsheets/d/e/2PACX-1vRpH4yhKRts7Co5tydhZhojIPTcTTybms1PqJ9j1tmSBzzPLoU2U9XjUWDwiKYxnE6gMLayl71rpGC8/pubhtml?gid=493251550&single=true	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/PL_GRedlarski-DistrictsSum.yml
prolix	prolix-euclidean	Loïc Pottier	Offsets obtained by correlations, best linear approximation of reproduction rates (using vaccination approximation) by least euclidean distance, and linear prediction.	https://cp.lpmib.fr/medias/covid19/_synthese.html	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/prolix-euclidean.yml
Robert Walraven	RobertWalraven-ESG	Robert Walraven	Multiple skewed gaussian distribution peaks fit to raw data	http://rwalraven.com/COVID19	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/RobertWalraven-ESG.yml

Supplement: Team metadata

Swiss Data Science Center / University of Geneva	SDSC_ISG-TrendModel	Ekaterina Krymova, Dorina Thanou, Benjamin Bejar Haro, Tao Sun, Gavin Lee, Elisa Manetti, Christine Choirat, Antoine Flahault, Guillaume Obozinski	The Trend Model predicts daily cases and deaths using linear extrapolation on the linear or log scale of the underlying trend estimated by a robust LOESS seasonal-trend decomposition model.	https://renkulab.shinyapps.io/COVID-19-Epidemic-Forecasting/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/SDSC_ISG-TrendModel.yml
Statgroup19	Statgroup19-richards	Pierfrancesco Alaimo Di Loro, Fabio Divino, Alessio Farcomeni, Giovanna Jona Lasinio, Antonello Maruotti, Marco Mingione, Gianfranco Lovison	Richards' curve based generalized growth model	https://statgroup19.shinyapps.io/Covid19App/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/Statgroup19-richards.yml
Statgroup19	Statgroup19-spatialrichards	Pierfrancesco Alaimo Di Loro, Fabio Divino, Alessio Farcomeni, Giovanna Jona Lasinio, Antonello Maruotti, Marco Mingione, Gianfranco Lovison	Richards' curve based generalized growth model taking into account spatial dependence	https://statgroup19.shinyapps.io/Covid19App/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/Statgroup19-spatialrichards.yml
Universidad Carlos III de Madrid	UC3M-EpiGraph	David E. Singh, Miguel Guzman Merino, Maria Cristina Marinescu, Jesus Carretero, Alberto Cascajo Garcia	Agent-based parallel simulator that models individual interactions extracted from social networks and demographical data.	https://www.arcos.inf.uc3m.es/epigraph/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UC3M-EpiGraph.yml
University of Ljubljana, Faculty of Health Sciences Team	ULZF-SEIRC19SI	Janez Zibert	SEIHR model extended with compartments for hospitals, intensive care units, asymptomatic cases, separate submodels for vaccinated and unvaccinated, divided to 5 age subgroups of population	https://apps.lusy.fri.uni-lj.si	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ULZF-SEIRC19SI.yml
UMass-Amherst	UMass-MechBayes	Dan Sheldon, Graham Gibson, Nick Reich	Bayesian compartmental model with observations on cumulative case counts and cumulative deaths. Model is fit independently to each state. Model includes observation noise and a case detection rate.	https://github.com/dsheldon/covid	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UMass-MechBayes.yml
UNED	UNED-PreCoV2	José L. Aznarte, César Pérez, José Almagro, Pedro Álvarez, Álvaro Ortiz, Fernando Blat	Bayesian time series models with ARIMA noise and fixed transfer functions for each input.	https://precov2.org	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UNED-PreCoV2.yml

Supplement: Team metadata

University of Perugia / University of Milano-Bicocca / Università della Svizzera Italiana	UpgUmibUsi-MultiBayes	Francesco Bartolucci, Fulvia Pennoni, Antonietta Mira	Bayesian Dirichlet-Multinomial models for counts of patients in mutually exclusive and exhaustive categories such as hospitalized in regular wards and in intensive care units, deceased and recovered	https://github.com/francescobartolucci/ARMultinomial	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UpgUmibUsi-MultiBayes.yml
University of Southern California	USC-SIkJalpha	Ajitesh Srivastava, Frost Tianjian Xu	A heterogeneous infection rate model with human mobility for epidemic modeling. Our model adapts to changing trends and provide predictions of confirmed cases and deaths.	https://scc-usc.github.io/ReCOVER-COVID-19	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/USC-SIkJalpha.yml
University of Virginia, Biocomplexity COVID-19 Response Team	UVA-Ensemble	Aniruddha Adiga, Lijing Wang, Srinivasan Venkatramanan, Akhil Sai Peddireddy, Benjamin Hurt, Przemyslaw Porebski, Bryan Lewis, Madhav Marathe, Jiangzhou Chen, Anil Vullikanti	An ensemble of multiple methods such as auto-regressive (AR)models with exogenous variables, Long short-term memory (lSTM) models,Kalman filter and PatchSim (an SEIR model).	https://biocomplexity.virginia.edu/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UVA-Ensemble.yml

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Materials Design Analysis Reporting (MDAR) Checklist for Authors

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For all that apply, please note **where in the article** the information is provided. Please note that we also collect information about data availability and ethics in the submission form.

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For commercial reagents, provide supplier name, catalogue number and RRID, if available.		N/A

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Short novel DNA or RNA including primers, probes: Sequences should be included or deposited in a public repository.		N/A

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Experimental study design (statistics details) *		
For in vivo studies: State whether and how the following have been done	**Indicate where provided: section/figure legend. If it could have been done, but was not, write "not done"**	**N/A**
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Randomisation		N/A
Blinding		N/A
Inclusion/exclusion criteria		N/A

Sample definition and in-laboratory replication	**Indicate where provided: section/figure legend**	**N/A**
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Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.		N/A

Dual Use Research of Concern (DURC)	**Indicate where provided: section/submission form**	**N/A**
If study is subject to dual use research of concern regulations, state the authority granting approval and reference number for the regulatory approval.		N/A

Analysis:

Attrition	Indicate where provided: section/figure legend	N/A
Describe whether exclusion criteria were pre-established. Report if sample or data points were omitted from analysis. If yes, report if this was due to attrition or intentional exclusion and provide justification.	Exclusion criteria were pre-established. Forecasts were omitted where they corresponded to an observed data point that was retrospectively adjusted in official reporting statistics by >5%.	

Statistics	Indicate where provided: section/figure legend	N/A
Describe statistical tests used and justify choice of tests.		N/A

Data availability	Indicate where provided: section/submission form	N/A
For newly created and reused datasets, the manuscript includes a data availability statement that provides details for access (or notes restrictions on access).	Abstract, Discussion	
When newly created datasets are publicly available, provide accession number in repository OR DOI and licensing details where available.	Code and data DOI: 10.5281/zenodo.6895700	
If reused data is publicly available provide accession number in repository OR DOI, OR URL, OR citation.	Katharine Sherratt, Hugo Gruson, Helen Johnson, Rene Niehus, Bastian Prasse, Frank Sandman, Jannik Deuschel, Daniel Wolffram, Sam Abbott, Alexander Ullrich, Graham Gibson, Evan L Ray, Nicholas G Reich, Daniel Sheldon, Yijin Wang, Nutcha Wattanachit, Lijing Wang, Jan Trnka, Guillaume Obozinski, ... Sebastian Funk. (2022). European Covid-19 Forecast Hub (v2022.07.21) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.6874754	

Code availability	Indicate where provided: section/figure legend	N/A

For any computer code/software/mathematical algorithms essential for replicating the main findings of the study, whether newly generated or re-used, the manuscript includes a data availability statement that provides details for access or notes restrictions.	Abstract, Discussion	
Where newly generated code is publicly available, provide accession number in repository, OR DOI OR URL and licensing details where available. State any restrictions on code availability or accessibility.	URL: https://github.com/covid19-forecast-hub-europe/euro-hub-ensemble DOI: 10.5281/zenodo.6895700 License: MIT	
If reused code is publicly available provide accession number in repository OR DOI OR URL, OR citation.		N/A

Reporting:

The MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives.

Adherence to community standards	**Indicate where provided: section/figure legend**	**N/A**
State if relevant guidelines (e.g., ICMJE, MIBBI, ARRIVE, STRANGE) have been followed, and whether a checklist (e.g., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	EPIFORGE guidelines for reporting epidemic forecasting; checklist provided in Supplement	

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* We provide the following guidance regarding transparent reporting and statistics; we also refer authors to Ten common statistical mistakes to watch out for when writing or reviewing a manuscript.

Sample-size estimation

- You should state whether an appropriate sample size was computed when the study was being designed
- You should state the statistical method of sample size computation and any required assumptions
- If no explicit power analysis was used, you should describe how you decided what sample (replicate) size (number) to use

Replicates

- You should report how often each experiment was performed
- You should include a definition of biological versus technical replication
- The data obtained should be provided and sufficient information should be provided to indicate the number of independent biological and/or technical replicates
- If you encountered any outliers, you should describe how these were handled
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- High-throughput sequence data should be uploaded before submission, with a private link for

reviewers provided (these are available from both GEO and ArrayExpress)

Statistical reporting

- Statistical analysis methods should be described and justified
- Raw data should be presented in figures whenever informative to do so (typically when N per group is less than 10)
- For each experiment, you should identify the statistical tests used, exact values of N, definitions of center, methods of multiple test correction, and dispersion and precision measures (e.g., mean, median, SD, SEM, confidence intervals; and, for the major substantive results, a measure of effect size (e.g., Pearson's r, Cohen's d)
- Report exact p-values wherever possible alongside the summary statistics and 95% confidence intervals. These should be reported for all key questions and not only when the p-value is less than 0.05.

Group allocation

- Indicate how samples were allocated into experimental groups (in the case of clinical studies, please specify allocation to treatment method); if randomization was used, please also state if restricted randomization was applied
- Indicate if masking was used during group allocation, data collection and/or data analysis

Characterising information gains and losses when collecting multiple epidemic model outputs

Sherratt K, Srivastava A, Ainslie K, Singh DE, Cublier A, Marinescu MC, Carretero J, Garcia AC, Franco N, Willem L, Abrams S, Faes C, Beutels P, Hens N, Müller S, Charlton B, Ewert R, Paltra S, Rakow C, Rehmann J, Conrad T, Schütte C, Nagel K, Abbott S, Grah R, Niehus R, Prasse B, Sandmann F, Funk S. Characterising information gains and losses when collecting multiple epidemic model outputs. Epidemics. 2024 Mar 27;47:100765. doi: 10.1016/j.epidem.2024.100765.

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RESEARCH PAPER COVER SHEET

Please note that a cover sheet must be completed for each research paper included within a thesis.

SECTION A – Student Details

Student ID Number	1701639	**Title**	Ms
First Name(s)	Katharine		
Surname/Family Name	Sherratt		
Thesis Title	Collaborative outbreak modelling for decision support: evaluating trade-offs from multi-model combination		
Primary Supervisor	Sebastian Funk		

If the Research Paper has previously been published please complete Section B, if not please move to Section C.

SECTION B – Paper already published

Where was the work published?	Epidemics		
When was the work published?	2024		
If the work was published prior to registration for your research degree, give a brief rationale for its inclusion	PhD by Publication		
Have you retained the copyright for the work?*	**Yes**	Was the work subject to academic peer review?	**Yes**

*If yes, please attach evidence of retention. If no, or if the work is being included in its published format, please attach evidence of permission from the copyright holder (publisher or other author) to include this work.

SECTION C – Prepared for publication, but not yet published

Where is the work intended to be published?	
Please list the paper's authors in the intended authorship order:	
Stage of publication	Choose an item.

SECTION D – Multi-authored work

For multi-authored work, give full details of your role in the research included in the paper and in the preparation of the paper. (Attach a further sheet if necessary)	I first contributed building and maintaining the European Scenario Hub infrastructure. My work included adapting and developing software for collecting, validating, and visualising scenario projections, and producing all documentation. This was supported by Hugo Gruson with supervision from Sebastian Funk and collaboration with the US Scenario Hub. During the project, I led four rounds of collaborative scenario modelling. I facilitated stakeholder workshops to codesign scenarios, and designed procedures for collecting metadata, visualising, and narratively synthesising results from the scenario projections. For this paper, I conceived and conducted the research questions and conducted all analysis, visualisations, and drafting, supervised by Sebastian Funk. First, I led work analysing trajectories for policy relevant characteristics, including writing code and creating visualisations, as part of the development of the Scenario Hub. This aspect of the work was also reviewed by the team at the ECDC. Second, I planned and conducted all analysis and visualisation for work comparing an ensemble from trajectories to the Vincentised ensemble from quantile summaries. In later drafts I added the Linear Opinion Pool ensemble, drawing on code developed by Emily Howerton and Evan Ray. The third aspect of this work was to ensemble scenario trajectories using a weighting procedure based on ongoing evaluation against observed data. I developed all code and visualisation for this element of the work. I led all draft writing, revisions, and paper management. We sought a first round of review from all those who had contributed to the Scenario Hub before submission for peer review. Code contributions are at: https://github.com/epiforecasts/multi-model-information

SECTION E

Student Signature	Katharine Sherratt
Date	14 June 2024

Supervisor Signature	Sebastian Funk
Date	14 June 2024

Epidemics 47 (2024) 100765

Contents lists available at ScienceDirect

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journal homepage: www.elsevier.com/locate/epidemics

Characterising information gains and losses when collecting multiple epidemic model outputs

Katharine Sherratt [a,*], Ajitesh Srivastava [b], Kylie Ainslie [c,d], David E. Singh [e], Aymar Cublier [e], Maria Cristina Marinescu [f], Jesus Carretero [e], Alberto Cascajo Garcia [e], Nicolas Franco [g], Lander Willem [h], Steven Abrams [h,i], Christel Faes [i], Philippe Beutels [h], Niel Hens [h,i], Sebastian Müller [j], Billy Charlton [j], Ricardo Ewert [j], Sydney Paltra [j], Christian Rakow [j], Jakob Rehmann [j], Tim Conrad [k], Christof Schütte [k], Kai Nagel [j], Sam Abbott [a], Rok Grah [l], Rene Niehus [l], Bastian Prasse [l], Frank Sandmann [l], Sebastian Funk [a]

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ABSTRACT

Background: Collaborative comparisons and combinations of epidemic models are used as policy-relevant evidence during epidemic outbreaks. In the process of collecting multiple model projections, such collaborations may gain or lose relevant information. Typically, modellers contribute a probabilistic summary at each time-step. We compared this to directly collecting simulated trajectories. We aimed to explore information on key epidemic quantities; ensemble uncertainty; and performance against data, investigating potential to continuously gain information from a single cross-sectional collection of model results.

Methods: We compared projections from the European COVID-19 Scenario Modelling Hub. Five teams modelled incidence in Belgium, the Netherlands, and Spain. We compared July 2022 projections by incidence, peaks, and cumulative totals. We created a probabilistic ensemble drawn from all trajectories, and compared to ensembles from a median across each model's quantiles, or a linear opinion pool. We measured the predictive accuracy of individual trajectories against observations, using this in a weighted ensemble. We repeated this sequentially against increasing weeks of observed data. We evaluated these ensembles to reflect performance with varying observed data.

Results: By collecting modelled trajectories, we showed policy-relevant epidemic characteristics. Trajectories contained a right-skewed distribution well represented by an ensemble of trajectories or a linear opinion pool, but not models' quantile intervals. Ensembles weighted by performance typically retained the range of plausible incidence over time, and in some cases narrowed this by excluding some epidemic shapes.

Conclusions: We observed several information gains from collecting modelled trajectories rather than quantile distributions, including potential for continuously updated information from a single model collection. The value of information gains and losses may vary with each collaborative effort's aims, depending on the needs of

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projection users. Understanding the differing information potential of methods to collect model projections can support the accuracy, sustainability, and communication of collaborative infectious disease modelling efforts.

1. Background

During outbreaks of infectious disease, it is critical to account for the uncertainty of future disease incidence in order for public health decision-makers to fully evaluate risk (Zelner et al., 2021; Li et al., 2017). Infectious disease modellers use a variety of approaches to meet this demand for information. A common challenge is the representation of multiple sources of uncertainty, both within each model as well as across separate model projections (McCabe et al., 2021; Swallow et al., 2022). In recognising this challenge, infectious disease modelling has seen an increasing emphasis on both probabilistic modelling methods, together with collaborative approaches to modelling (Bracher et al., 2021a; Reich et al., 2022).

Probabilistic infectious disease models can address the challenge of uncertainty by simulating the complex and changing real-world process of disease transmission. Modellers must handle stochasticity in transmission dynamics, often using observed data to estimate model parameters and latent trajectories that are themselves uncertain. Each such model can generate any number of simulated trajectories, and modellers choose at what point to conclude there are sufficient iterations to reach a stable distribution of possible outcomes. The output of these simulations can then be summarised to calculate quantities of interest, such as weekly incidence of infections or cases.

When creating models to characterise the future, modellers have often drawn a distinction in the meaning of uncertainty between forecast compared to scenario projections (Lipsitch et al., 2011). Forecasts are predictions of future epidemic trajectories, and the probabilities assigned to different outcomes quantify the belief of the forecaster that these may or may not happen. In addition to potential fundamental limits to predictability, forecasts are usually reliable for, at best, a few generations of transmission (Sherratt et al., 2023) because of unmodelled factors affecting future transmission such as behavioural or policy changes, heterogeneity in transmission risk, or the emergence of new variants of different transmissibility or severity.

In contrast, scenarios are projections attuned to a particular context by being conditioned on specific factors whose futures may not be quantitatively predictable, such as options for policy interventions (Runge et al., 2023; Rhodes et al., 2020). Probabilities of future outcomes as stated by scenario models should be interpreted as valid only under the specific circumstances given by the scenario but not otherwise, without specifying any probability of the scenario itself occurring. Because of this difference, forecasts can be evaluated by confronting them with future data as it becomes available, while this evaluation is more challenging for scenarios where predictive performance will always depend on a combination of adequacy of the chosen assumptions (e.g. on pathogen biology, human behaviour and government policy), with adequacy of the model in reflecting these assumptions.

Infectious disease modelling collaborations aim to bring together models that project the future using diverse methods (Reich et al., 2022). Each collaboration sets a clearly defined target for projections, communicates this target to multiple independent modellers, and collects model results in a standardised format. This standardisation allows for a like-for-like comparison of varying modelling methods' results and accompanying uncertainty. Ensemble methods can then combine results across models. Typically, this creates a more comprehensive and robust projection (Ray et al., 2020) or reflection of expert judgement (Shea et al., 2020).

Formal, large-scale modelling collaborations have, so far, been used for influenza, Ebola, Zika, dengue fever, and COVID-19 (Reich et al., 2022). In the case of COVID-19, a number of policy-facing research groups have set up collaborations to collate forecasts and scenarios (Borchering, 2021; Cramer et al., 2021; Funk et al., 2020; Sherratt et al., 2023), and there is a substantial effort towards expanding the practice of ensemble projections of infectious disease spread and burden. Ongoing work evaluating these efforts has focused on assessing the output of past and current ensemble modelling projects. This has included evaluating differing performance among individual models (Viboud et al., 2018; Bracher et al., 2021b; Cramer et al., 2022), and a variety of methods for creating ensembles from multiple models (Howerton et al., 2023; Ray et al., 2020; Sherratt et al., 2023; Taylor and Taylor, 2021).

The standardised format in which model projections are collected is key to meeting such projects' aims of comparing information from multiple models. The most common approach to this is to collect descriptive statistics from each model at each given time step. In this format, each modeller submits values across a pre-specified set of quantiles in order to represent uncertainty in their projection. The benefits of this system include that it should accurately represent an underlying distribution of outcomes while being storage-efficient (Genest, 1992), it is not restricted to probabilistic models producing simulations, and it allows for quantitative evaluation against observed data (Bracher et al., 2021a). Various methods for subsequent combination from quantiles depend on the view taken of uncertainty between and across model projections (Howerton et al., 2023).

However, using quantile intervals separately across each time step may lose information pertinent to epidemic decision making. As a quantile representation provides a summary across trajectories at each time step, it has no theoretical continuity through the time-series. This does not permit aggregation over time to calculate cumulative totals or means, and may misrepresent time-series characteristics including epidemic peak size or timing (Juul et al., 2021). Whilst some of these can be remedied by also collating quantiles of cumulative quantities, these still lose some of the temporal information contained in the full joint probability distribution across all future time points.

An alternative method of collecting output from multiple probabilistic models is to collect the individual simulated trajectories produced by each modeller. Each simulated trajectory comprises a single value for each time step, with modellers contributing some number of these trajectories. Each trajectory retains its own time-series characteristics, and these can therefore be summarised across different models. Collecting trajectories also creates potential for the analysis and combination of each trajectory independently from the originating model's total output. One option could include comparing each trajectory to observed data as it becomes available, even after the time of collecting model outputs. This would enable creating an ensemble projection that is conditioned on the observed accuracy of each individual trajectory. As further observed data become available, this ensemble could be updated to create a single combined projection that continuously reflects the changing performance of each trajectory. This would act similarly to methods of particle filtering in continuously conditioning on past behaviour.

We aim to explore aspects of information gains and losses from these two methods of collecting multiple model results. We contrast collecting a set of simulated trajectories, against collecting a summary at quantile intervals of those trajectories. We use the setting of the European COVID-19 Scenario Hub, where the use of quantile summaries was replaced in mid-2022 by collecting trajectories. These trajectories represent random samples from the collection of all possible trajectories of each model consistent with a given scenario and the data available up to the time at which the simulation was generated.

In this work, we assess the impact of the collection method when seeking information about policy-relevant epidemic characteristics, including cumulative totals, timing of peaks, and the extent of

uncertainty across multiple models. We then explore the information gained by the ability to compare modelled epidemic trajectories to observed data as this becomes available over time. We use this to create a multi-model ensemble which weights across all available trajectories by their past accuracy. This demonstrates the potential to continuously gain information from only a single cross-sectional collection of model results. Understanding the potential sources of information gains and losses when collecting multiple model projections may support improving the accuracy, reliability, and communication of collaborative infectious disease modelling efforts.

2. Methods

2.1. Study setting

In this work we use projections from Round 2 of the European COVID-19 Scenario Modelling Hub (Taylor and Taylor, 2021). The European COVID-19 Scenario Hub was launched in March 2022 to reflect demand for the ECDC to support longer term European policy planning. It used the existing US Scenario Hub (Borchering, 2021) as a basis for Hub infrastructure and methods. Modelling teams were recruited by word of mouth to join a series of collaborative workshops, approximately fortnightly from March through June 2022. In these sessions both policy-focussed colleagues from the ECDC and modelling-focussed researchers co-developed a set of four scenarios. Each scenario represented a combination of two possible epidemiological and policy changes that could impact the incidence of COVID-19 across Europe in the medium term.

Teams were asked to project the incidence of COVID-19 infections, cases, deaths, and hospitalisations in 32 European countries over the next year. To facilitate comparison across models, we identified and agreed a common set of key assumptions and parameters to be used by all models in each scenario as well as standard data sets to which to compare the model outputs where available. Modellers uploaded projections to a Github repository, and we summarised results across models, with a focus on targets with three or more different models. Over 2022 this process was repeated four times to explore a variety of different scenarios. In total nine separate teams submitted projections, with six teams contributing to each round.

Over June 2022 (Round 2), we specified four scenarios (A-D) as: an autumn second booster campaign among the population aged over 60 (scenarios A/C), or over 18 (scenarios B/D); and future vaccine effectiveness as 'optimistic' (equivalent to the effectiveness as of a booster vaccine against the Delta SARS-CoV-2 variant; scenarios A/B); or 'pessimistic' (as against variants Omicron BA.4/BA.5/BA.2.75; scenarios C/D). Modellers were asked to start their projections from 24th July 2022, meaning that even if data were available beyond this date they were not to inform calibration of the model. Modellers were asked to submit up to 100 simulations, each reflecting a trajectory of weekly incidence of reported cases and deaths over time for a given projection target. Modellers were informed that data presented on the Johns Hopkins University dashboard was to be used for future comparison to data (Dong et al., 2020). In practice some of the models were not calibrated to reported cases and therefore used symptomatic cases as a proxy (see model details in Supplement). Simulations were to represent random samples from the distribution of simulation trajectories consistent with the given scenario that each modelling team produced. We have published full scenario details including shared parameters, all teams' projections, and summary results online (European COVID-19 Scenario Hub).

This work specifically focuses on contrasting the sampled simulated trajectories with their representation in time-specific quantiles. We collected raw data in the form of up to 100 trajectories from each model for each projection target. We used these data to retrospectively create a marginal fixed-time quantile representation of results from each model and target. Following the current submission procedure across COVID-19 Modelling Hubs for an individual model, we calculated a median and 22 further quantiles for each week using the values of the trajectories in that week, separately for each scenario. We processed all data in R with code available online (Sherratt and Funk, 2024).

2.2. Characterising potential information gains and losses

First we considered information about key epidemic characteristics. At the time the projections were in production, discussion with the ECDC modelling team led to an interest in: estimates of incidence over time; cumulative values over different periods; and number of distinct peaks, size, and timing of peak incidence over the projection period.

When projections were available, we estimated these characteristics from the simulated trajectories. We summed incidence over time to produce a cumulative total from each trajectory. We assessed the size of the expected burden of each target relative to a known threshold by comparing the cumulative projected total to the cumulative total of the preceding year. We identified peaks in each simulated trajectory as the local maxima in a sliding window of five weeks, using the ggpmisc R package (Aphalo, 2023). We chose a sliding window of five weeks to capture each distinct peak while avoiding detecting noise in each trajectory. We summarised across the individual peaks detected in each trajectory using quantiles at each weekly time-step, to produce a range indicating possible peak timing and maximum values across all trajectories. We produced a real-time report of this summary at the time that projections became available in July 2022.

In further retrospective analysis, we compared the use of a standard unweighted ensemble to express uncertainty across multiple models in the two representations. We created an ensemble projection from first combining all individual simulated trajectories with equal weight for each scenario, location, and outcome target. Next, we took model-specific quantiles from each model's distribution of trajectories at each time point, for each scenario, location, and outcome target. We used each set of quantiles to create linear opinion pool ensembles (LOP), which use linear extrapolation between the given quantiles to estimate the cumulative distribution function in order to then randomly sample trajectories to aggregate, again with equal weight; and a quantile-average ensemble, which takes the median across the different models' values at each quantile and time step. The LOP and quantile-average ensembles have both been used to produce ensemble projections across multiple epidemiological forecasts (Howerton et al., 2023; Ray et al., 2020; Sherratt et al., 2023). To assess the difference in uncertainty across the two ensembles, we compared the mean of the values at each quantile across all time points, outcomes and scenarios.

Lastly, we evaluated the performance of each simulated trajectory against proximity to observed data, and used this to weight an ensemble of trajectories (as above). To measure performance, we calculated the mean absolute error (MAE) for each trajectory, where the MAE is the average of the difference from observed data across all available time points for a single projection. We created a weighted ensemble from all trajectories for a country (not further separating by scenario or model) using the inverse MAE for each trajectory as a weight. To calculate weighted quantiles we used a Harrell Davis weighted estimator (Harrell and Davis, 1982) from the *cNORM* R package (v3.0.2) (Lenhard et al., 2018). As above, we calculated 23 quantiles including the median to express uncertainty.

We repeated this process to create a sequence of ensembles with changing weights over time. We created the first weighted ensemble after 4 weeks of observed data, and then created consecutive ensembles with weights re-calculated weekly to use up to the maximum available 29 weeks of observed data (to 11 March 2023). This showed varying lengths of projections repeatedly conditioned on simulated trajectories' performance against increasing data over time.

We evaluated the predictive performance of these sequences of weighted ensembles. We transformed forecasts and observed data to a logarithmic scale, as this allows a more consistent evaluation across

varying magnitudes and better reflects the exponential nature of epidemic processes (Bosse et al., 2023). We then calculated the weighted interval score for each forecast, as a quantitative performance measure that evaluates across both the accuracy and the dispersion of probabilistic forecasts (Bracher et al., 2021a). In the same way we evaluated the unweighted ensemble of trajectories described above, and used this as a relative baseline with which to compare the effect of weighting individual trajectories on ensemble performance.

3. Results

A total of six modelling teams contributed projections for various targets to the European COVID-19 Scenario Hub in Round 2. Here we focus on multi-model comparison and include only projection targets with three contributing models. These targets included 52 weeks' case and death incidence for the Netherlands and Belgium, and 41 weeks' case incidence for Spain.

Five teams contributed projections for these targets. Three teams used compartmental models, one an agent-based model, and one a machine learning method (see Supplement). Four models generated 100 simulated trajectories, and one 96 trajectories (implying a slightly smaller weight to this model in trajectory-based aggregates). In total, we consider 294,816 data points from 5920 trajectories, where each data point is the estimated weekly incidence in a simulated trajectory of an outcome in a target country and scenario over up to one year (Fig. 1.i).

Aggregating across simulated trajectories from multiple models allowed access to information about various epidemic characteristics. These included cumulative totals, and peak size and timings (see contemporaneous report reproduced in the Supplement). By summarising across the peaks of each individual trajectory, we were able to create an estimate of uncertainty around the size and timing of peaks for each target. We were also able to summarise cumulative outcomes. For example, across all 5920 trajectories for all targets and scenarios, 10% saw a cumulative total exceeding the preceding year. These epidemic characteristics could not be meaningfully estimated from the same results summarised into quantiles.

We compared information loss in the aggregation of simulated trajectories into ensemble projections (Fig. 1). We compared an ensemble taken from all trajectories (Fig. 1.ii) with a linear opinion pool (not shown), and the quantile-average ensemble (Fig. 1.iii). We noted that a linear opinion pool ensemble produces near-identical results to taking an ensemble directly from trajectories. Across all projection targets, we observed substantially increased uncertainty in an ensemble that aggregated either directly from trajectories, or via linear opinion pool, compared to a quantile-average ensemble. This represented the wider variety of epidemic shapes projected by different models. For example, the credible interval of projections for Spain included high autumn-winter incidence, while for Belgium gave greater credibility to multiple peaks of incidence. These were not observed in the interval projections of an ensemble derived from models' quantiles.

We quantified the range of uncertainty between each ensemble by comparing the mean of values at each quantile across all time points and

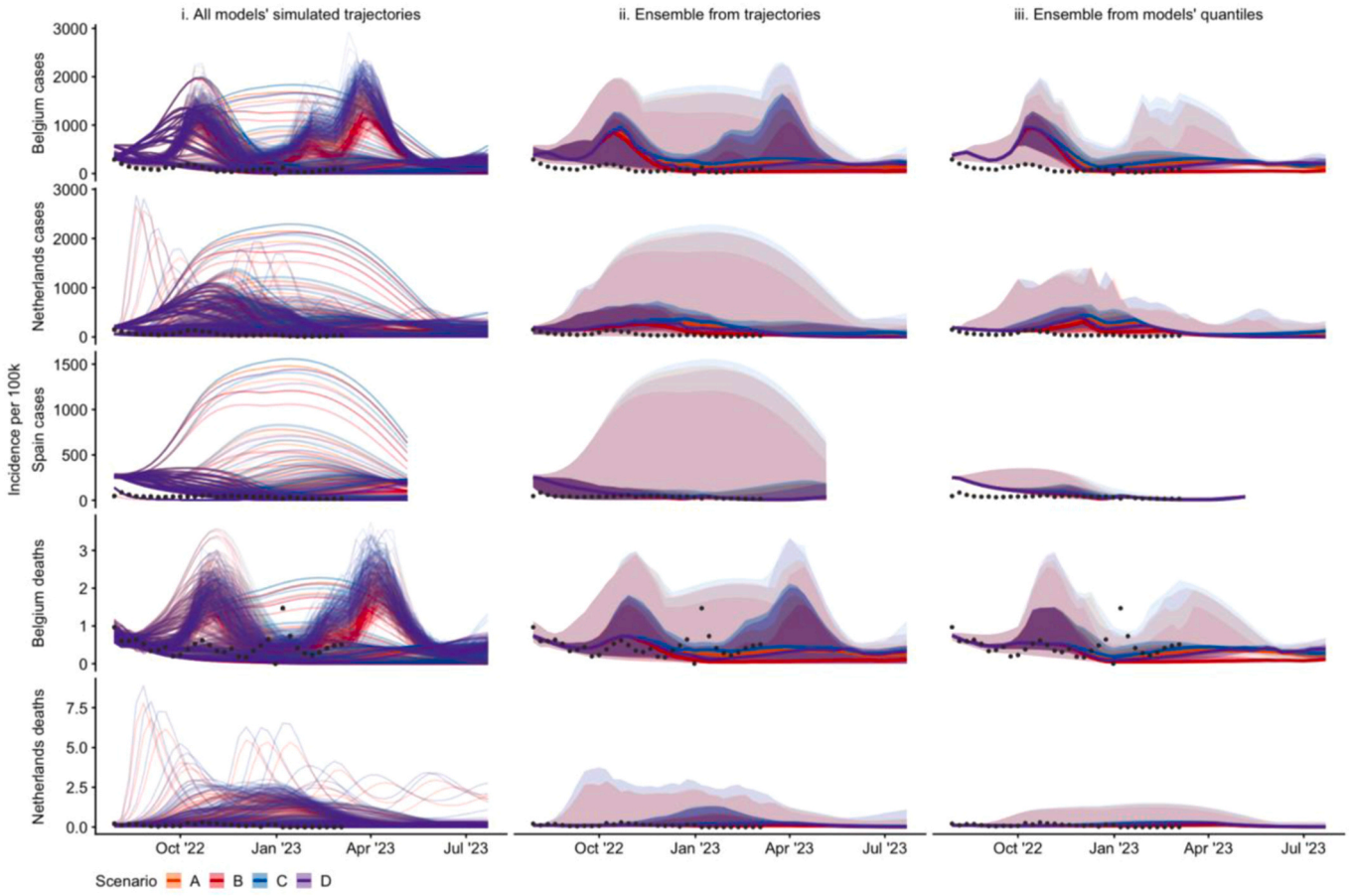

Fig. 1. Projections of incidence per 100,000 population, by country (row) and aggregation method (column) showing median, 50%, and 99% probabilistic intervals (increasingly shaded ribbons), for each scenario, using: i) no ensemble method (100 simulated trajectories per model, or 96 in case of one of the models); ii) quantile intervals of the distribution across all simulated trajectories; ii; a median across each model's projections at a given quantile interval. We do not show the linear opinion pool ensemble here as results are near-identical to the ensemble drawn directly from trajectories (ii)). Scenarios included: an autumn second booster vaccine campaign among population aged 18+ (scenarios B & D) or 60+ (scenarios A & C); where vaccine effectiveness is 'optimistic' (effectiveness as of a booster vaccine against Delta; scenarios A & B) or 'pessimistic' (as against BA.4/BA.5/BA.2.75; scenarios C & D). See Supplement for further detail on individual models' trajectories.

scenarios (supplementary figure 1A). All ensembles produced similar values around the centre of the distribution, with no noticeable difference between the median values of each projection. However, across all five targets we observed that an ensemble based on either simulated trajectories, or an LOP ensemble, produced sharply increasing uncertainty between the 90–98% intervals. For example, at the upper 98% probability interval, ensemble projections for cases in Spain averaged nearly six times higher incidence when drawn directly from trajectories compared to when drawn from a median of three models' quantiles (respectively averaging 1016 and 173 weekly new cases per 100,000 population).

We then considered an ensemble of individual trajectories each weighted against a sequentially increasing amount of observed data (Fig. 2). We note that models used a variety of methods and may have been calibrated to alternative data sources (see Supplement). In comparison to the unweighted ensemble (shown in grey), we observed reduced uncertainty across weighted ensemble projections. Compared to conditioning on data up to 16 weeks before, adding 8 weeks of additional data in weighting case projections reduced the upper 98% bound of uncertainty by at least 5% and up to 30% on average (supplementary figure 1B). The accuracy-weighted contribution of each trajectory to an ensemble varied substantially between models and targets, and over time. For example, in Spain each trajectory's weight remained stable after mid December 2022, reflecting the data by effectively downweighting those trajectories projecting sustained high incidence over winter (see Fig. 1i).

We used this information to create consecutive weekly ensembles, with weights updating as increasing observed data became available to measure trajectories' accuracies. In the combined (weighted interval) score, forecasts using weighted trajectories generally performed similarly to the unweighted equivalent, with a median relative WIS among the weighted ensembles of 0.99 (IQR: 0.89–1.05; supplementary figure 2).

When using the full 31 weeks of available data, a weighted ensemble performance improved compared to projections made without weighting on accuracy (with a median relative WIS across targets of 0.77

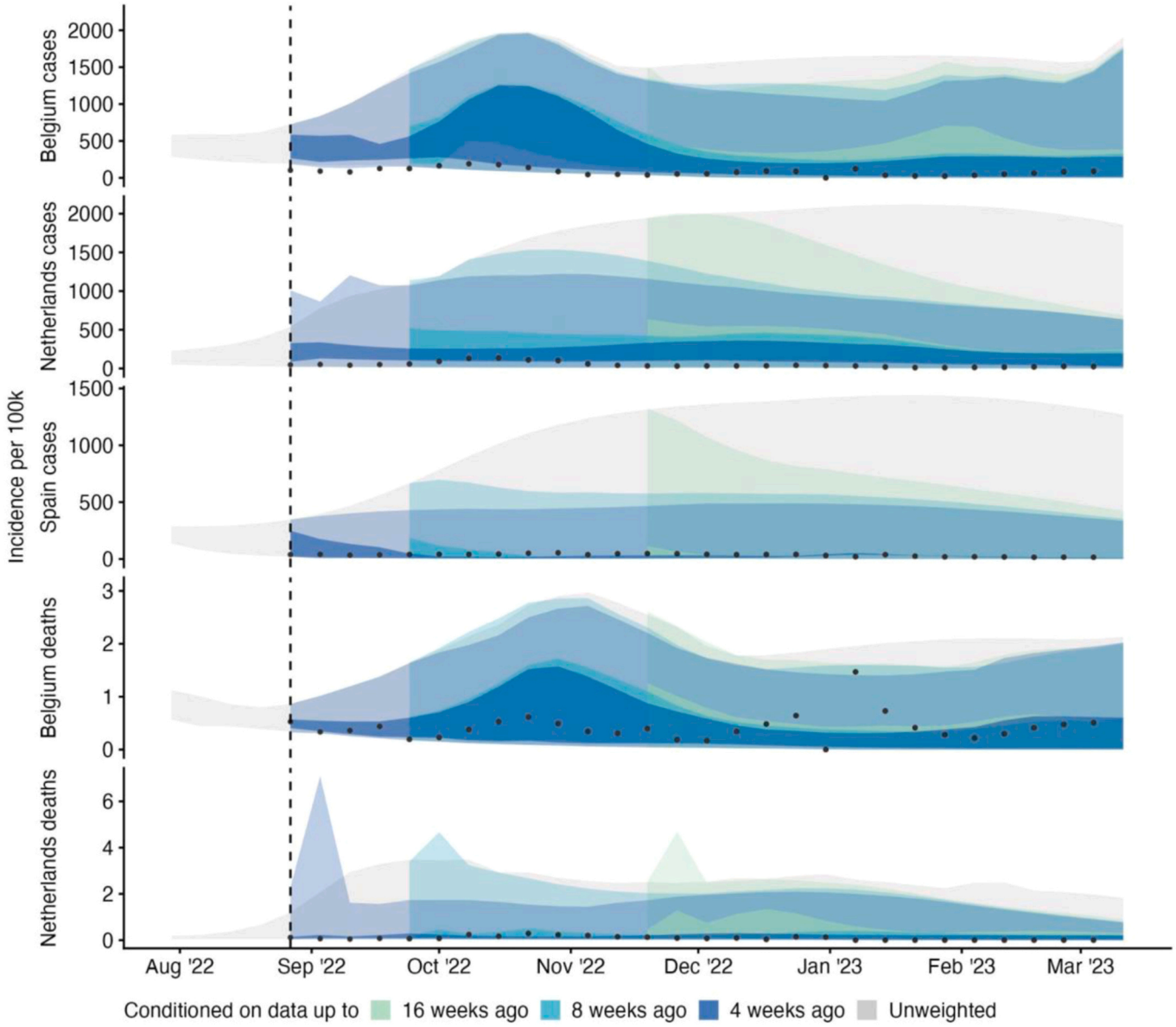

Fig. 2. Ensemble forecasts of incidence by target, using no weighting (grey ribbon), or 4, 8, and 16 weeks ahead of available data, with available data increasing weekly over time (coloured ribbons); showing 50% and 99% credible intervals. Each simulated trajectory started from 30 July 2022 and was weighted using its inverse mean absolute error against available data. We used at least 4 and up to 31 weeks of this observed accuracy data.

compared to the baseline of 1). However, this improvement was not linearly correlated with increasing data, and the relationship varied by target (Fig. 3). Weighted forecasts that used only a few data points of trajectories' accuracy performed similarly or poorly compared to the unweighted ensemble.

However, among three targets, using more accuracy data gave a stable or consistently improved performance (after 10 weeks for cases in Belgium and 17 weeks for cases and deaths in the Netherlands). This was also true of cases in Spain, with worse performance compared to the unweighted ensemble when using up to 9 weeks of data, and improving and then better relative performance after 17 weeks of data, until performance worsened once more after 27 weeks of accuracy data. In contrast, forecasts of deaths in Belgium were better with fewer weeks of accuracy data, and weighting with between 14 and 26 weeks' data produced a worse performance than the unweighted ensemble of trajectories.

4. Discussion

A significant part of the value of collaborative infectious disease modelling projects comes from the standardisation of model output across varying numbers of model teams, methods, and simulations. We compared two methods of collecting information from multiple models' projections of an epidemic. We took three scenario models for each of five projection targets, and contrasted collecting a sample of up to 100 simulated trajectories against collecting quantile intervals of those trajectories at each time step.

We found that collecting simulated trajectories enabled analysis of trajectory shapes, peaks, and cumulative total burden. We observed that trajectories contained a right-skewed probabilistic distribution, which meant that ensembles either directly from trajectories, or using a linear opinion pool method, increasingly diverged from the quantile-average ensemble in projecting the outer upper limit of the probabilistic distribution. We also found that collecting trajectories could be used to create a competitively performing ensemble based on continuous predictive performance.

The common practice of collecting a standardised set of quantile intervals has several advantages. Firstly, combining across a set of quantiles should accurately represent the underlying distribution (Genest, 1992), and we observed that the linear opinion pool (based on a combination of quantiles) produced a near-identical ensemble as that created directly from combining individual trajectories. This suggests that the LOP ensemble may be the best choice for reflecting the widest range of uncertainty in settings where model results are only collected in quantiles, while noting that in order to create a LOP ensemble quantiles of cumulative rather than incident quantities need to be collected (Howerton et al., 2023). Furthermore, our results suggest little information about uncertainty is lost when using quantile outputs to compare the central estimates from different models. This is a useful validation for collecting multiple model results in any format when the purpose is short-term situational awareness.

Further advantages include where collecting quantile outputs also allows for a broader range of modelling methods, including quantile regression, that directly create quantile outputs rather than a joint distribution over time. Additionally, a single set of quantiles can be held in comma-separated value (csv) files of easily manageable size, requiring minimal technical knowledge of big data storage solutions or processing. This has been important in the past given a lack of readily available skills or investment in software for emergency outbreak settings. However this argument weakens with mounting evidence that this type of underresourcing hampers outbreak response (Sherratt et al., 2024; Rivers et al., 2020).

An alternative method for multiple model collection is directly collecting models' trajectories, with the advantage of retaining each trajectory's time-dependence. We observed greater availability and flexibility of accessing information from this method in contrast to collecting quantile distributions. This was evident when comparing the tails of multiple distributions in a quantile-average ensemble, assessing

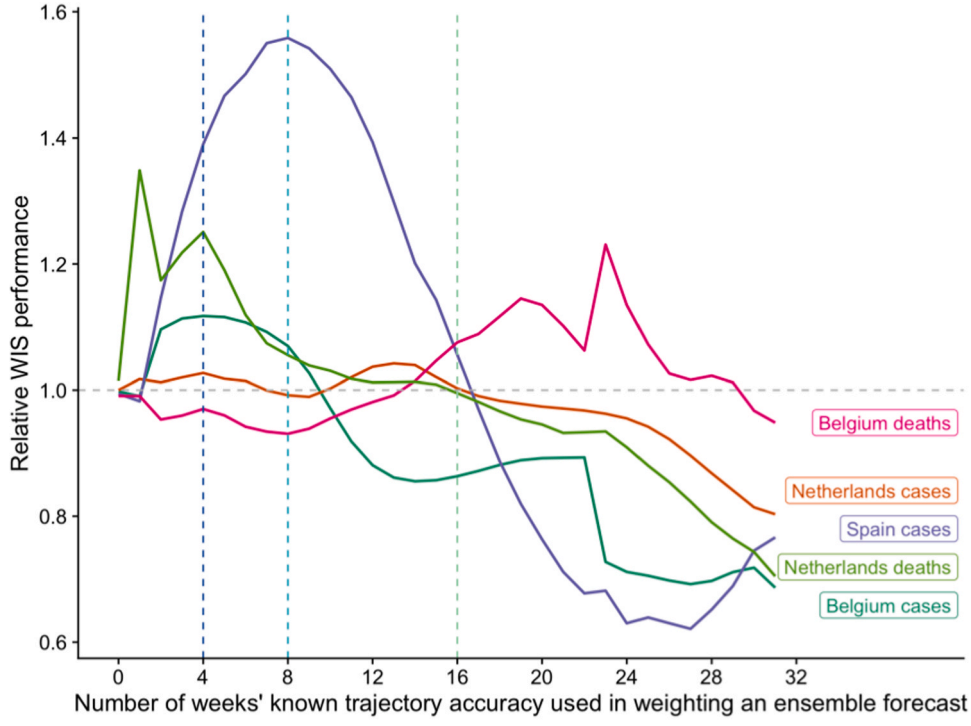

Fig. 3. Predictive performance of weighted ensembles by projection target. Weighted ensembles were created using a weighted median, where the weight of each trajectory was determined by its previous accuracy in predicting between 0 and 31 weeks of observed data (x axis). The performance of each ensemble is measured by the weighted interval score (WIS); a lower WIS score indicates better performance of the weighted ensemble than the simple unweighted median ensemble of all trajectories (reference line at 1).

the number of projected waves or the risk of crossing a specific threshold such as the burden in the preceding year, or in reevaluating projections against reported data. These analyses could also be conducted after collecting model outputs, making the method of collecting trajectories more flexible to the needs of one or multiple end-users. In particular these areas of information are more likely to be relevant to longer term preparedness and mitigation. As a result, we suggest the impact of information gains and losses from model collection may differ depending on the aim of a multi-model comparison.

Our findings comparing quantile with trajectory model outputs are compatible with ongoing work addressing issues from the loss of epidemic shape. From point forecasts, recent forecasting work has created an ensemble from multiple point forecasts in terms of similarity to canonical curve shapes (Srivastava et al., 2022). From probabilistic models it is also possible to create an ensemble of many trajectories using the centrality of each curve as a weight in a curve boxplot (Juul et al., 2021).

We have also demonstrated the potential for unique information gains when collecting simulated trajectories by assessing their performance against observed data. By conditioning the weight of each trajectory in an ensemble on subsequently observed data, we were able to create an ensemble that excluded entire trajectories, or epidemic curves, based on dependence to unrealised events. This typically either matched or reduced the uncertainty of an unweighted equivalent ensemble, and in some settings performed better overall than the unweighted equivalent.

This suggests an additional way in which collaborative modelling efforts can respond to changing outbreak dynamics and policy needs. For our setting, model results had originally been created based on a set of four scenarios relevant to policy decisions to be made in spring/summer 2022. However, given the complex dynamics of disease transmission, no predefined future scenario is likely to accurately predict eventual reality. Among four scenarios with deliberately contrasting assumptions, most of these assumptions will be disproven by observation over time. Meanwhile, when scenario modelling outputs are collected as quantiles at each time-point, they lose their time-dependence and thus cannot be interpreted except in the light of an increasingly obsolete scenario context.

By focussing on individual model simulations in this work, we were able to abstract away from the context in which model results were created. We weighed each trajectory using only its past accuracy against observed data, regardless of the modelling technique, original scenario, or parameter values from which it arose. From this we created an ensemble that did not reflect any particular scenario assumptions, but only the time-varying accuracy of each trajectory. This meant we were able to continue to use trajectories in an ongoing evaluation, increasing the useful life of the results from a single cross-sectional collection of multiple model output.

This could be particularly useful when repeated rounds of model collection are time-intensive or computationally expensive, such as for individual-based models, or where personnel resources are constrained such as in an ongoing outbreak with potentially many competing priorities. Whilst beyond the scope of this work, future work in this area could also investigate model weights in order to rank trajectories from each scenario by proximity to observed trajectories and potentially interpret this as proximity of the given scenario assumptions to reality.

We highlight several important limitations to our comparison of information gains and losses between methods of collecting model output. In the first part of this work, we represent an analysis of trajectories that reflects our work in real-time response to the needs of policy decision-making. We did not consider alternative approaches to time-series analysis that would likely make the analysis of trajectories more robust, for example in calculations of peaks or wave durations.

In our comparison to quantile distributions, our method of collecting simulated trajectories was not specifically designed for this comparative purpose, and as a result our findings are difficult to interpret. In this work we did not attempt to characterise how many samples might be sufficient to appropriately represent a probabilistic distribution in comparison to a quantile representation. For example, in some cases the collated trajectories were already subsampled from model runs conducted by individual teams. In contrast, in a situation with low sampling sizes of trajectories from each model, a quantile representation might provide a more stable representation.

We suggest that further work should characterise and standardise sampling techniques for model simulations in multi-model comparisons. Future study designs could focus on collating multiple representations (e.g. time-sliced quantiles and trajectories) from contributing teams directly for comparison, or collate arbitrary numbers of feasible simulated trajectories and re-weigh according to the number of simulations. Our work also demonstrates the importance of investing in and developing capacity to store and use simulation outputs rather than fixed-time quantile probabilities for well founded intercomparison modelling projects.

To conclude, we observed several information gains from collecting modelled trajectories rather than summarised quantile distributions. We highlight the potential to create continuous new information from a single collection of model output. Working from combined simulations offers the opportunity to explore creating ensembles by the shape of epidemic curve that can be updated over time, and for more detailed quantitative evaluations against observed data, such as in projected peaks or cumulative totals. We believe our findings apply whether projections are conditioned on the context of the present (as in forecasts), or on schematic futures (as in scenarios). However, the value of different information gains and losses may vary with the aims of each collaborative effort, depending on the requirements and flexibility required by projection users. Understanding potential information gains and losses when collecting model projections can support the accuracy, reliability, sustainability, and communication of collaborative infectious disease modelling efforts.

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CRediT authorship contribution statement

Rok Grah: Data curation, Writing – review & editing. **Ajitesh Srivastava:** Data curation, Writing – review & editing. **Kai Nagel:** Data

curation, Writing – review & editing. **Katharine Sherratt:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Christof Schütte:** Data curation, Writing – review & editing. **Tim Conrad:** Data curation, Writing – review & editing. **Sebastian Funk:** Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing. **Maria Cristina Marinescu:** Data curation, Writing – review & editing. **Frank Sandmann:** Data curation, Writing – review & editing. **Aymar Cublier:** Data curation, Writing – review & editing. **Bastian Prasse:** Data curation, Writing – review & editing. **David E. Singh:** Data curation, Writing – review & editing. **Rene Niehus:** Data curation, Writing – review & editing. **Kylie Ainslie:** Data curation, Writing – review & editing. **Jakob Rehmann:** Data curation, Writing – review & editing. **Nicolas Franco:** Data curation, Writing – review & editing. **Alberto Cascajo Garcia:** Data curation, Writing – review & editing. **Jesus Carretero:** Data curation, Writing – review & editing. **Philippe Beutels:** Data curation, Writing – review & editing. **Christel Faes:** Data curation, Writing – review & editing. **Steven Abrams:** Data curation, Writing – review & editing. **Lander Willem:** Data curation, Writing – review & editing. **Ricardo Ewert:** Data curation, Writing – review & editing. **Billy Charlton:** Data curation, Writing – review & editing. **Sebastian Müller:** Data curation, Writing – review & editing. **Niel Hens:** Data curation, Writing – review & editing. **Sam Abbott:** Writing – review & editing. **Christian Rakow:** Data curation, Writing – review & editing. **Sydney Paltra:** Data curation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code and data are available on Github at: epiforecasts/multi-model-information; DOI: https://doi.org/10.5281/zenodo.10891377 .

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.epidem.2024.100765.

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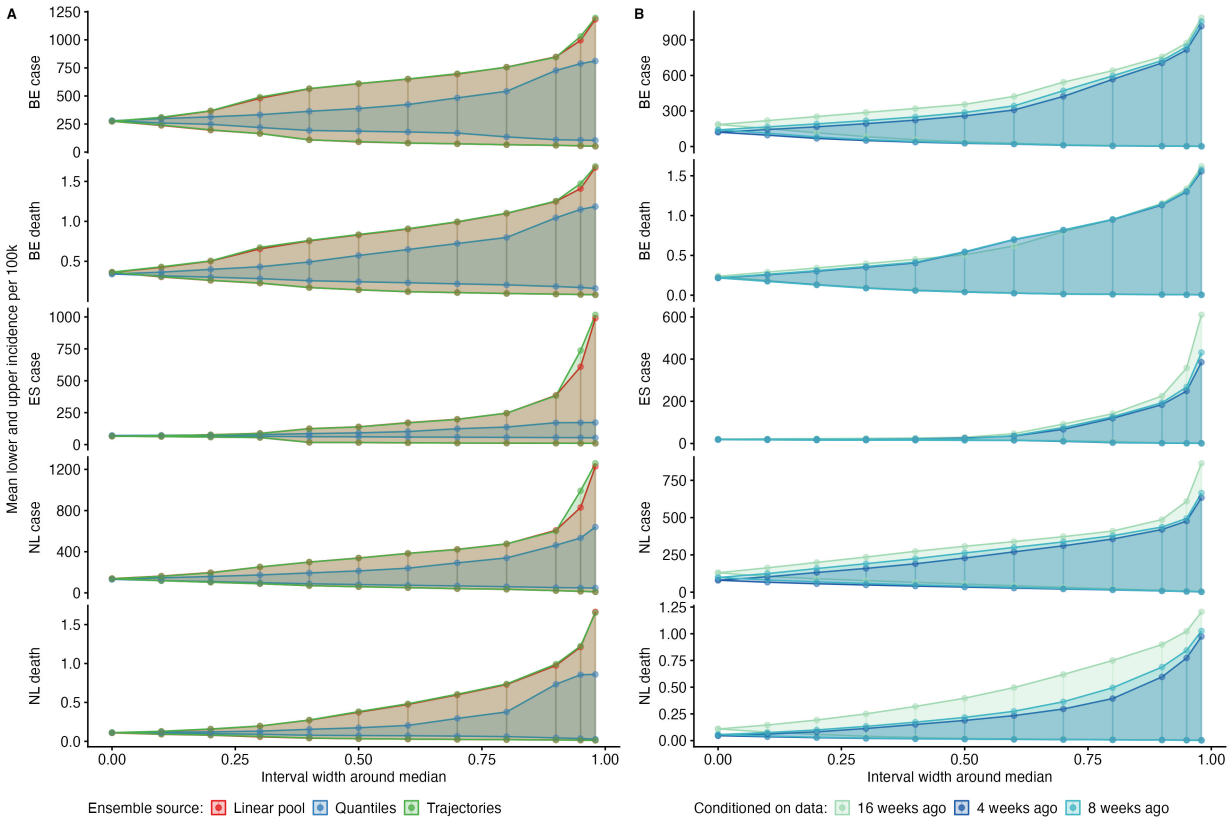
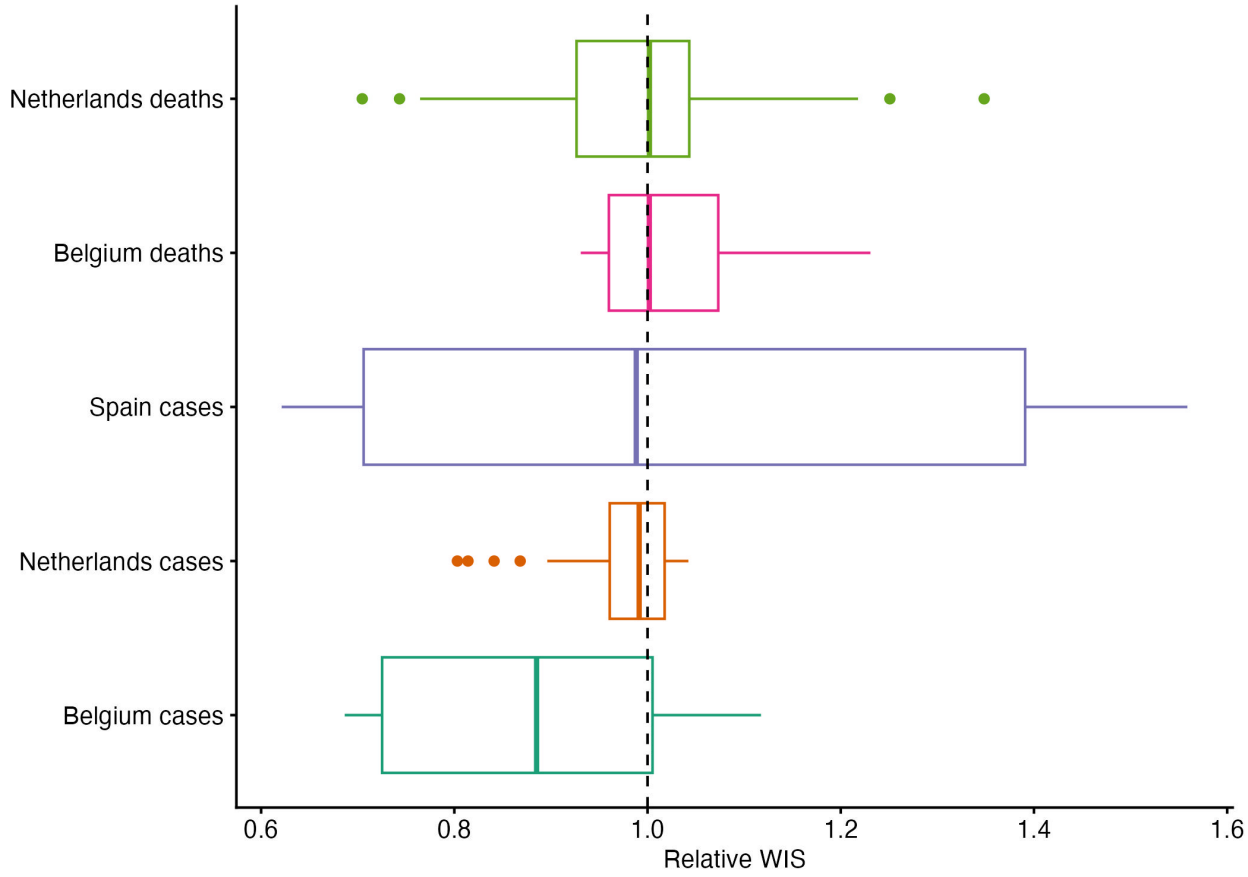
Supplementary Information

SI Figure 1



SI Figure 1. Mean central prediction intervals at increasing distances from the median. The 52-week mean of incidence per 100,000 population across all time points and scenarios, showing mean central prediction intervals at increasing distances from the median (interval width), by aggregation method (A) or weighting (B). The median estimate for each ensemble has 0 interval width (x-axis), with uncertainty increasing until an interval width at 0.98 represents the 1%-99% credibility interval around the median.

SI Figure 2



SI Figure 2. Distribution of forecast performance scores (relative WIS), of forecasts from model trajectories weighted using 0 through 31 weeks' available data. Performance is compared to an unweighted ensemble (reference line at 1).

SI Table 1

Team	Methods
ECDC	
ECDC-CM_ONE	Discrete-time, deterministic, mean-field SEIR-type compartmental model on metapopulation level. Population divided by age, vaccination status, and previous recovery; incl. seasonality, BA2 & behavior.
Dutch National Institute of Public Health and the Environment (RIVM)	
RIVM-vacamole	Deterministic, age-structured SEIR model, accounting for differences in susceptibility/infectiousness by age, seasonality, contact patterns, modes of vaccine protection, and waning immunity.
SIMID	
SIMID-SCM	Stochastic age-structured discrete time extended compartmental model
Universidad Carlos III de Madrid	

Team	Methods
UC3M-EpiGraph	Agent-based parallel simulator that models individual interactions extracted from social networks and demographical data.
University of Southern California	
USC-SIkJalpha	Uses SIKJalpha which models temporally varying infection, death, and hospitalization rates. Learning is performed by reducing the problem to multiple simple linear regression problems.

SI Table 1. Teams that contributed models to Round 2 of the European Scenario Hub, with self-described methods and links to further information. See also:

- Full model metadata, at: https://github.com/covid19-forecast-hub-europe/covid19-scenario-hub-europe/tree/main/model-metadata
- Information about each model's assumptions for Round 2, at: https://github.com/covid19-forecast-hub-europe/covid19-scenario-hub-europe/tree/main/model-abstracts/2022-07-24

Round 2 report

The following pages are the original website reporting for the European Scenario Hub Round 2 as of July 2022.

The report is currently (January 2023) available at: https://covid19scenariohub.eu/report2.html

Code to generate this report is available at: https://github.com/european-modelling-hubs/covid19-scenario-hub-europe-website/blob/main/report2.Rmd

Round 2
Scenarios

We asked teams of researchers across Europe to use quantitative models to project COVID-19 outcomes for 32 European countries over the next year. In order to explore different sets of assumptions about drivers of the pandemic, we asked teams to vary four sets of parameters. We can describe this in a 2x2 scenario specification:

	Age 60+ booster campaign • 2nd* booster recommended for 60+ • Uptake starts 15th September, and reaches 50% coverage by 15th December	**Age 18+ booster campaign** • 2nd* booster recommended for general population, ages 18+ • Uptake starts 15th September, and reaches 50% coverage by 15th December
Optimistic vaccine effectiveness • Increased booster vaccine effectiveness to that seen against **Delta variant**	Scenario A	Scenario B
Pessimistic vaccine effectiveness • Reduced booster vaccine effectiveness against infection from **BA.4/BA.5/BA.2.75 variants**	Scenario C	Scenario D

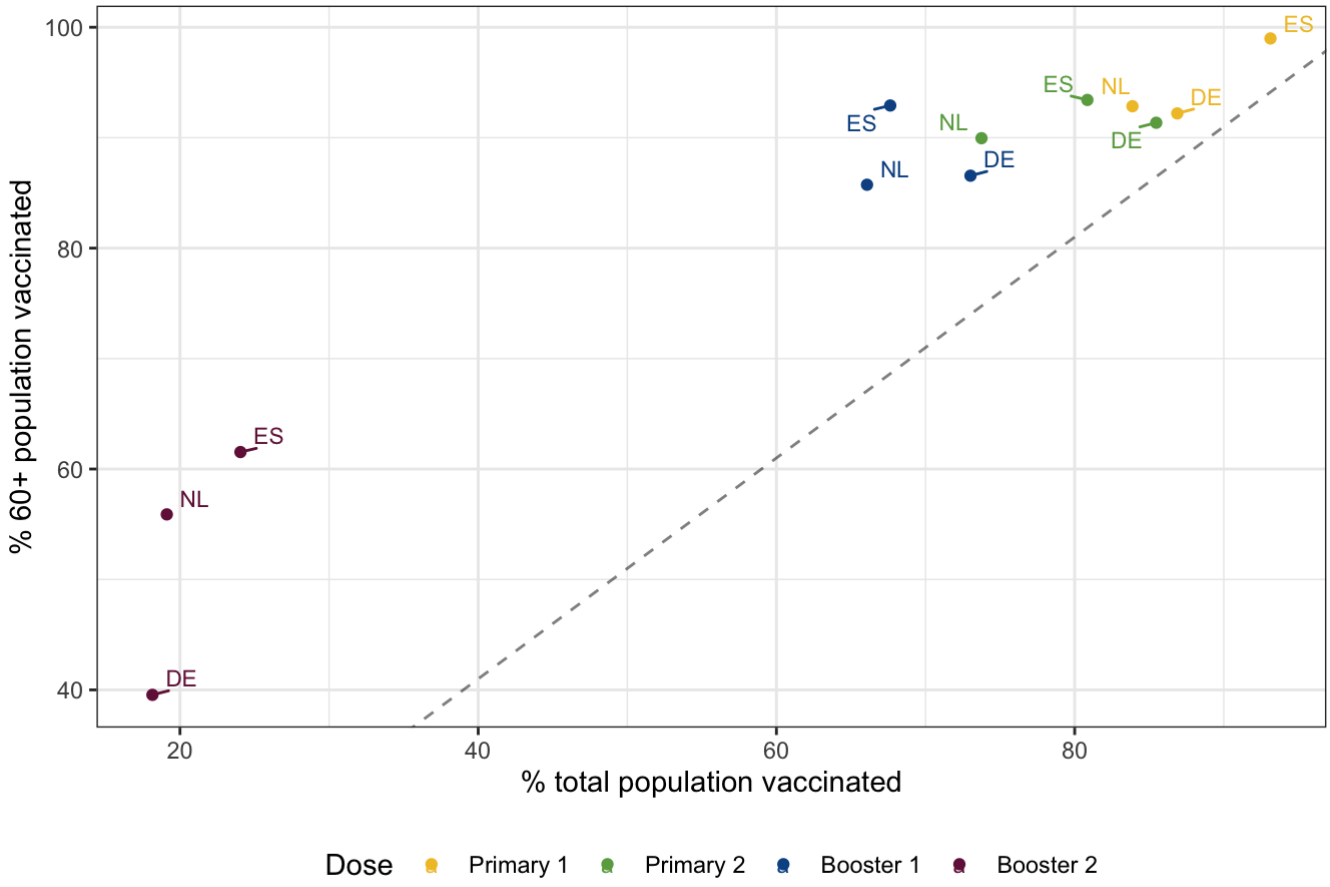
See also the full scenario details (https://github.com/covid19-forecast-hub-europe/covid19-scenario-hub-europe/wiki/Round-2) for more detail on the common set of assumptions teams used to create their models.

In Round 2, we asked modellers to start their projections from the 2022-07-24. Data after this date were not included, and as a result, model projections are unlikely to fully account for later information on the changing variants or behavioural patterns.

In this report we only show results from countries with at least 3 models.

Current situation

We consider vaccination rates in countries for which multiple teams of modellers contributed projections.



Vaccination % among 60+ population, relative to total population

Participating teams

6 models contributed scenario projections to Round 2.

Models

Participating teams by number of countries and horizon

Team	Countries	Weeks
USC-SIkJalpha	31	52
ECDC-CM_ONE	28	53
MODUS_Covid-Episim	1	53
RIVM-vacamole	1	53
SIMID-SCM	1	52
UC3M-EpiGraph	1	41

Countries

Number of independent model projections for each target variable and location

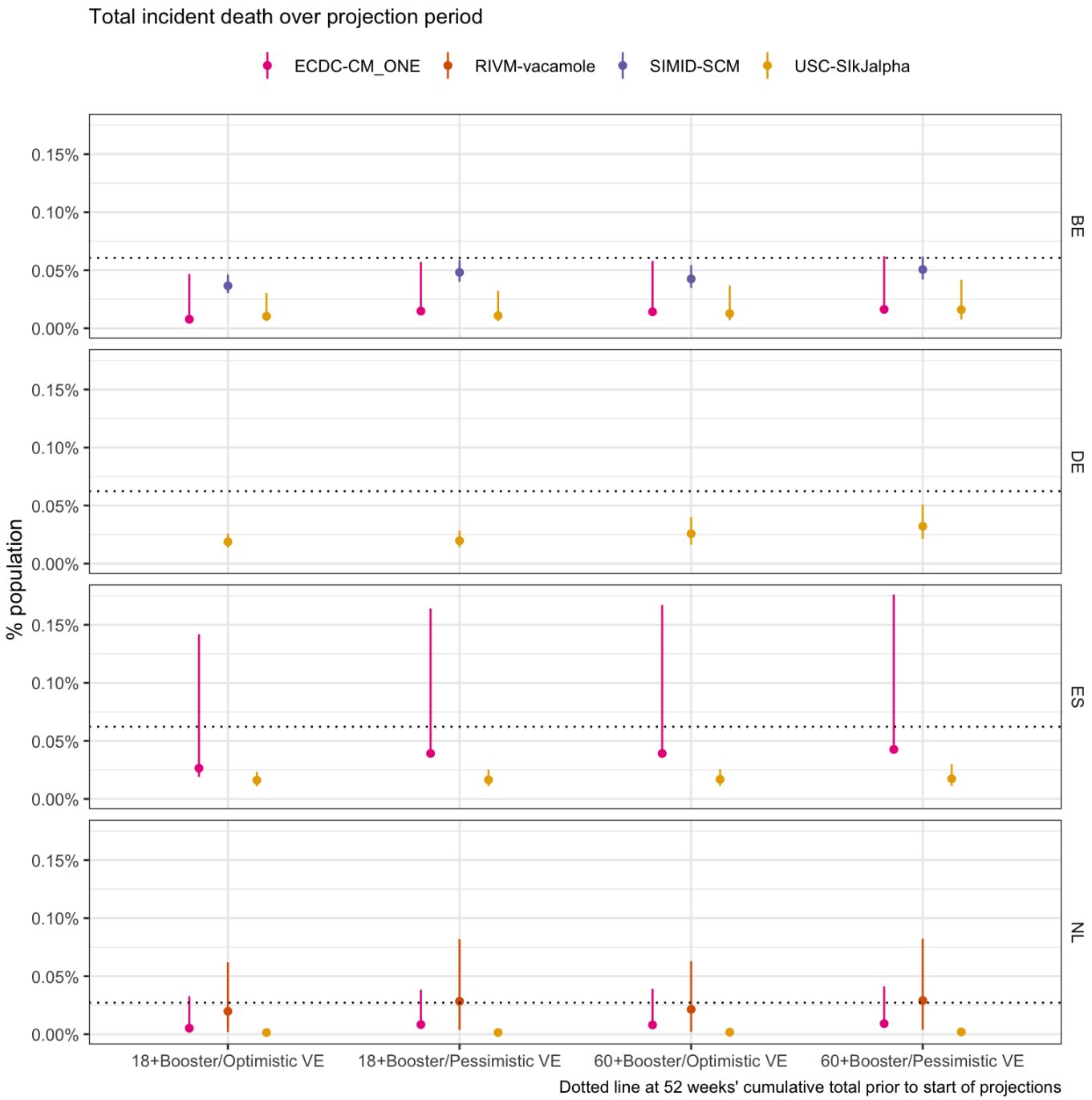
Code	Country	Infection	Case	Hosp	Icu	Death
BE	Belgium	1	3	2	1	3
DE	Germany	1	2	2	0	1
ES	Spain	1	3	2	0	2

Code	Country	Infection	Case	Hosp	Icu	Death
NL	Netherlands	1	3	2	1	3

Cumulative outcomes

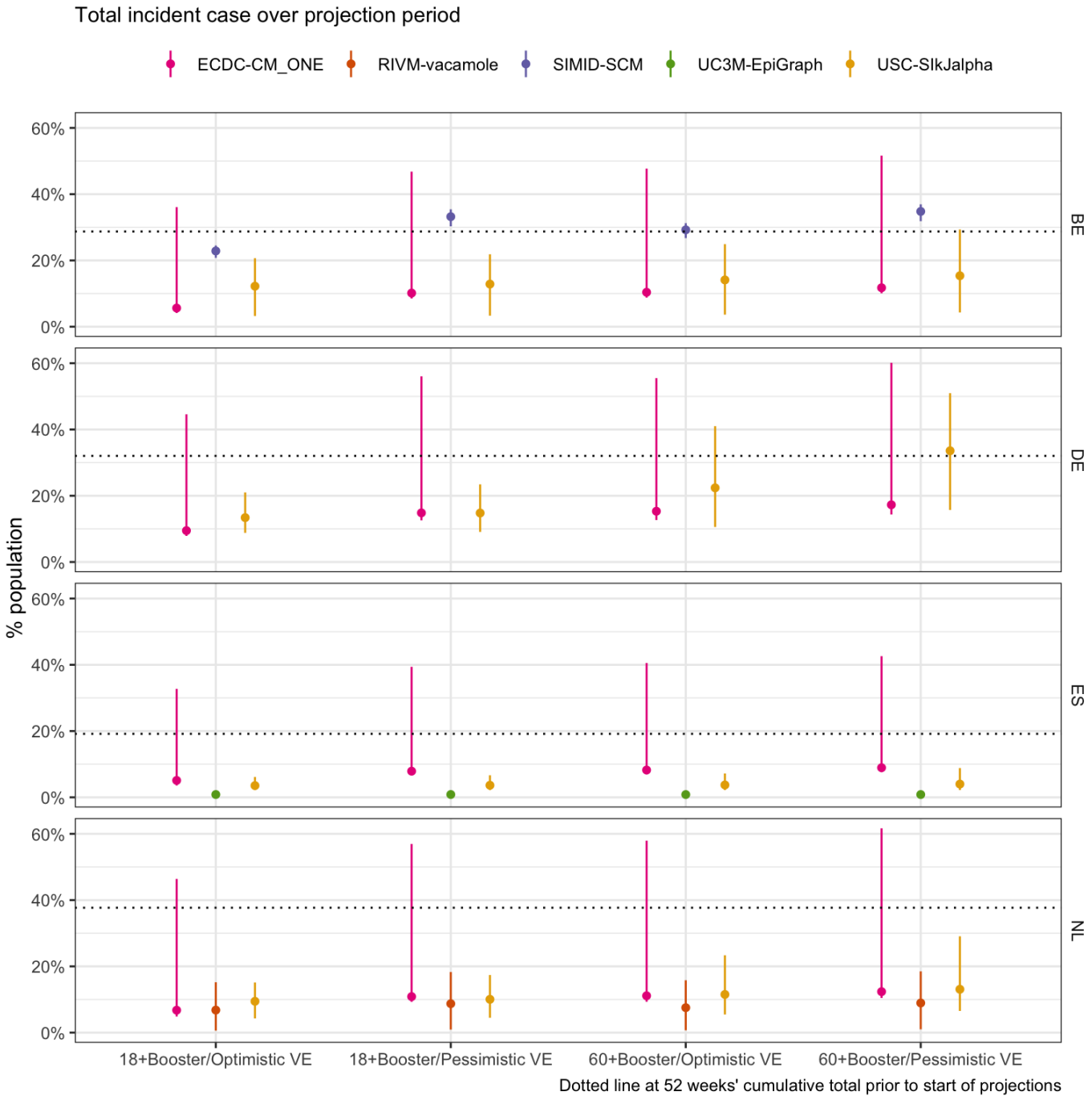
For each model and scenario, we compare the total number of outcomes over the entire projection period as a % of the total country population. We compared the cumulative number of projected outcomes to the cumulative total over one year before projections started (July 2021 to July 2022).

Death



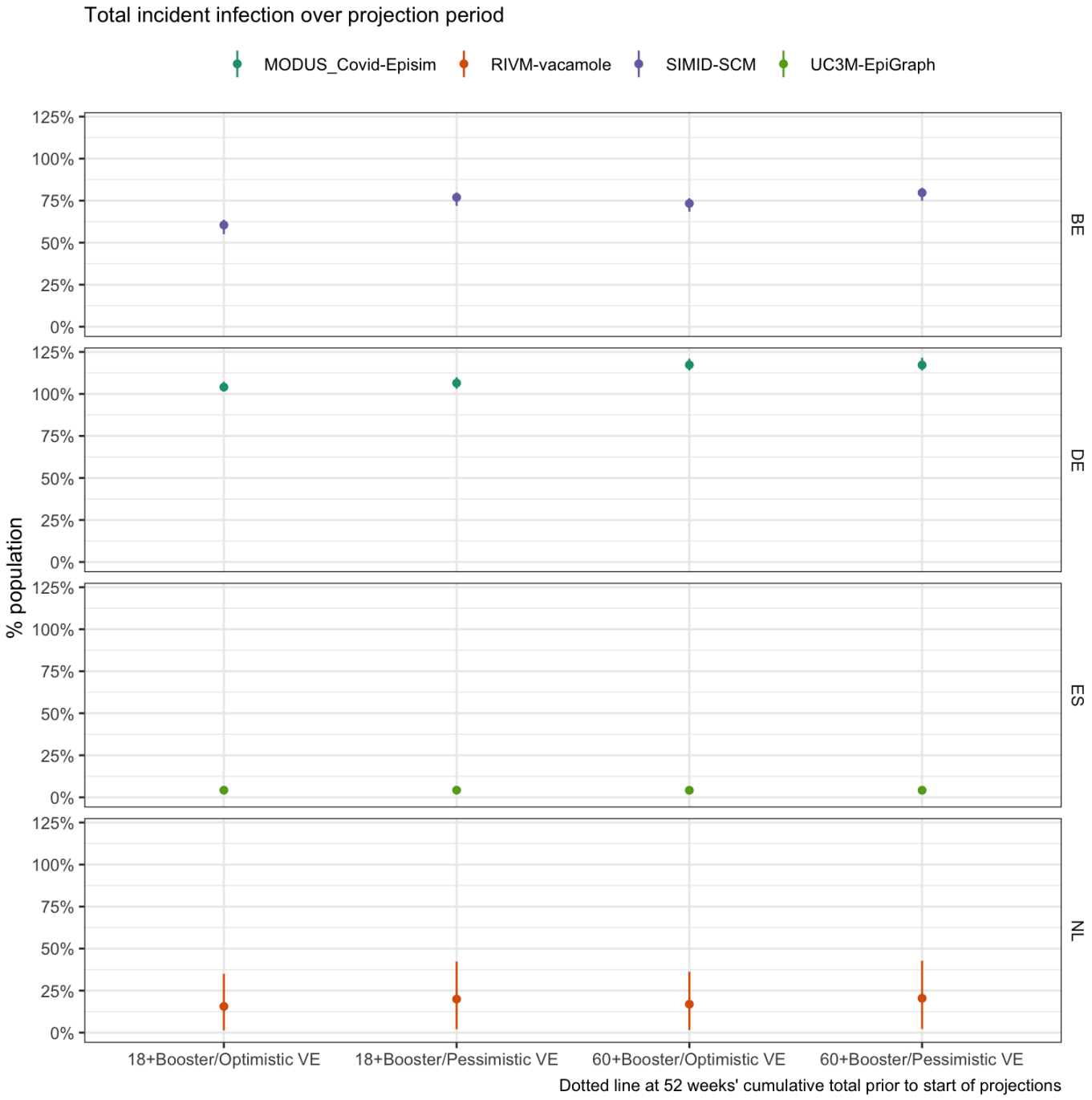
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Case



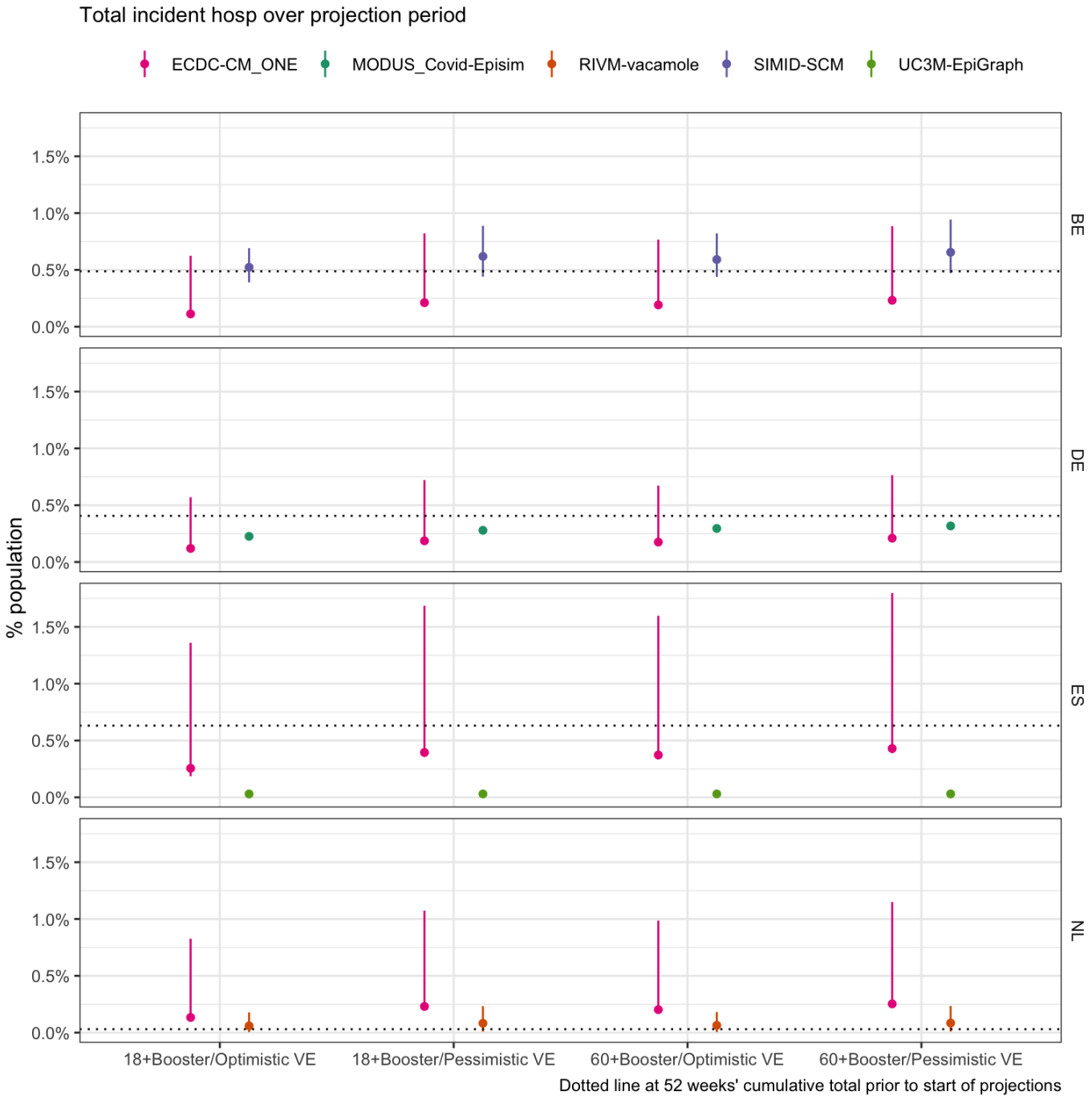
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Infection



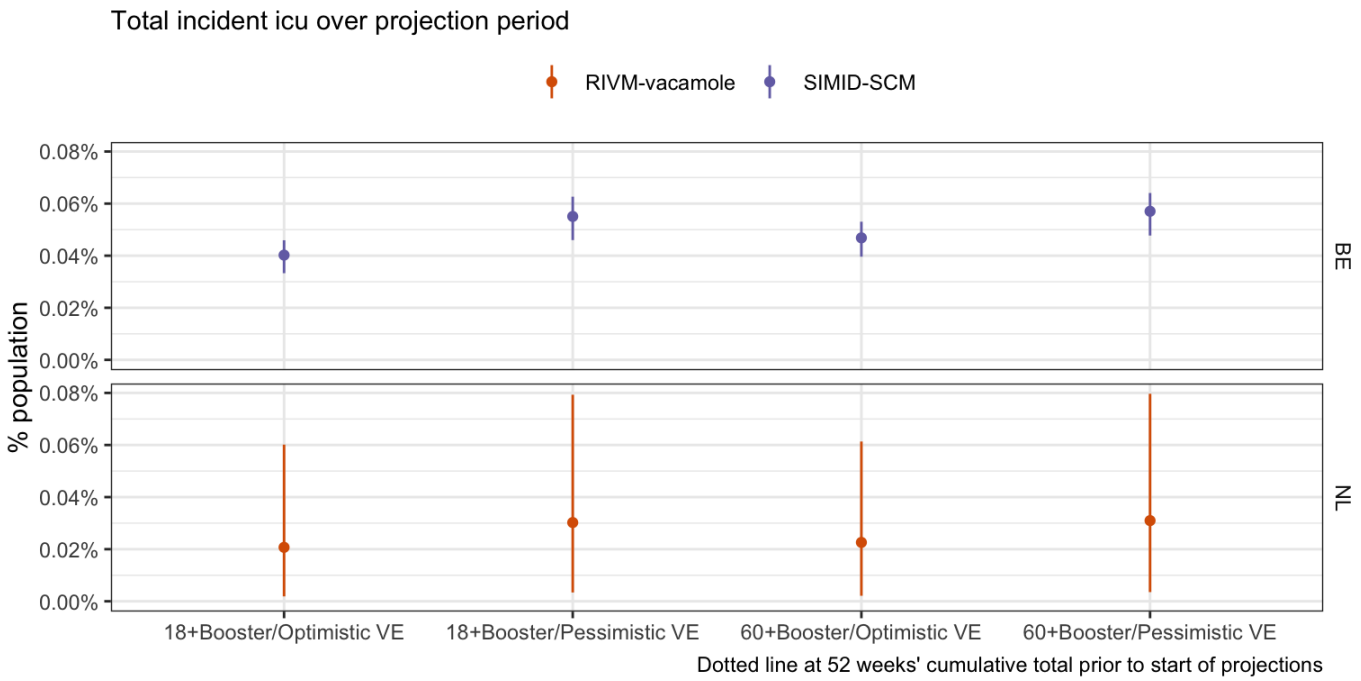
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Hosp



Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Icu



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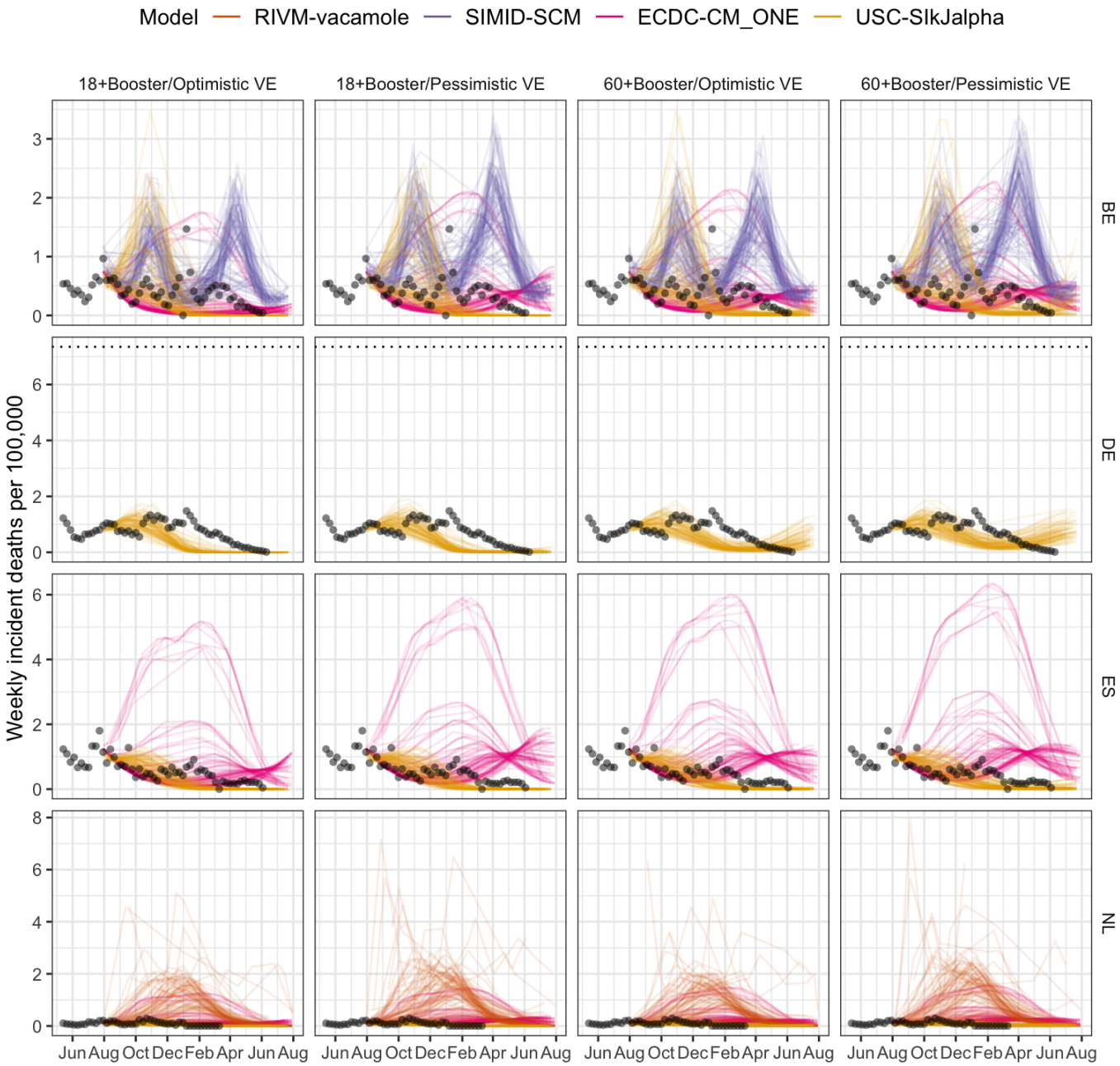
Incident outcomes

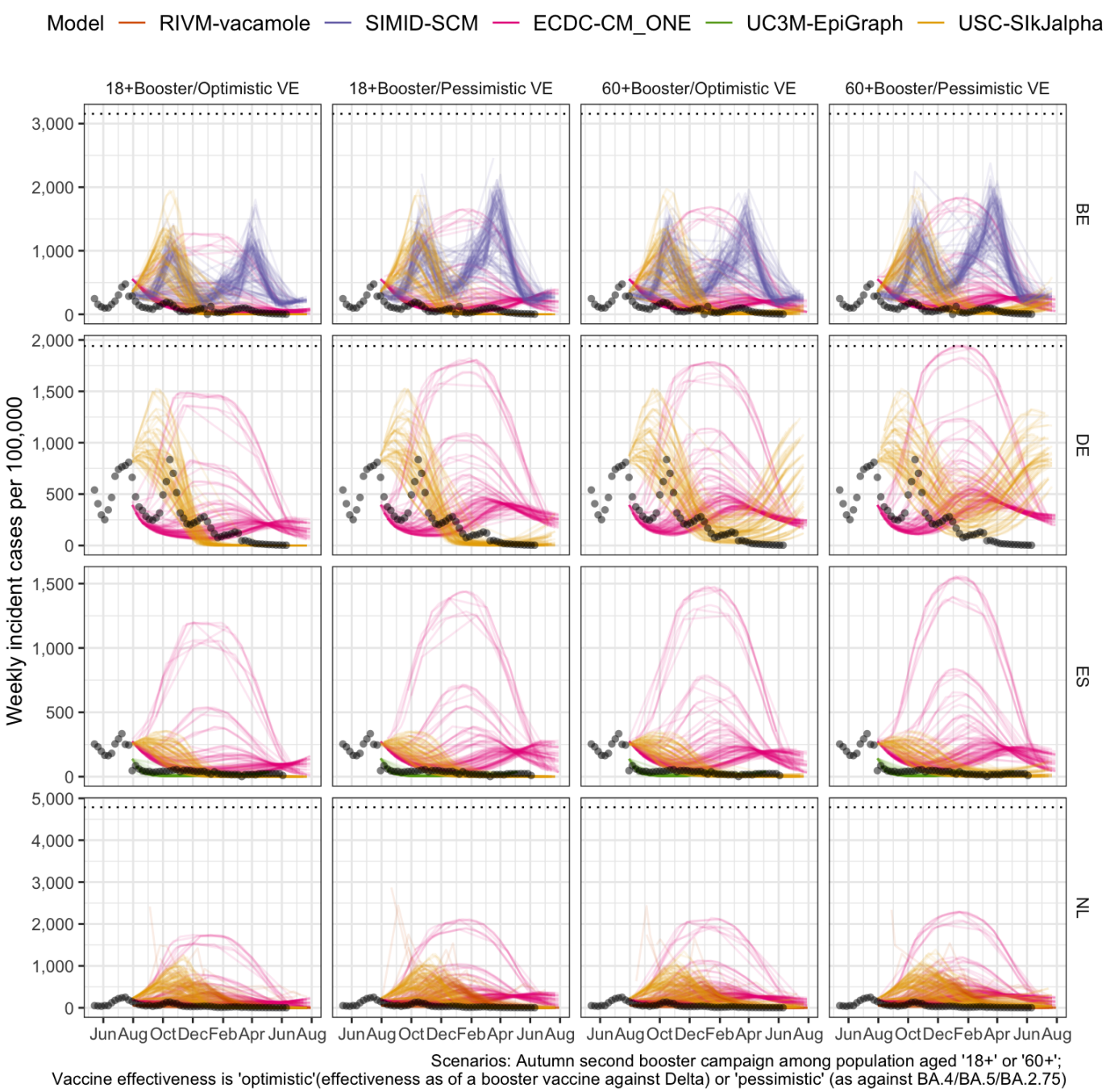
We explored the incidence of COVID-19 per 100,000 over the projection period and in terms of projected peaks in incidence. We summarised peaks both over the entire projection period, and over only the autumn-winter period (October through March); we considered (A) the timing and maximum weekly incidence of each peak, and (B) the total number of peaks.

Trajectories

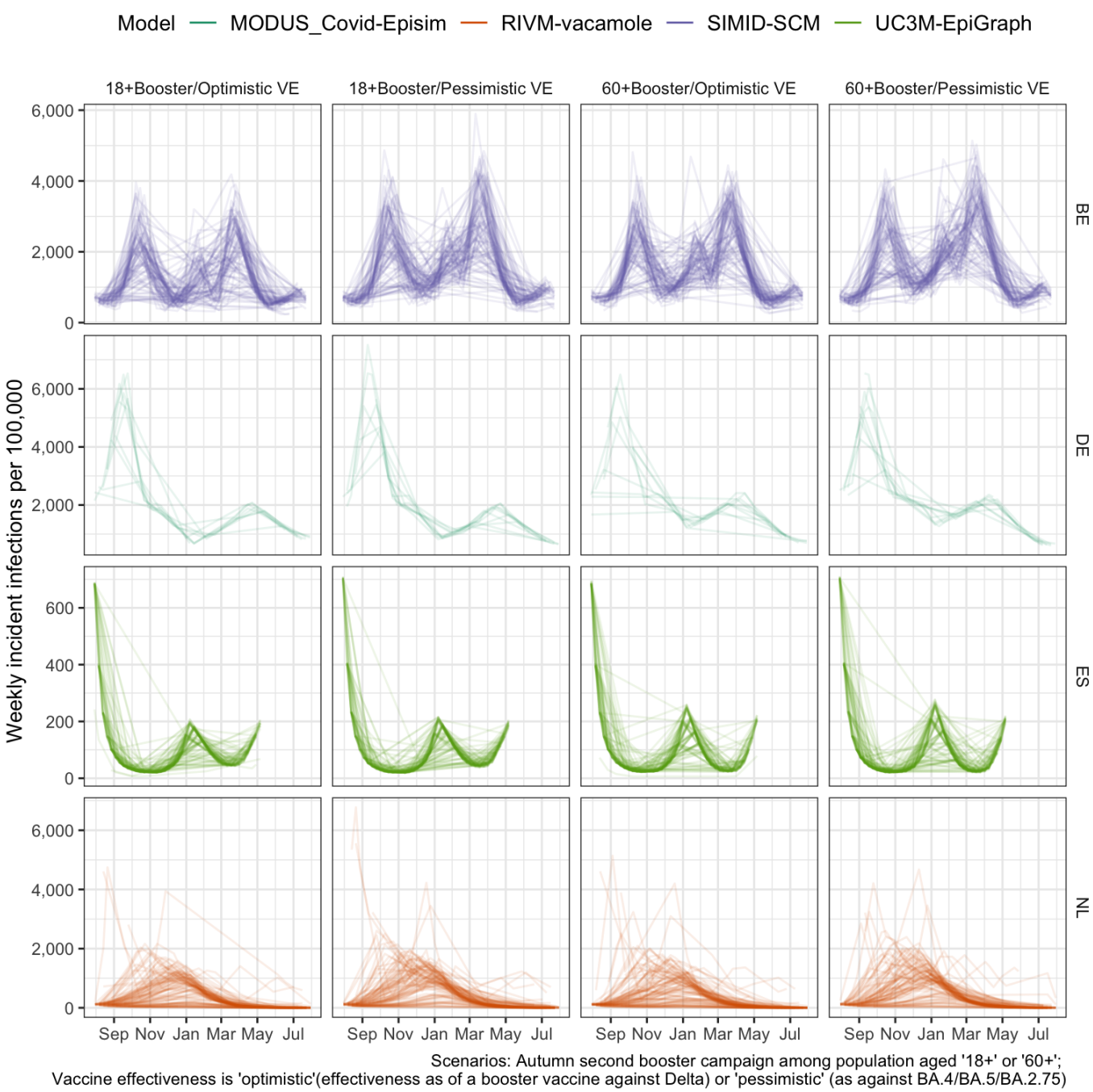
Death



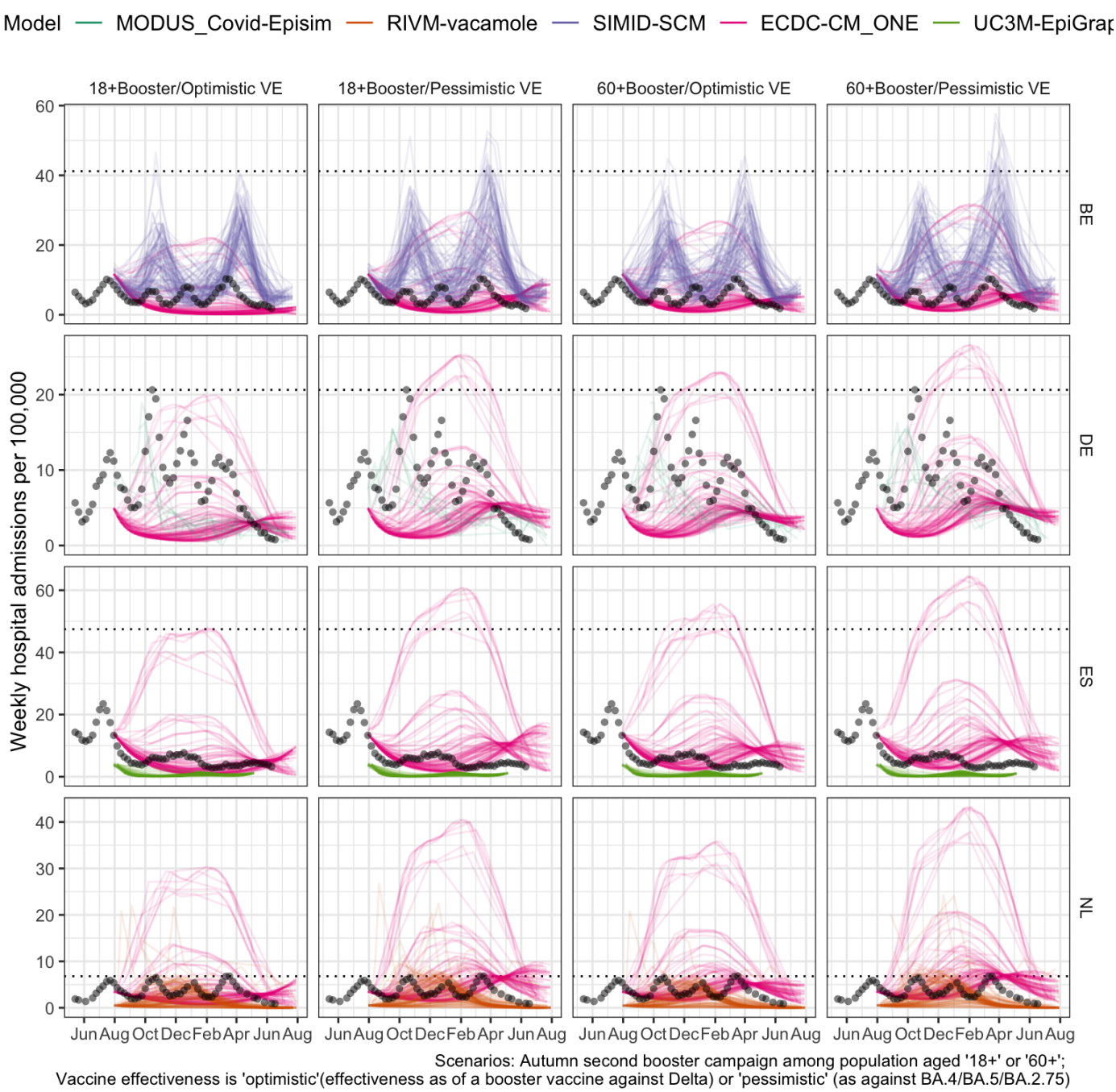
Case


Model
RIVM-vacamole
SIMID-SCM
ECDC-CM_ONE
UC3M-EpiGraph
USC-SIkJalpha
18+Booster/Optimistic VE
18+Booster/Pessimistic VE
60+Booster/Optimistic VE
60+Booster/Pessimistic VE
BE
DE
ES
NL
Weekly incident cases per 100,000
Jun Aug Oct Dec Feb Apr Jun Aug
Scenarios: Autumn second booster campaign among population aged '18+' or '60+';
Vaccine effectiveness is 'optimistic'(effectiveness as of a booster vaccine against Delta) or 'pessimistic' (as against BA.4/BA.5/BA.2.75)

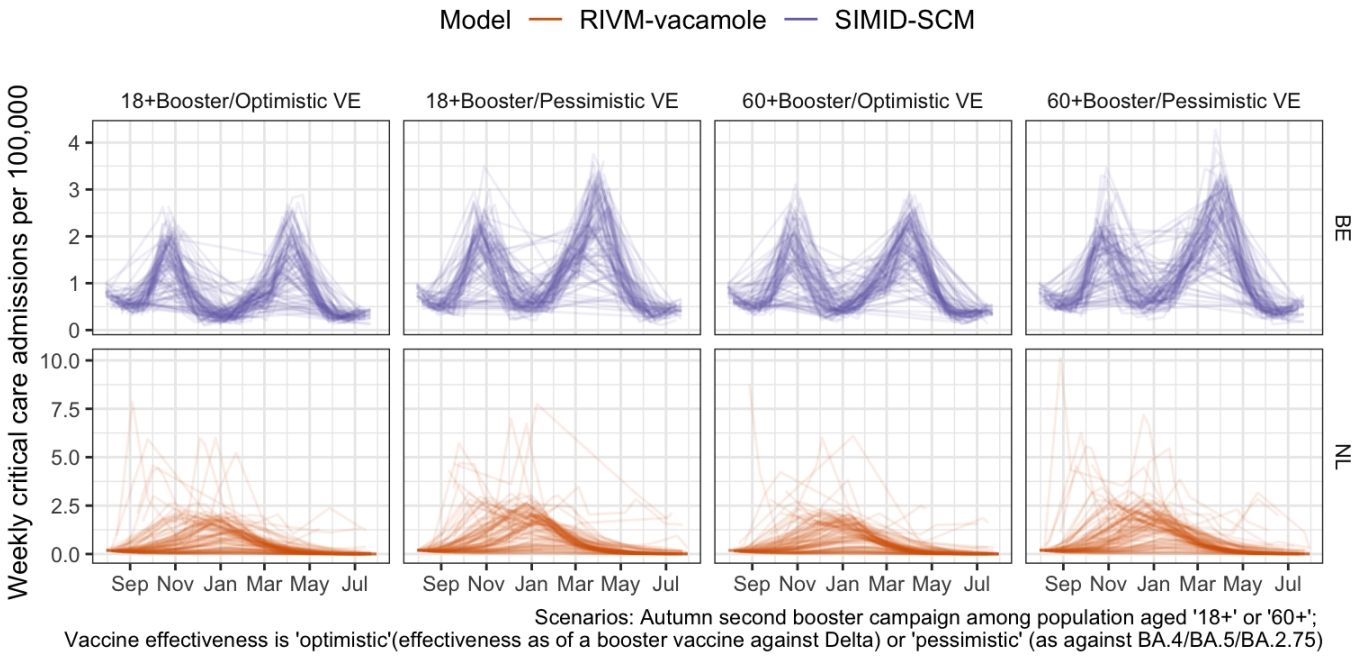
Infection


Model
MODUS_Covid-Episim
RIVM-vacamole
SIMID-SCM
UC3M-EpiGraph
18+Booster/Optimistic VE
18+Booster/Pessimistic VE
60+Booster/Optimistic VE
60+Booster/Pessimistic VE
Weekly incident infections per 100,000
BE
DE
ES
NL
Sep Nov Jan Mar May Jul
Scenarios: Autumn second booster campaign among population aged '18+' or '60+';
Vaccine effectiveness is 'optimistic'(effectiveness as of a booster vaccine against Delta) or 'pessimistic' (as against BA.4/BA.5/BA.2.75)

Hosp


Model
MODUS_Covid-Episim
RIVM-vacamole
SIMID-SCM
ECDC-CM_ONE
UC3M-EpiGrap
18+Booster/Optimistic VE
18+Booster/Pessimistic VE
60+Booster/Optimistic VE
60+Booster/Pessimistic VE
BE
DE
ES
NL
Weekly hospital admissions per 100,000
Jun Aug Oct Dec Feb Apr Jun Aug
Scenarios: Autumn second booster campaign among population aged '18+' or '60+';
Vaccine effectiveness is 'optimistic'(effectiveness as of a booster vaccine against Delta) or 'pessimistic' (as against BA.4/BA.5/BA.2.75)

Icu



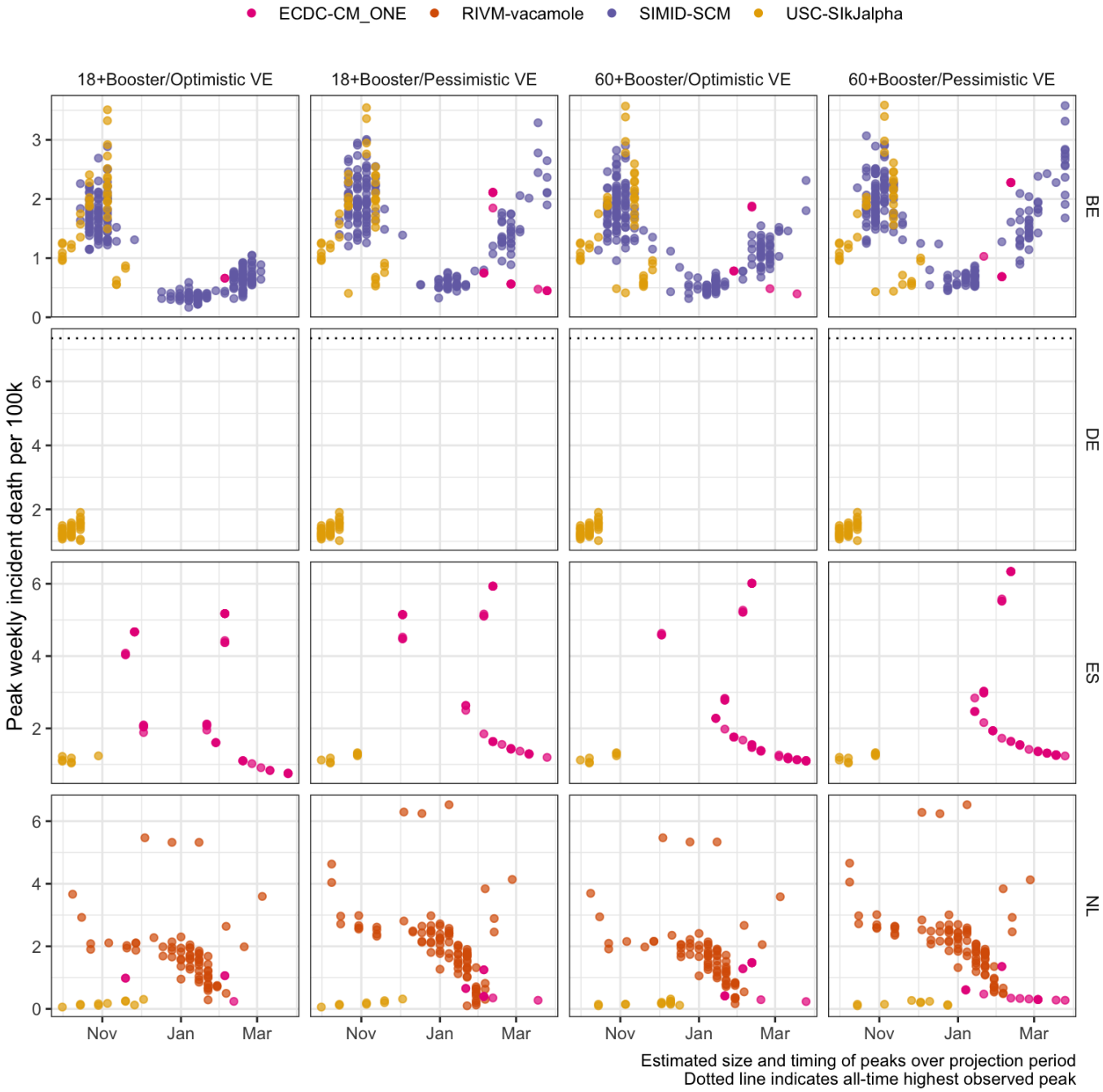
Peaks

Autumn-winter

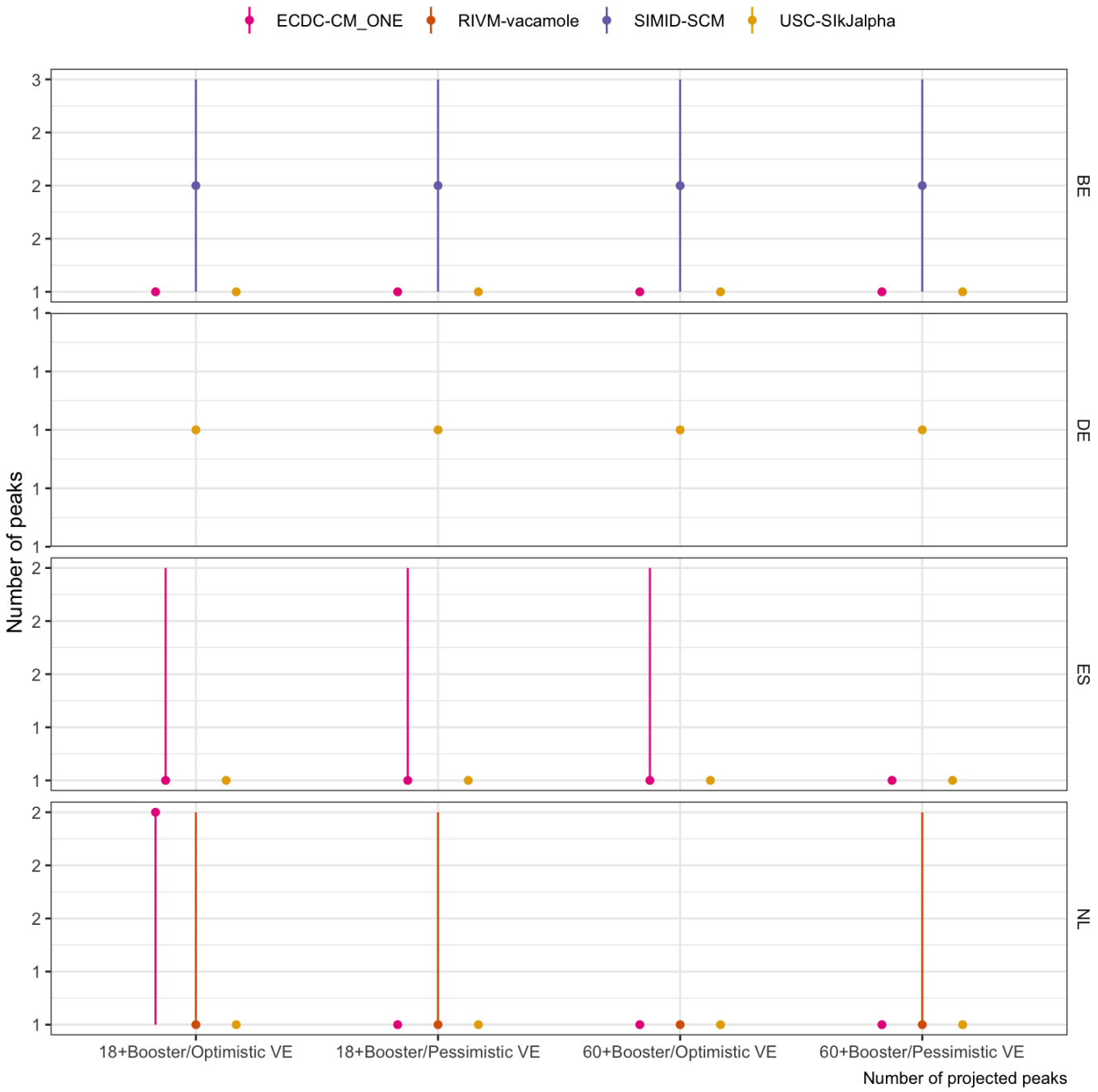
Projections over October 2022 through March 2023

Death

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



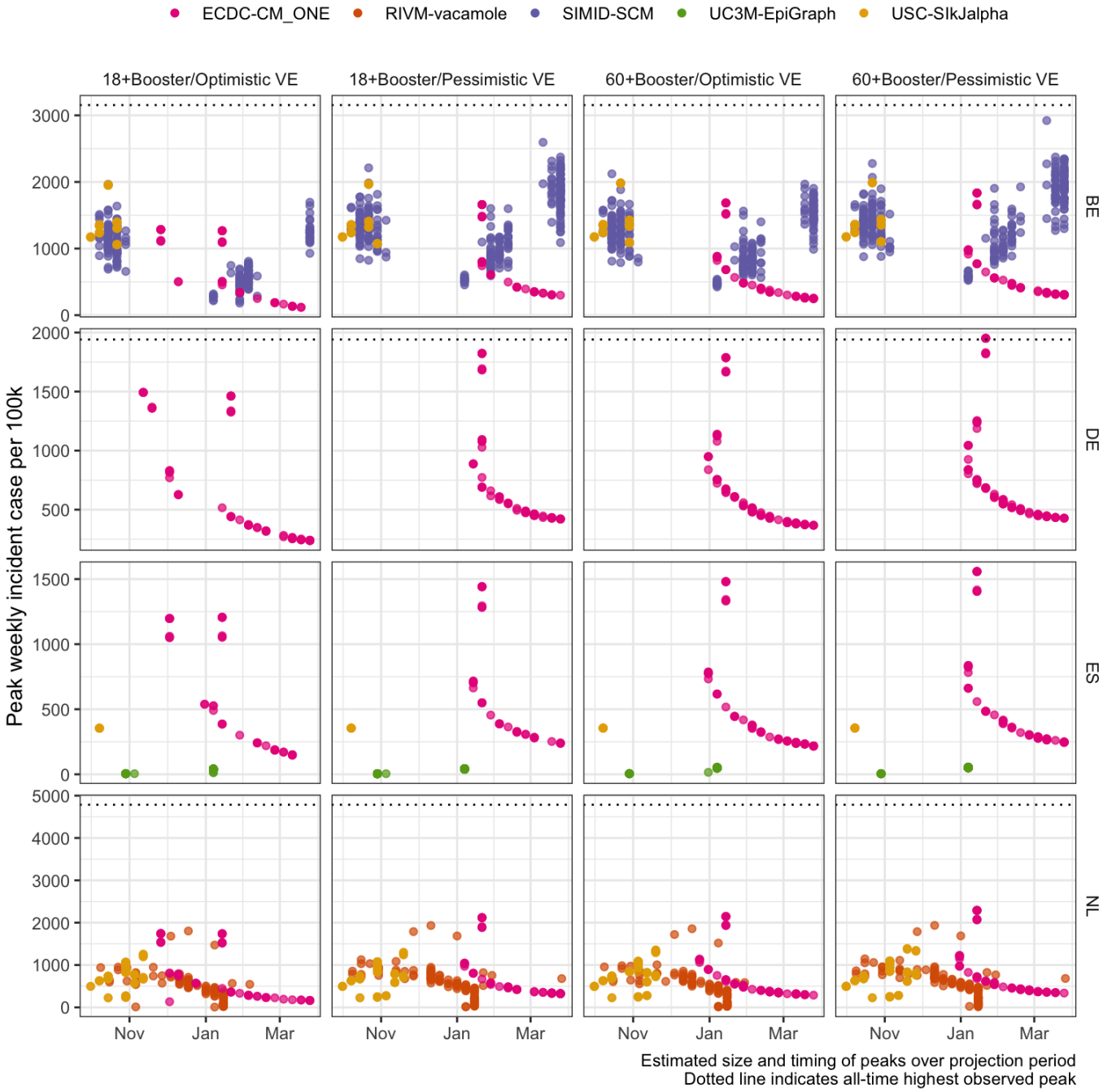
B. Projected number of peaks (median with 5-95% probability)



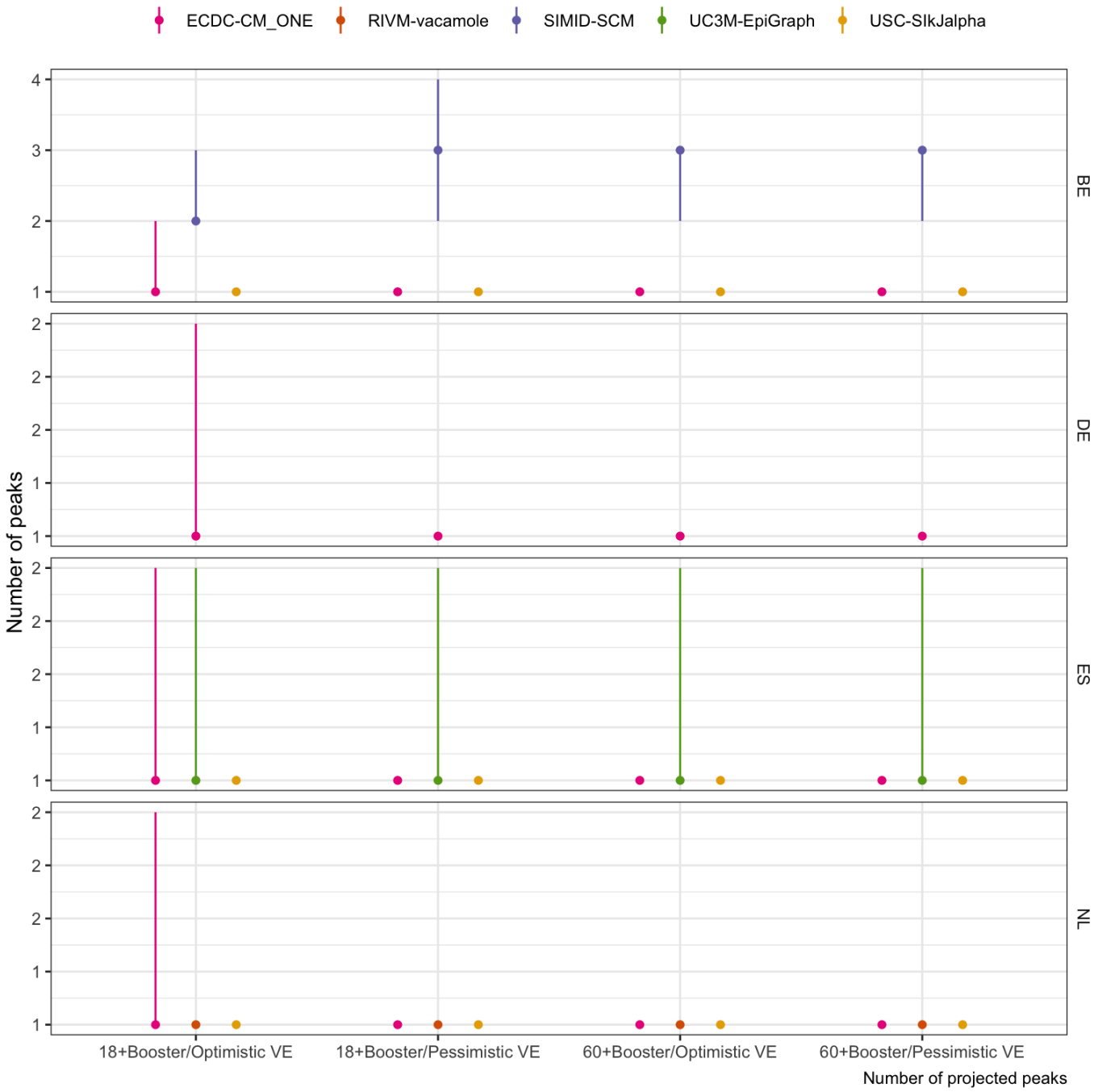
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Case

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



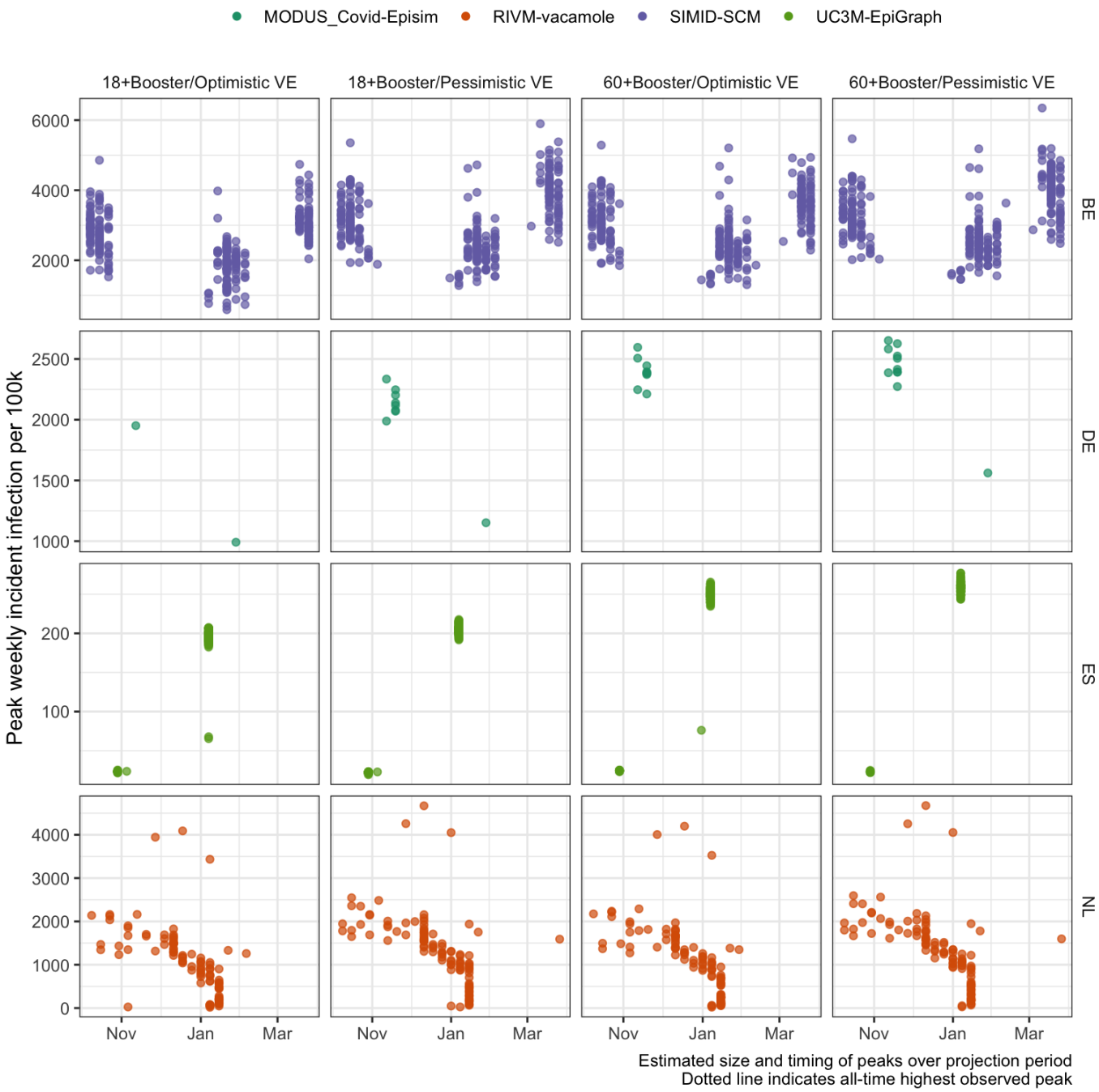
B. Projected number of peaks (median with 5-95% probability)



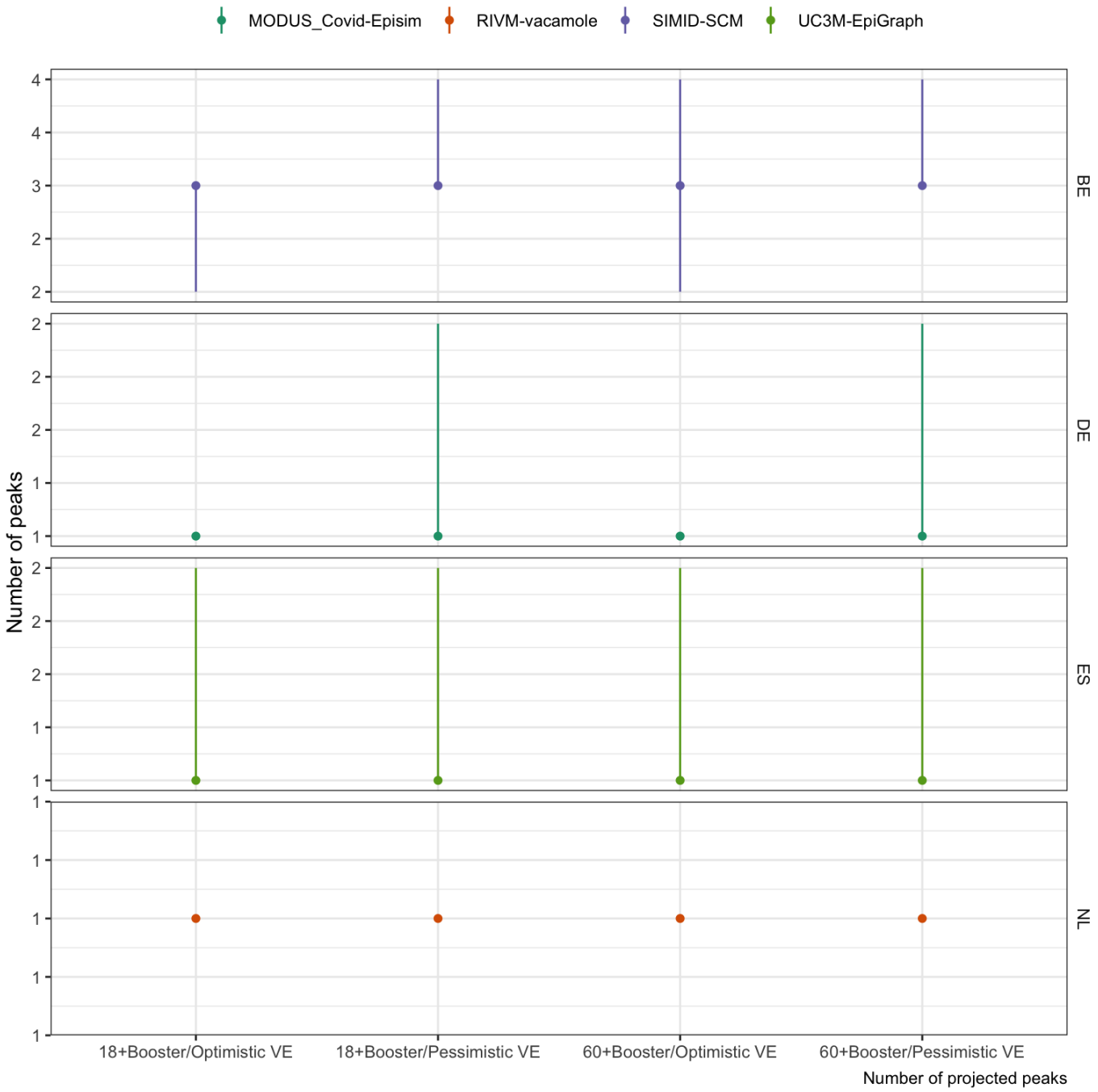
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Infection

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



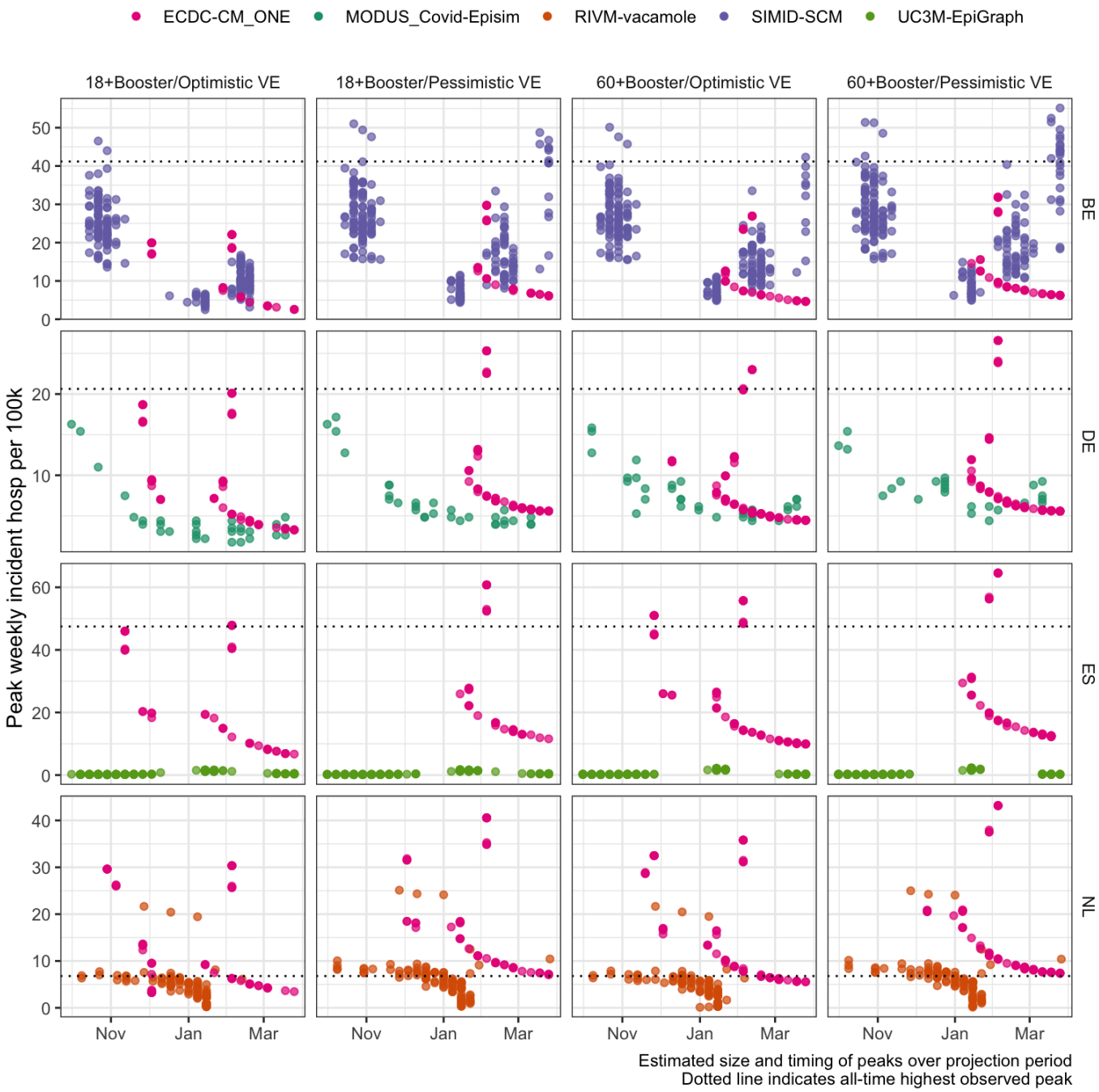
B. Projected number of peaks (median with 5-95% probability)



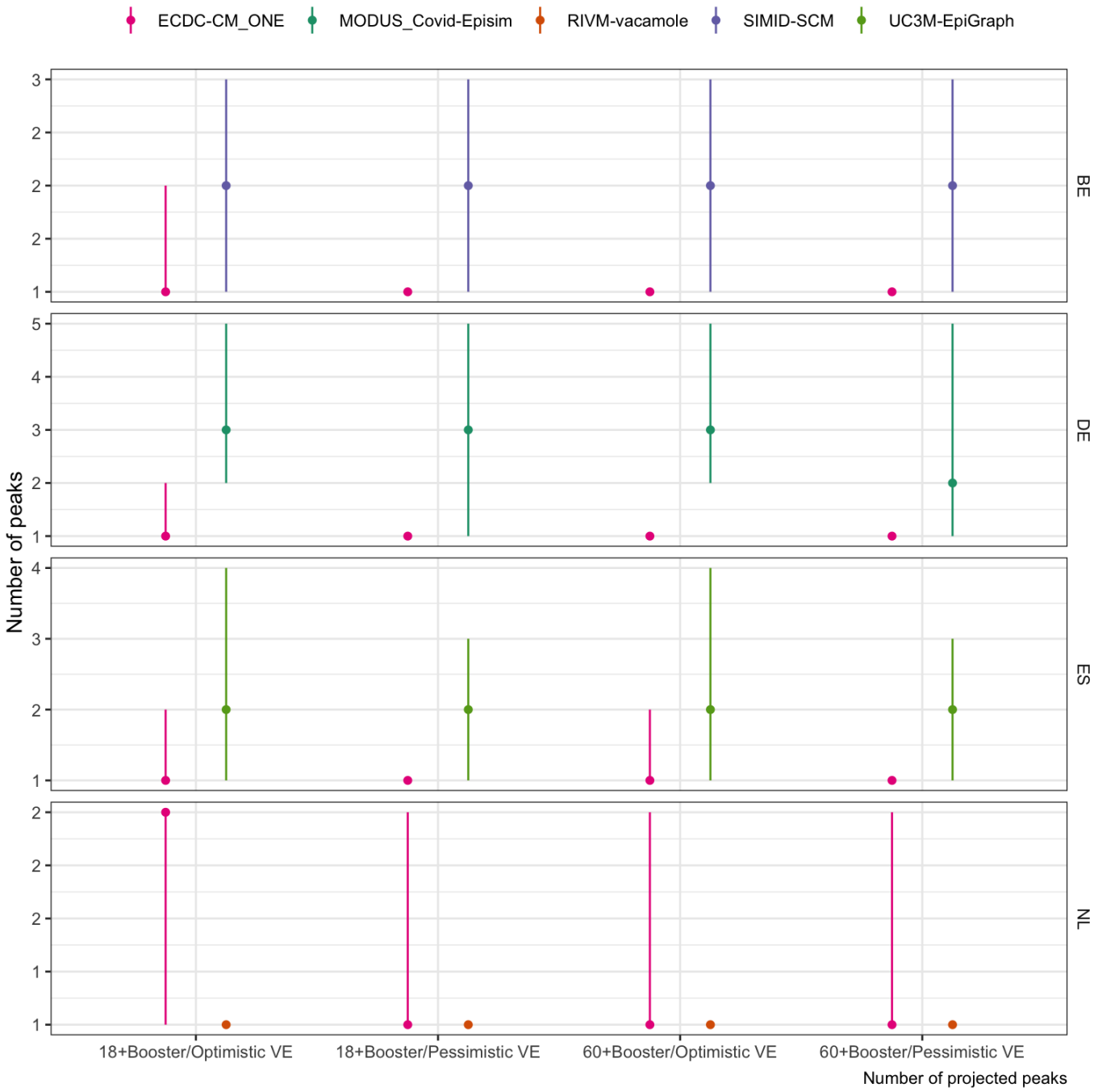
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Hosp

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



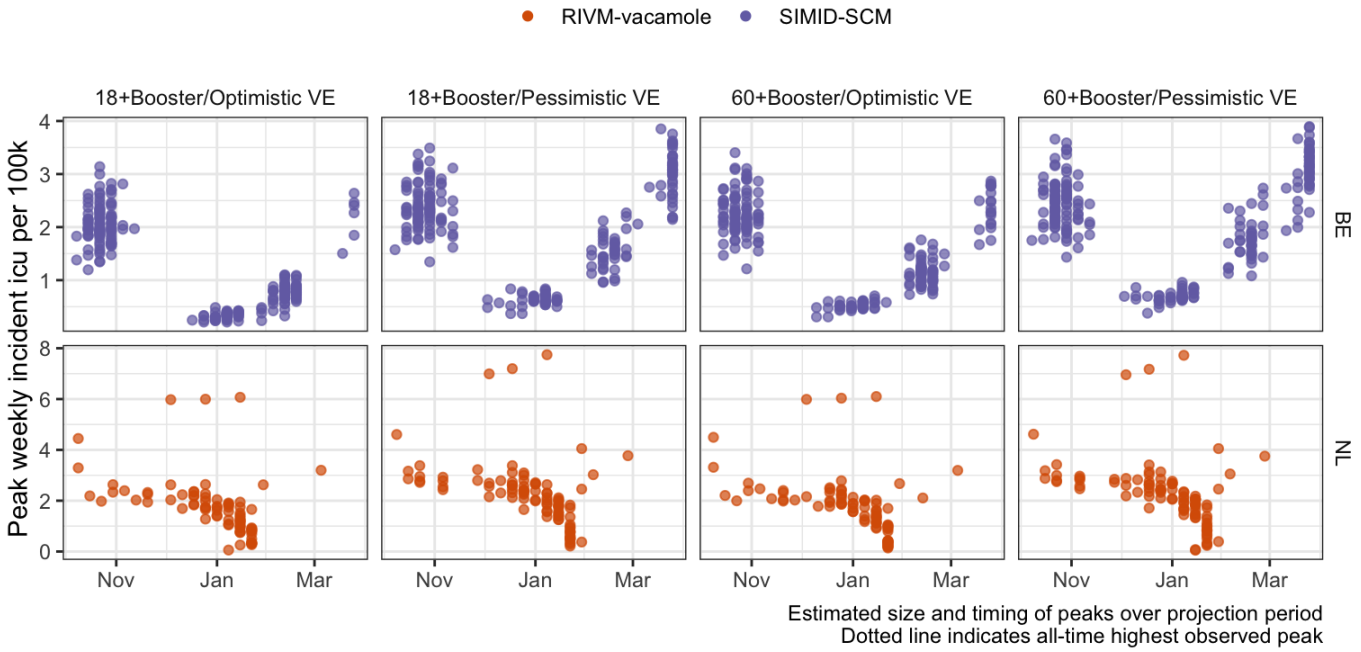
B. Projected number of peaks (median with 5-95% probability)



Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Icu

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



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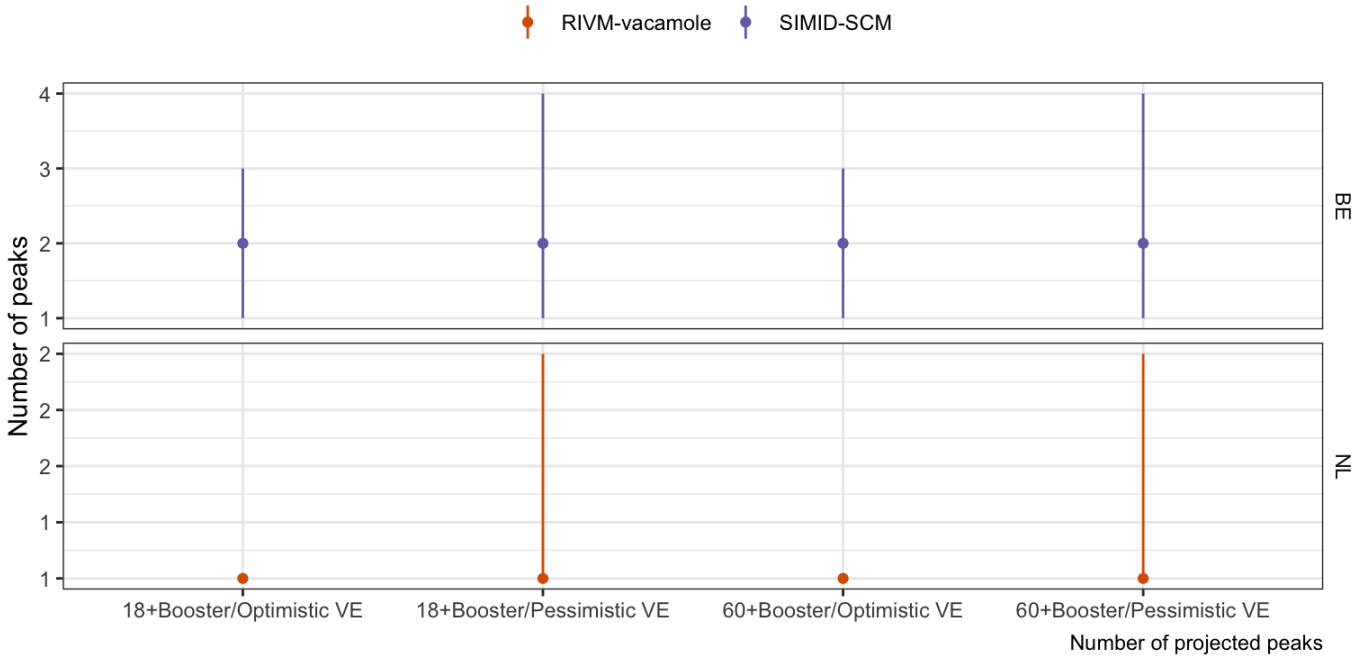


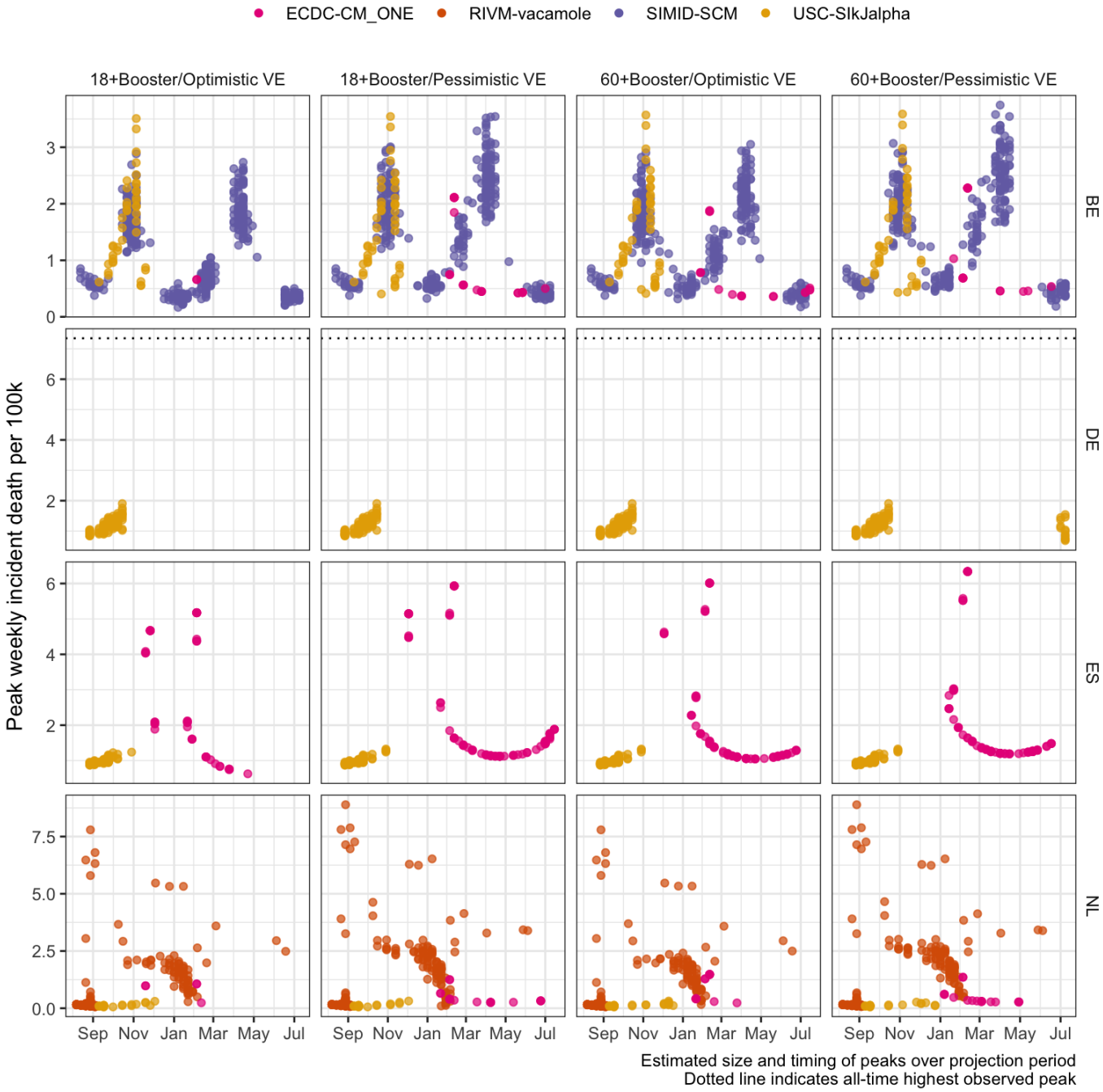
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Entire projection period

Projections over June 2022 through June 2023

Death

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



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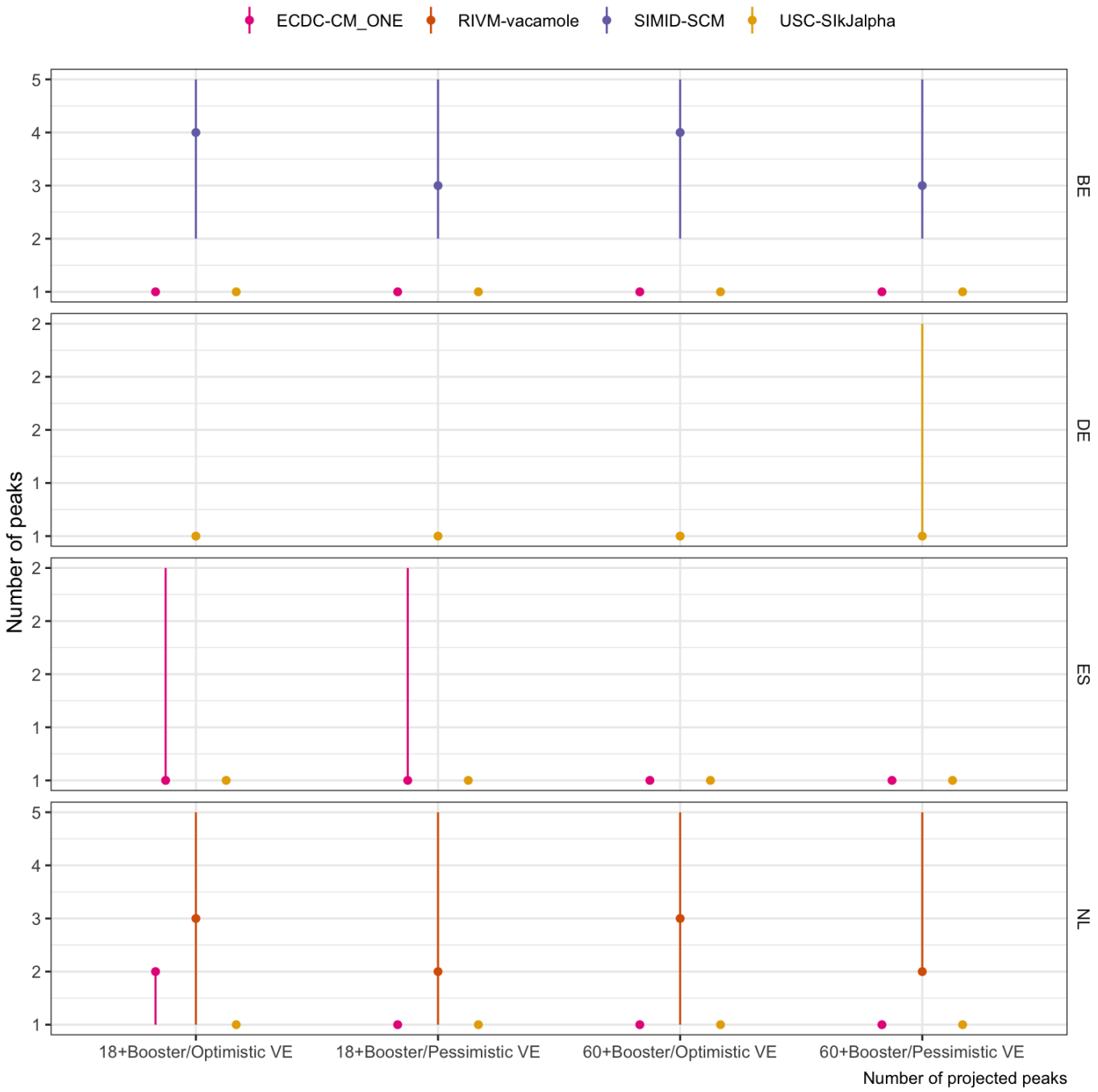
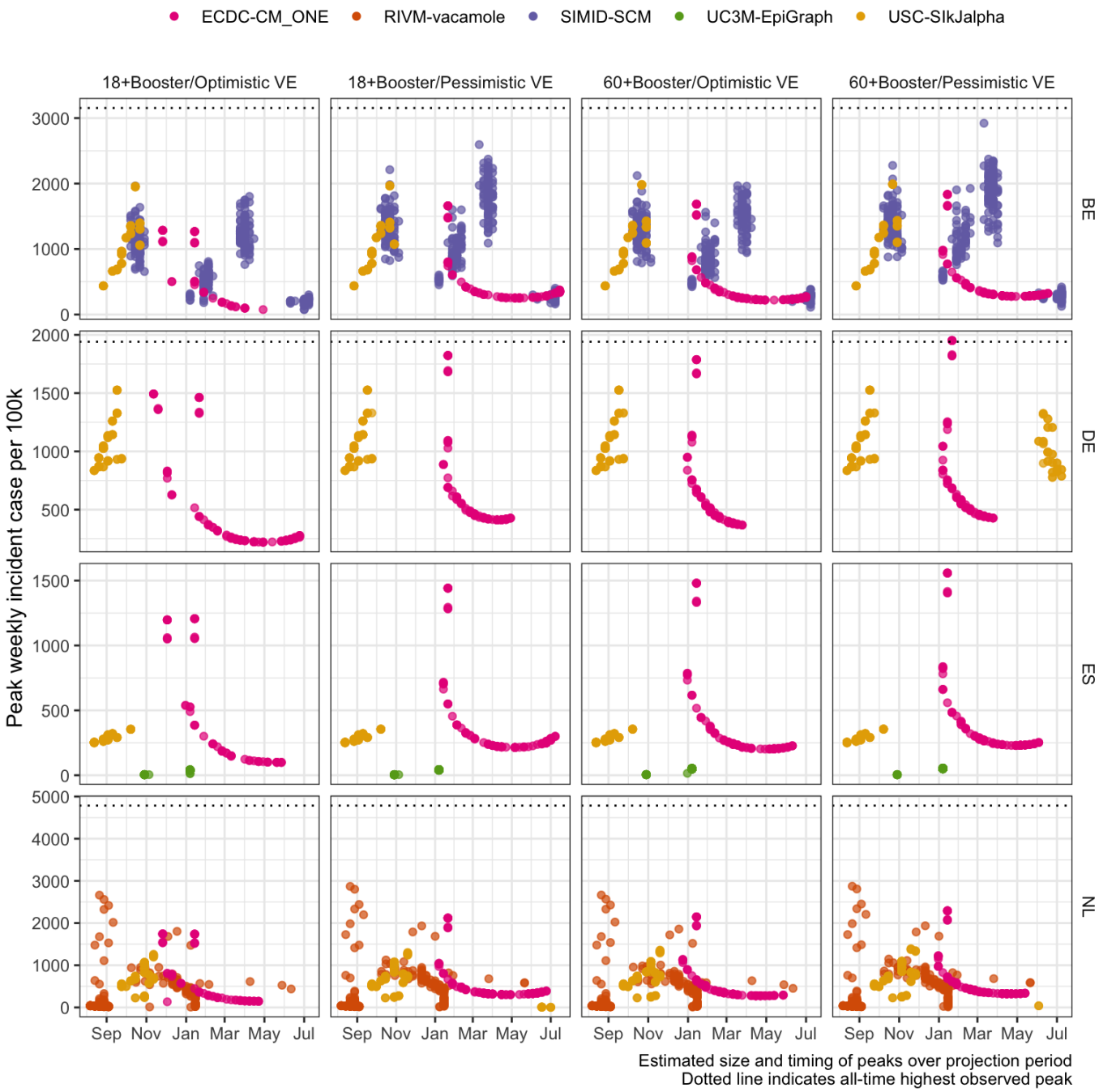


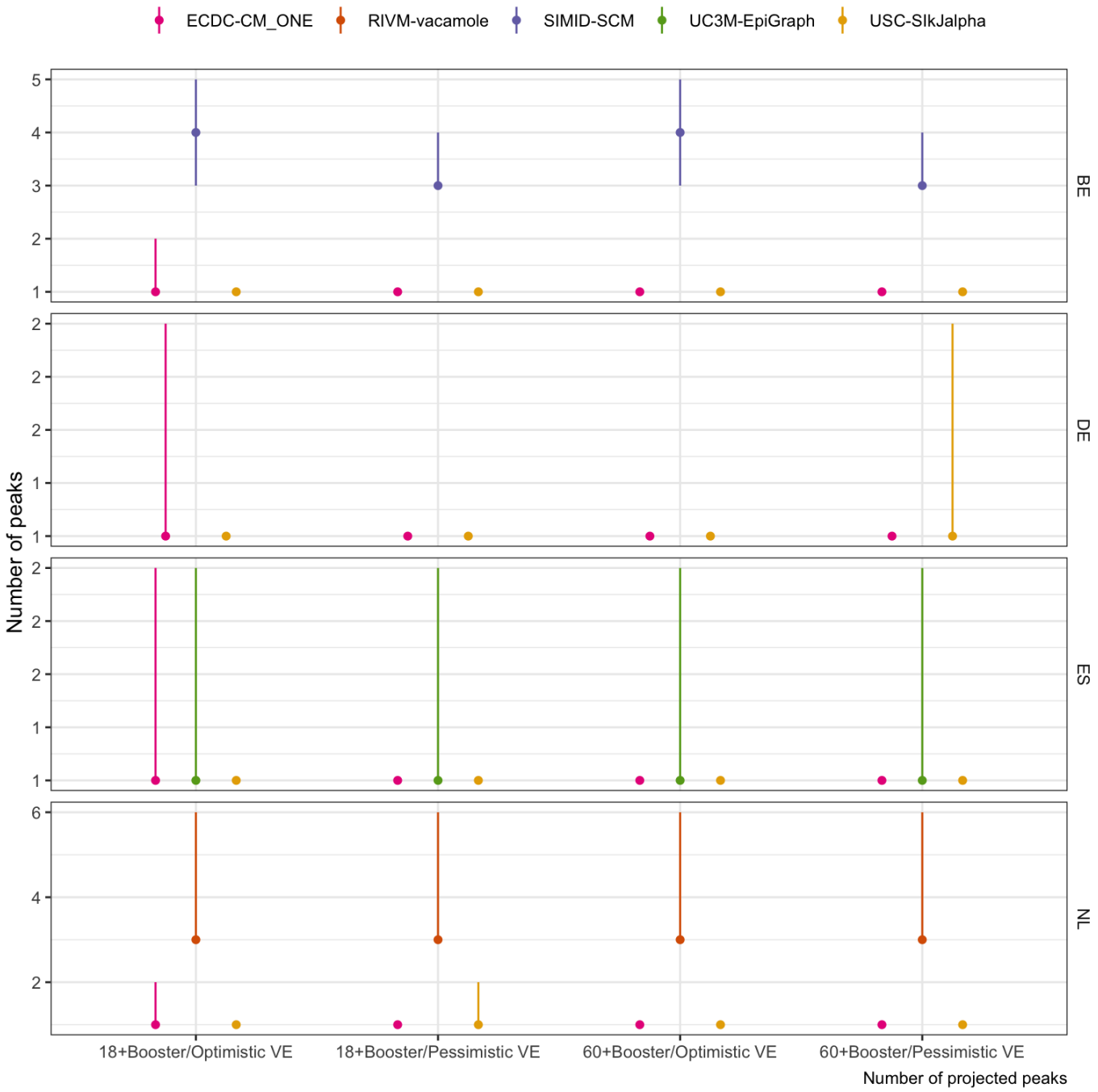
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Case

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



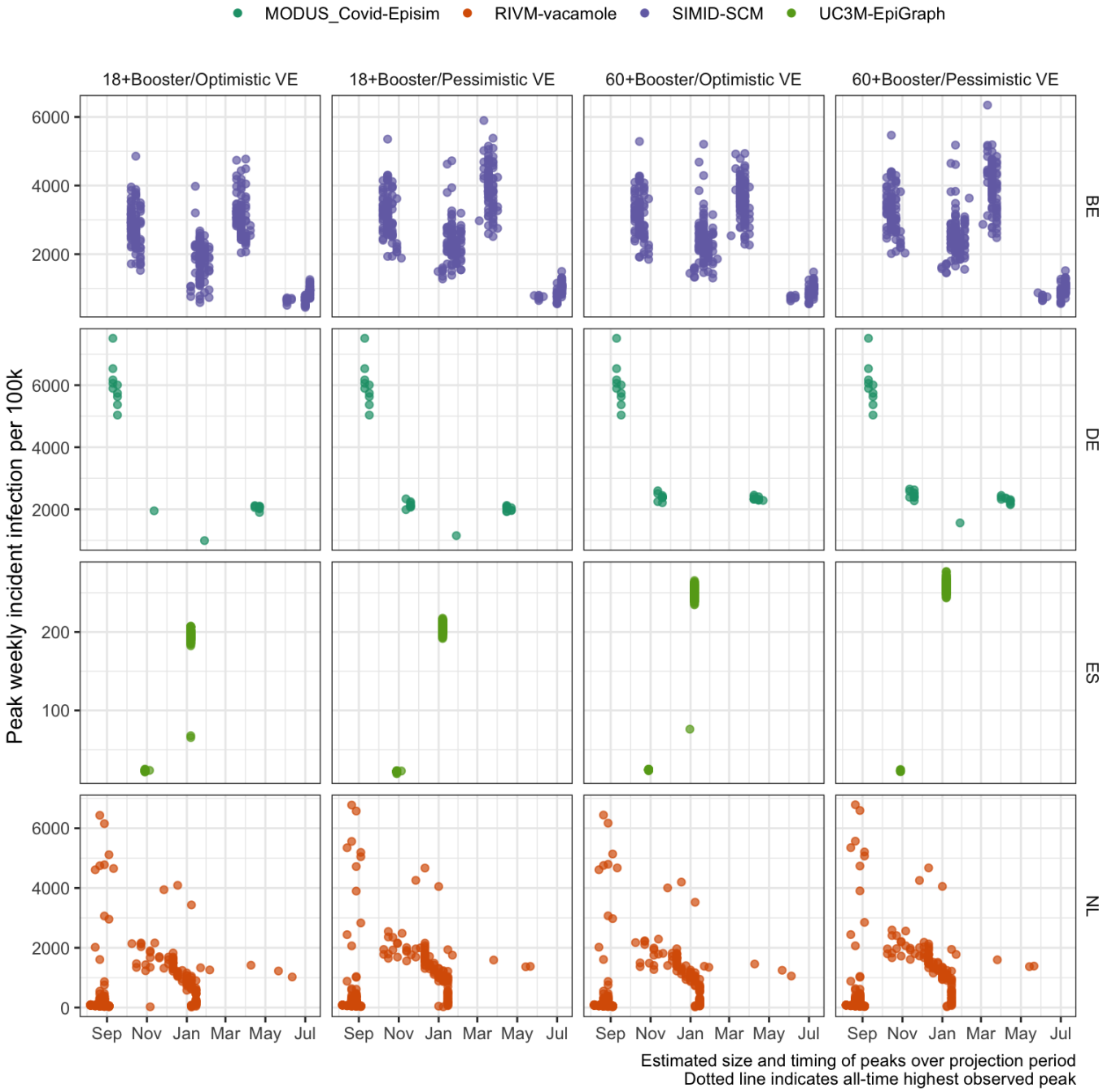
B. Projected number of peaks (median with 5-95% probability)



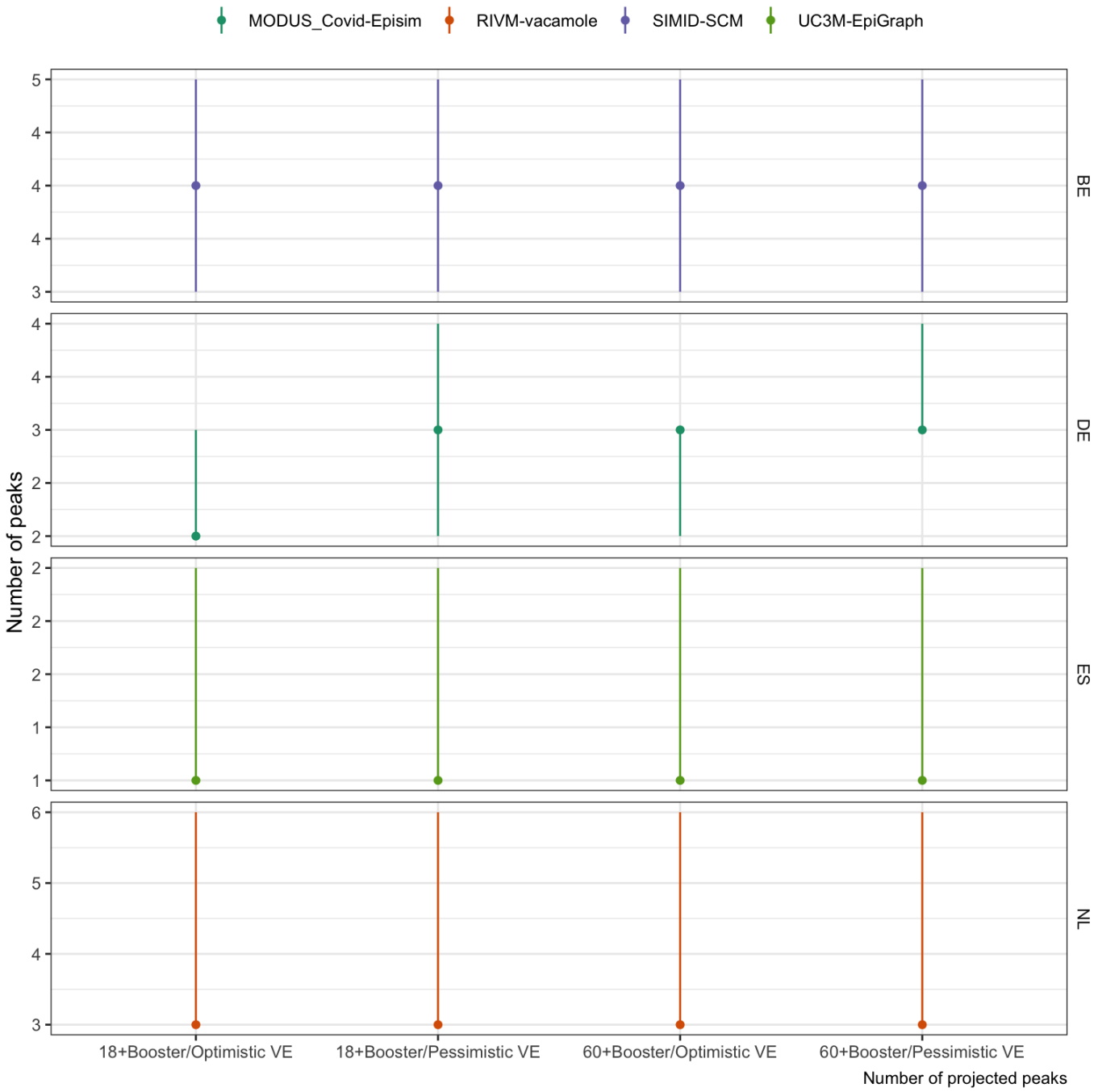
Scenarios: Autumn second booster campaign among population aged ‘18+’ or ‘60+’; Vaccine effectiveness is ‘optimistic’(effectiveness as of a booster vaccine against Delta) or ‘pessimistic’ (as against BA.4/BA.5/BA.2.75)

Infection

A. Size and timing of peaks. Boxplots show summary of the likely value at peak incidence (median and interquartile range); points show timing and size of peaks from independent sample simulations



B. Projected number of peaks (median with 5-95% probability)



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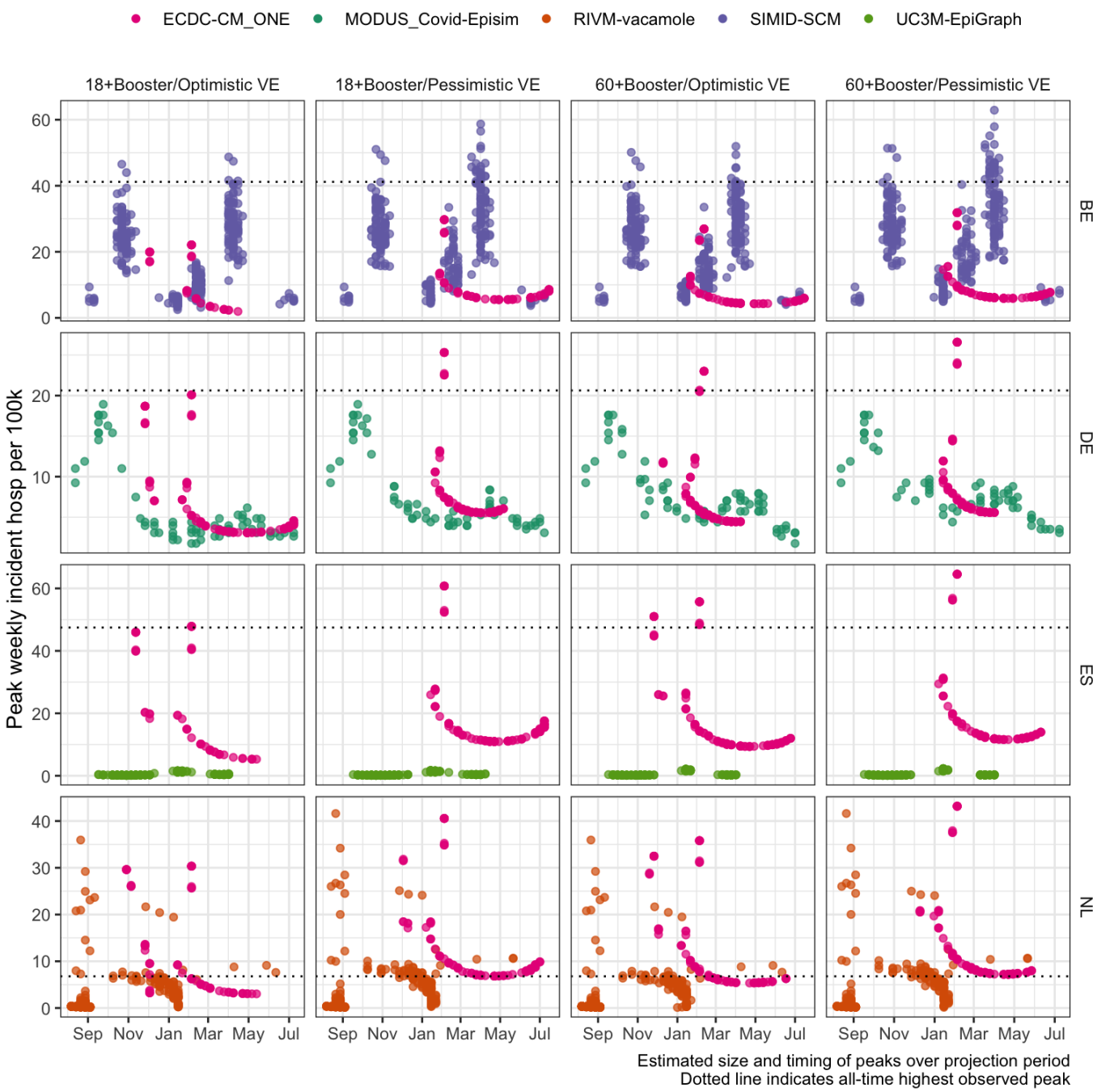

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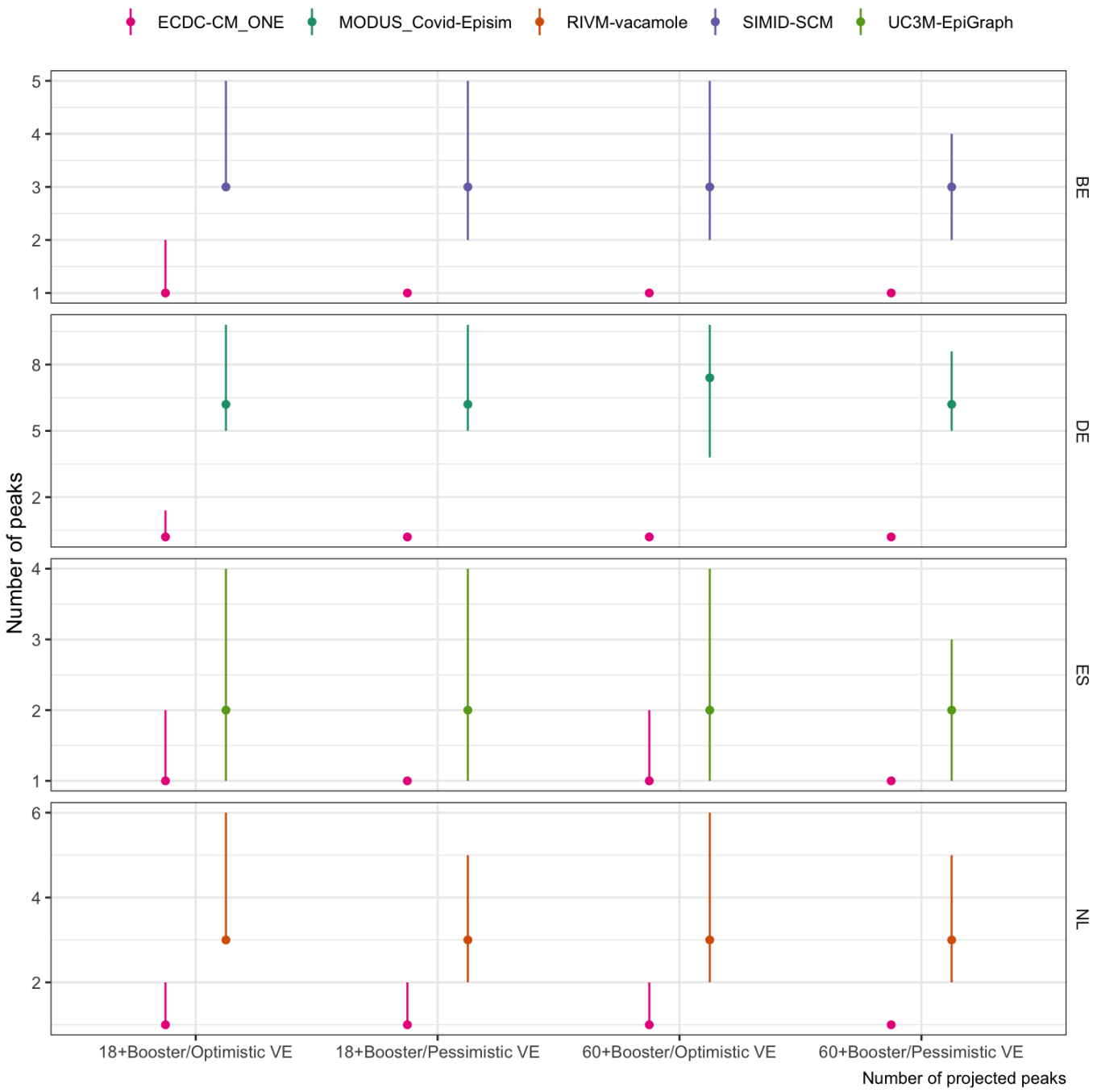

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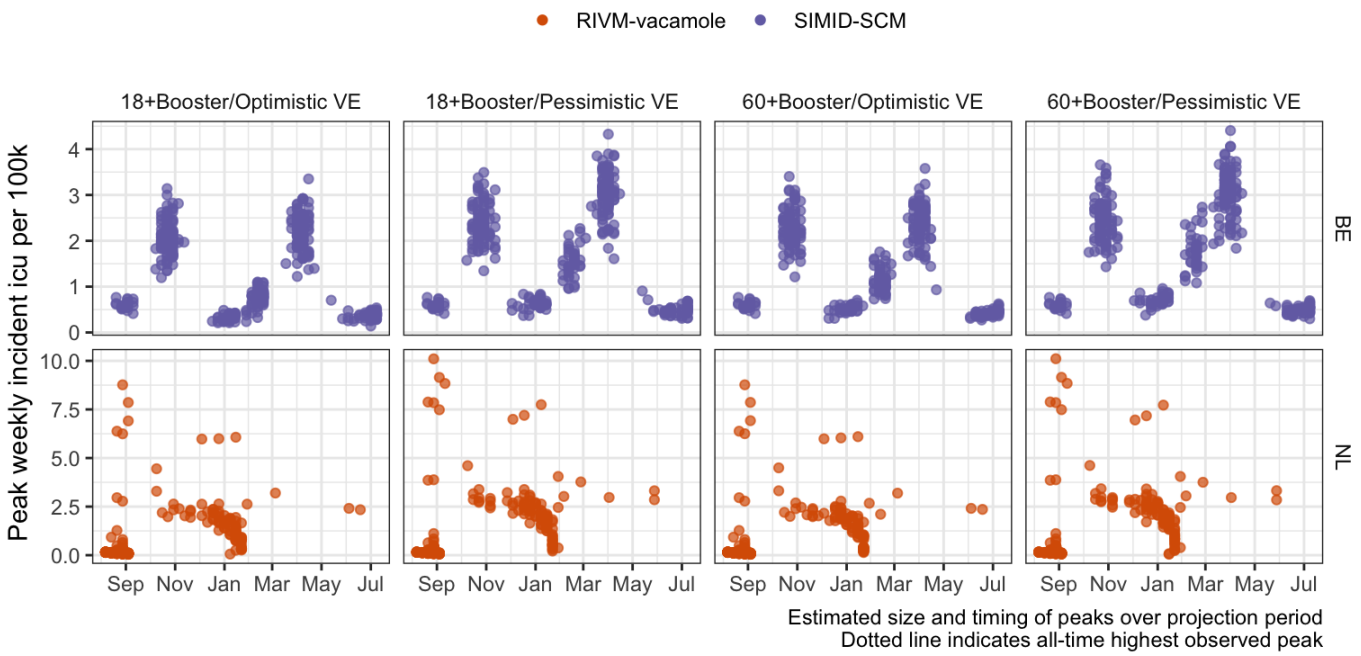

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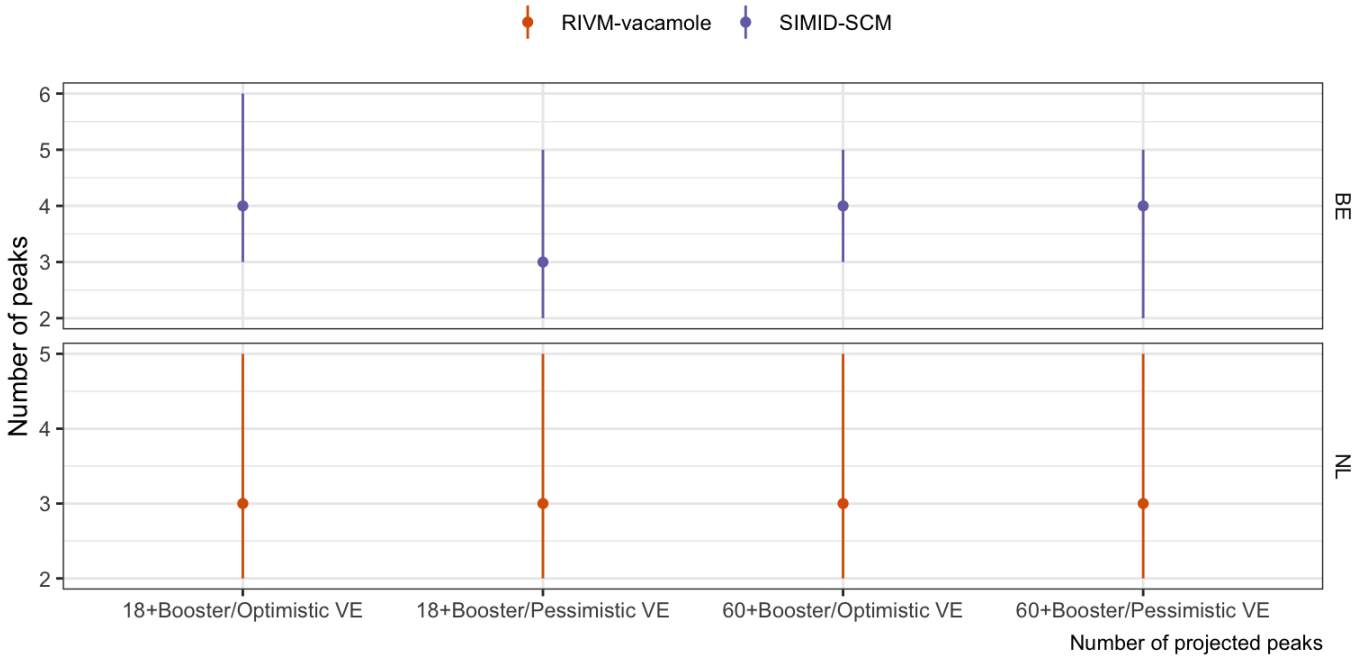

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Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic

Sherratt K, Carnegie AC, Kucharski A, Cori A, Pearson CAB, Jarvis CI, Overton C, Weston D, Hill EM, Knock E, Fearon E, Nightingale E, Hellewell J, Edmunds WJ, Villabona Arenas J, Prem K, Pi L, Baguelin M, Kendall M, Ferguson N, Davies N, Eggo RM, van Elsland S, Russell T, Funk S, Liu Y, Abbott S. Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic. Wellcome Open Res. 2024 Jan 8;9:12. doi: 10.12688/wellcomeopenres.19601.1.

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Surname/Family Name	Sherratt		
Thesis Title	Collaborative outbreak modelling for decision support: evaluating trade-offs from multi-model combination		
Primary Supervisor	Sebastian Funk		

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For multi-authored work, give full details of your role in the research included in the paper and in the preparation of the paper. (Attach a further sheet if necessary)	This work was developed in collaboration with Anna Carnegie (co-first author), Sam Abbott, and Yang Liu. The original concept of surveying modellers involved in the UK COVID-19 response was developed jointly and then led by Sam Abbott and Anna Carnegie. I provided feedback on the survey design, together with Yang Liu, and supported its promotion. We then jointly designed the workshop, with implementation led by Sam Abbott and Anna Carnegie. I reviewed this together with Yang Liu, participated in the workshop, and gathered materials after the workshop completed. I then led on synthesising the experiences and recommendations documented throughout. This was initially intended as a workshop report before developing further into the academic output presented here. In this phase of the work, I first led work gathering and digitising all workshop outputs, for example notes, mind maps, and dot plots from the day, converting these into usable data. I then led work ordering and synthesising these qualitative data into themes and recommendations, with review and feedback from Anna Carnegia, Sam Abbott and Yang Liu. Anna Carnegie then wrote an initial outline for a workshop report. I first reviewed this and then with joint agreement I led work to draft an academic paper reporting our findings. I led on the development of this write up, including the background, formalising the structure of the methods, writing the results and discussion. pI then managed several rounds of review from the core authorship group and those who participated in the workshop.

SECTION E

Student Signature	Katharine Sherratt
Date	14 June 2024

Supervisor Signature	Sebastian Funk
Date	14 June 2024

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RESEARCH ARTICLE

Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic [version 1; peer review: 2 approved]

Katharine Sherratt[1*], Anna C Carnegie[1*], Adam Kucharski[1], Anne Cori[2], Carl A B Pearson[1,3], Christopher I Jarvis[1], Christopher Overton[4-6], Dale Weston[7], Edward M Hill[8,9], Edward Knock[2], Elizabeth Fearon[10], Emily Nightingale[1], Joel Hellewell[11], W John Edmunds[1], Julián Villabona Arenas[1], Kiesha Prem[1,12], Li Pi[13], Marc Baguelin[1,2], Michelle Kendall[8], Neil Ferguson[2], Nicholas Davies[1], Rosalind M Eggo[1], Sabine van Elsland[2], Timothy Russell[1,11], Sebastian Funk[1], Yang Liu[1], Sam Abbott[1]

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Abstract

Background

The COVID-19 pandemic both relied and placed significant burdens on

Open Peer Review

Approval Status ✓ ✓

	1	2
version 1 08 Jan 2024	✓ view	✓ view

the experts involved from research and public health sectors. The sustained high pressure of a pandemic on responders, such as healthcare workers, can lead to lasting psychological impacts including acute stress disorder, post-traumatic stress disorder, burnout, and moral injury, which can impact individual wellbeing and productivity.

Methods

As members of the infectious disease modelling community, we convened a reflective workshop to understand the professional and personal impacts of response work on our community and to propose recommendations for future epidemic responses. The attendees represented a range of career stages, institutions, and disciplines. This piece was collectively produced by those present at the session based on our collective experiences.

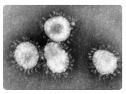
Results

Key issues we identified at the workshop were lack of institutional support, insecure contracts, unequal credit and recognition, and mental health impacts. Our recommendations include rewarding impactful work, fostering academia-public health collaboration, decreasing dependence on key individuals by developing teams, increasing transparency in decision-making, and implementing sustainable work practices.

Conclusions

Despite limitations in representation, this workshop provided valuable insights into the UK COVID-19 modelling experience and guidance for future public health crises. Recognising and addressing the issues highlighted is crucial, in our view, for ensuring the effectiveness of epidemic response work in the future.

Keywords
modelling, COVID-19, pandemic response



This article is included in the Coronavirus (COVID-19) collection.

1. **Srikanth Umakanthan**, The University of the West Indies at Saint Augustine Faculty of Medical Sciences, Saint Augustine, Trinidad and Tobago
2. **Robert Moss**, The University of Melbourne, Melbourne, Australia

Any reports and responses or comments on the article can be found at the end of the article.

Corresponding author: Katharine Sherratt (katharine.sherratt@lshtm.ac.uk)

Author roles: Sherratt K: Data Curation, Formal Analysis, Investigation, Methodology, Writing – Original Draft Preparation, Writing – Review & Editing; **Carnegie AC**: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Writing – Original Draft Preparation, Writing – Review & Editing; **Kucharski A**: Investigation, Writing – Review & Editing; **Cori A**: Investigation, Writing – Review & Editing; **Pearson CAB**: Investigation, Writing – Review & Editing; **Jarvis CI**: Investigation, Writing – Review & Editing; **Overton C**: Investigation, Writing – Review & Editing; **Weston D**: Investigation, Writing – Review & Editing; **Hill EM**: Investigation, Writing – Review & Editing; **Knock E**: Investigation, Writing – Review & Editing; **Fearon E**: Investigation, Writing – Review & Editing; **Nightingale E**: Investigation, Writing – Review & Editing; **Hellewell J**: Investigation, Writing – Review & Editing; **Edmunds WJ**: Investigation, Writing – Review & Editing; **Villabona Arenas J**: Investigation, Writing – Review & Editing; **Prem K**: Investigation, Writing – Review & Editing; **Pi L**: Investigation, Writing – Review & Editing; **Baguelin M**: Investigation, Writing – Review & Editing; **Kendall M**: Investigation, Writing – Review & Editing; **Ferguson N**: Investigation, Writing – Review & Editing; **Davies N**: Investigation, Writing – Review & Editing; **Eggo RM**: Investigation, Writing – Review & Editing; **van Elsland S**: Investigation, Writing – Review & Editing; **Russell T**: Investigation, Writing – Review & Editing; **Funk S**: Investigation, Writing – Review & Editing; **Liu Y**: Investigation, Methodology, Supervision, Writing – Original Draft Preparation, Writing – Review & Editing; **Abbott S**: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Supervision, Writing – Original Draft Preparation, Writing – Review & Editing

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Introduction

The response to the COVID-19 pandemic necessitated a multi-pronged approach, with infectious disease transmission modelling playing a key role in informing strategy and policy decisions[1,2]. Input from UK modellers was mostly channelled through weekly meetings of the Scientific Pandemic Influenza Group on Modelling, Operational subgroup (SPI-M-O) feeding into the Scientific Advisory Group for Emergencies (SAGE)[3]. This advisory group, drawing on expertise from the academic, and public health sectors, developed planning scenarios and short-to-medium term forecasts and projections, routinely estimated key parameters such as the reproduction number (a proxy for transmissibility), conducted routine data analysis, as well as authoring ad-hoc reports on modelling results relevant to the ongoing pandemic in the UK[4,5]. Some of these analyses resulted in academic papers along with those produced by the wider UK modelling community (e.g. 6,7).

The high-pressure environment and daunting responsibilities of those at the frontlines of pandemic response have been shown to exert significant psychological tolls. Notably, healthcare workers (HCWs) involved in infectious disease outbreaks, including COVID-19, have been shown to experience profound and enduring psychological impacts. These include acute stress disorder, post-traumatic stress disorder (PTSD), burnout, as well as moral injury[8–10]. Moral injury refers to a specific form of distress that stems from guilt, anxiety, and loss of trust when actions or roles conflict with one's deeply held moral beliefs. These psychological impacts not only diminish individual wellbeing but can also considerably affect worker productivity, with lasting effects that can linger for years, as exemplified by the 2002/2003 SARS epidemic[11,12].

However, the experiences and challenges faced by non-healthcare responders to the pandemic, such as those involved in modelling and research, have received comparatively less attention[8,10]. Stressors such as high workloads, long hours, tight deadlines, and harassment from the public and press during the COVID-19 response had the potential to cause both visible and invisible impacts. These include mental health impacts, exhaustion, social isolation, compromised career progression in academia, and moral injury.

The experiences of modelling responders have not been systematically discussed but are indirectly reflected in issues of staff retention and burnout across institutions. With the aim of bridging this gap, on March 28th, 2023, we organised a one-day workshop to create a space for collective reflection and strategising improvements for future epidemic responses. This paper seeks to provide an outline of the workshop proceedings, the collective themes that emerged from our discussions, and synthesise our suggested actions into a set of priority recommendations to enhance future epidemic responses.

Methods

Our approach

We employed an iterative, participatory approach to both design and run a reflective workshop with members of the UK modelling community in order to facilitate the summarisation of our collective experiences. In the interest of clearly relaying the proceedings and results emanating from the workshop, we use the term 'participants' to refer to attendees (i.e. ourselves, including the organisers) in the remainder of the methods and the results.

Workshop design

We aimed to ensure the content of the reflective session captured the needs of the individuals at the forefront of the UK modelling response. To inform the content of the workshop, the session organisers (SA and ACC), alongside two additional members of the UK modelling community, solicited informal feedback from individuals involved in the COVID-19 response. This feedback included the personal and professional ramifications of participating in COVID-19 response work, along with the obstacles to effective response work and strategies to address them.

We then engaged an external facilitator to assist in planning the agenda and guiding participants throughout the session. This aimed to ensure unbiased management of discussions and to enable participants to express themselves openly in a safe and supportive environment. To select an appropriate facilitator, we sought input from the broader scientific community and chose an individual with a track record of successfully delivering similar events.

Initial discussion topics were developed by the session organisers in consultation with the external facilitator, drawing on anecdotal evidence from conversations with other modellers who were involved in the COVID-19 response. Further feedback was solicited from two members (KS and YL) of the UK modelling community who were not directly involved in the organisation process. This resulted in a set of discussion topics that addressed the concerns and interests of the community.

Participants

We aimed to include a diverse range of participants involved in the UK COVID-19 modelling response, encompassing researchers and professional services staff. A brief expression of interest form was disseminated by the session organisers to the UK modelling community via organisation mailing lists, personal networks (aiming to also reach those who may have transitioned away from the infectious disease modelling field), and social media channels to ensure representation across different levels of seniority. We invited all those who expressed an interest to attend. We provided a small travel fund for participants on a first come first served basis for those travelling from across the UK.

Workshop structure

Participant arrival and introduction

Upon arrival, the facilitator encouraged participants to engage with flipchart papers displaying "snapshot" questions with attendees providing their responses using stickers. These were:

1. Do you think sufficient action is currently being taken to improve future outbreak responses to the standard you think is acceptable?

2. Who is responsible for ensuring people are supported, and appropriately credited for their work?

3. Summarise your pandemic experience in one word.

See the supplementary information for the multi-choice answers[13].

At the formal start time, the facilitator opened with an overview of the day's agenda, establishing expectations and a code of conduct for participants. The Chatham House Rule ("share the information you receive, but do not reveal the identity of who said it"[14]) were introduced to ensure that individuals would not be identified, while allowing for the synthesis of outputs. A co-organiser (SA) shared their personal pandemic timeline (see the supplementary information[13]), setting the stage for the first exercise.

Iterative discussions of experience

Participants divided themselves into pairs, after being encouraged to work with someone they would not usually interact with. They were asked to discuss their individual pandemic timelines for 15 minutes each, while the partner asked questions based on those we developed when designing the workshop, listened, and asked follow up questions. The following questions were provided:

1. What was your pandemic timeline? What were the highs and lows?

2. What was your experience of pandemic work like?

3. What were some of the things that helped assist you to do effective research during the outbreak response?

4. Do you think team science was appropriately supported over the pandemic?

5. Has your employer or the wider community taken action to help mitigate any of the personal or professional costs/challenges you identified? What more can be done?

6. Do you think there were barriers to doing effective and sustainable COVID-19 outbreak response work? If so, what were they?

7. What has been done and what more can be done to reduce any barriers to effective outbreak response work in the future?

We also provided suggested follow-on questions which are available in the supplementary information[13].

Pairs then formed groups of four to identify common themes from their one-to-one reflections using post-it notes. The group was then brought together and themes were summarised and organised into headline categories. This approach maintained anonymity for the participants, while capturing their reflections in a summarised form. As a combined group we then further discussed these topics, leading to the identification of six major themes.

Synthesising recommendations

The latter portion of the workshop focussed on pinpointing recommendations for action. Participants were presented with primary categories derived from the morning's discussions. Participants were then divided into new groups, with each group assigned a theme. Each group was tasked with developing recommendations and potential implementers. Participants could move between themes and contribute their thoughts. These recommendations, along with actionable steps and suggestions for those responsible for implementing, were then exhibited on a wall for group review. Finally, attendees used dot stickers to identify priorities, allowing a visual representation of the group's consensus.

We (ACC, KS, YL, SA) then reviewed the contents of the group discussions based on the post-it notes, whiteboards, and recommendation board created during the session. Two authors (ACC and KS) independently digitised the output, and four authors (ACC, KS, YL, SA) independently reviewed results. We then came to a consensus on the common themes of participants' experiences, using the major themes identified by participants as a guide, and priority recommendations for stakeholders. We prepared an initial draft and shared this with participants. Finally, we integrated feedback, ensuring that the insights derived from the workshop were preserved.

Results

Outputs from the workshop

Summary of attendees

The event was attended by 27 individuals, including 25 research staff and two professional services staff. Staff attended from five higher education institutions (London School of Hygiene & Tropical Medicine, Imperial College London, University of Warwick, Liverpool University, the University of Oxford), and the UK Health Security Agency (UKHSA). The majority of attendees were based in London. Participants represented various career stages, including early, mid-career, and senior academics and professionals. Among the attendees were multiple members of SPI-M-O and SAGE.

Snapshot reflections

In response to the initial snapshot question, "Do you think sufficient action is currently being taken to improve response work to a standard you think is acceptable?", the participants expressed an overwhelmingly negative view (17/18). In the second snapshot question, "Who should ensure that individuals are adequately supported and credited for their response work" (this question allowed multiple answers), participants suggested this responsibility was shared among stakeholders. Affirmative responses were more common on the panel listing smaller groups than on the panel listing larger organisations, indicating that respondents considered themselves (9/56), line managers (16/56), and research groups (12/56) more responsible for this task compared to larger organisations such as institutions (6/56), academic funders (5/56), and the "system more generally" (8/56). The second panel displaying the larger organisations was situated to the right of the first panel, which may have resulted in decreased visibility. Photos of the panels are available in the supplementary information[13].

Each attendee was asked to summarise their pandemic experience in one word using post-it notes (see supplementary information). Positive responses were: *"exciting", "valuable"*, and "*engrossing"*. Neutral responses were: *"intense", "unprecedented", "ambiguous", "focussed", "hectic", "repetitive",* and "*surreal"*. Negative responses were: *"hard", "austere", "stressful", "lonely", "harrowing", "frustration"*, and "*exhausting"*. Multiple participants added stickers (indicating support) to both "*stressful"* and "*exhausting"*.

Themes from paired and group discussions

In the paired and small group discussions, several topics emerged into which participants' perspectives were grouped. These included (reproduced verbatim from those listed on the day): *"societal impact", "mental health", "life outside", "emotions", "personal", "team spirit", "institutional structures", "work process", "work feeling/support", "work pressure", "(negativity about)" "positives"*, *"career direction"*, *"rewards"*, *"access and privilege"*, *"bad stuff"*, "*COVID-19 modeller-specific experiences"*, and *"general experiences"*. In the following session, participants then refined these themes to leave the following: *"institutional factors", "mental health", "life/personal", "work process", "career direction", and "social impact"*. See the Supplementary Information for the full list of participants' points[13].

After the workshop, we reviewed individual post-it notes and further refined these themes to leave: *funding and institutional support*; *recognition, rewards, and access*; *team and work dynamics*; *non-academic contributions*; and *personal impacts*. The themes emerging from the group discussions are synthesised, stratified by these themes below. We indicate direct quotes from individual authors using quotation marks and italics.

Funding and institutional support

Lack of institutional support: Insufficient institutional support for those involved in the COVID-19 modelling response was a common issue among participants. Many felt that they were not protected by their institutions during the response or in its aftermath; for example, when receiving aggression from some sectors of the media and general public. Additionally, groups highlighted the lack of processes to respond in an emergency while protecting psychological safety. This included the need for training for managers and teams, and wellbeing procedures and human resources policies.

Contract insecurity and inflexible funding rules: The precarity of short-term contracts due to heavy reliance on external grant funding was highlighted, along with implicit pressures to underestimate personnel time in funding applications to meet budget thresholds, adhere to eligibility criteria and achieve cost recovery targets. The importance of providing sufficient and sustainable personnel funding was stressed, with this including academic and professional services roles such as project managers, administrators, communications professionals, technicians, and software engineers.

Recognition, rewards and access

Inadequacy of reward metrics: Credit attribution mechanisms were a recurring concern. Participants emphasised that there are currently insufficient frameworks to reward the nature of response work itself. Hurdles in receiving recognition for work included contributing to confidential reports where involvement was unable to receive external acknowledgement. In particular, it was noted that outputs such as software tools and policy reports do not fit within the traditional academic credit structure. Similarly, participants recognised that promotions, paper authorship, and grant Principal Investigator (PI) positions were not designed to promote collaborative team working. This was identified as a problem for both the general wellbeing of researchers and the quality of the science produced. The unequal, and individual-focussed, credit structures that persisted throughout the pandemic were also discussed, with senior or well-connected researchers being identified as receiving the majority of recognition. Participants noted *"rewards not attributed equally",* and that *"institutions got awards, not individuals (not all key players)"*. This uneven reward system was seen as contributing to a competitive culture, which was identified as a problematic aspect of response work and academia more widely.

Access to decision-making spaces: Individuals had different access to policy-making spaces which did not always reflect where or how their work was used. As a result, some individuals who lacked access reported feeling left behind when it came to updates relevant to their work. There was a general consensus that there should be more transparency regarding these forums for those involved in producing the work presented.

Team and work dynamics

Insufficient capacity: Participants highlighted issues with *"not being able to say no"* and the *"pressure [that] came in waves. 'Not again...'"* These issues contributed to poor working practices within teams, including insufficient capacity and reliance on one or two individuals to perform key tasks. In turn, this made it more challenging for these individuals to maintain a work-life balance. The highly pressurised and reactive nature of response work meant that there was not always space for teams to reflect on the effectiveness of routine aspects of the response, including whether academic groups were the best placed to perform this work. In addition, despite a need for additional capacity, working in highly reactive 'response mode' made it difficult to properly onboard new starters and hand over responsibility of tasks and projects where resources were available to do this. There was reference to other professions more adapted to response work, such as the military and emergency services, suggesting there may be learning to be gained from these sectors.

Competing demands and barriers to progression: Individuals faced challenges in balancing competing demands of and distinguishing between 'response' work and research. Some individuals sacrificed otherwise beneficial opportunities, such

as teaching. Although response work created some opportunities for career progression, these were distributed unequally relative to contribution. Access to these opportunities depended on several factors, including career stage, and relative privilege (which is the differential access to resources, opportunities, and advantages some groups have compared to others). We note that privilege is often invisible to those who have it, and recognizing one's own relative privilege is a key part of understanding and addressing social inequalities.

Collaborative working: Participants cited the positive experience of collaborative working and camaraderie within teams – academic, professional, and hybrid. However, as the pandemic progressed, there was a sense that the egalitarian working structures, which some felt were put in place at the start of the pandemic, faded: *"Shift from egalitarian structure to pre-existing hierarchies"*. Meanwhile, with close working relationships and the intensely personal impact of the COVID-19 response, professional disagreements sometimes took on an unusually emotional tone.

Non-academic contributions

Role of professional services staff: Participants highlighted the significance of integrating professional services roles into research teams, mentioning that these staff played crucial roles in response-related tasks. Participants pointed out that professional services roles, especially administrative positions like project managers, are frequently deemed ineligible costs in grant applications. Similar to academic staff, many individuals in these roles work on short-term contracts. Consequently, these positions were often under-resourced and experienced high turnover.

Public health agency workers: Participants emphasised the importance of strengthening the collaboration between academics and public health agencies, with the aim of fostering knowledge and skills exchange both during, prior to, and after responses. The importance of a bidirectional exchange was highlighted, with academics having the opportunity to learn about the practical challenges faced by public health agencies, while public health staff would benefit from access to the latest research findings. Participants called for more opportunities to facilitate these exchanges, such as joint workshops, shared working spaces, and dedicated training sessions.

Personal impacts

Public recognition: The COVID-19 pandemic brought the infectious disease modelling field public recognition and scrutiny. Participants acknowledged the personal responsibility that came with this visibility, while valuing the significance of their work. While friends and family gained deeper understanding of their work, some highlighted the challenge of work and life becoming intertwined. Participants referred to the *"surreal level of public and media interest (good or bad)"* and the idea that *"work and the world were one and the same. Neither was an escape from the other."*

Mental ill health and burnout: Participants across organisations and seniority levels reported prioritising work over their health and wellbeing, leading to extreme levels of overwork, burnout, and associated mental health effects, including depression and anxiety. The experience was common among attendees at all levels and career types, with recognition that this can creep up over time and not enough has been done to mitigate against it. Some participants expressed guilt and a sense of 'survivor bias' from being able to remain within academia, having witnessed friends and colleagues leave the field. One post-it note summed up the feeling of *"trading off career versus health and everything"*. People were reluctant to reach out to managers or colleagues for support. With close working relationships, the personal challenges faced by colleagues inevitably impacted the wider team. No strategies were identified by participants as having been in place to address these issues during the pandemic response or having been implemented more recently.

Commitments outside of work: Several participants highlighted the challenges they faced in balancing high-intensity roles with personal obligations during the pandemic response. They shared experiences of coping with loss and caregiving responsibilities, which were particularly difficult for those whose partners were also involved in the response. Certain groups faced heightened challenges; for example, women often bore a disproportionate burden of caregiving tasks, early career researchers tended to have less stable domestic situations, and non-UK nationals experienced difficulties such as visa concerns or being separated from their home countries.

Recommendations

The strategies collectively proposed at the workshop spanned societal impact, mental health, career direction, work processes, personal life, and institutional policy. Over ninety suggestions were made for possible actions by research teams, employers, and funding entities. The full list of recommendations is available in the supplementary information[13].

Priority recommendations

Participants distilled a set of priority recommendations to enhance the support and sustainability of epidemic response work. These directives tackle crucial facets affecting the well-being and efficiency of those engaged in pandemic response. Example actions for implementing these recommendations are italicised below each recommendation (see the full list of suggested actions from the workshop in the supplementary information[13]).

1. Acknowledge, and reward, impactful response work at institution, funder, and research community levels.

> *Funding bodies refine impact measures to credit all forms of output produced during, and required for, response work; institutions standardise incorporating response-driven work into criteria for doctoral theses and promotion.*

2. Encourage routine interaction between academia and public health agencies, including consistently reviewing the role of each during epidemic responses.

> *Government bodies and research institutes create sustainable dual positions recruiting from both sectors.*

3. Ensure response teams are well-staffed, well-resourced, stable, and provided psychological support.

> *Research teams establish sustainable team-building and training programmes with long term support from funders during non-response periods to ensure individuals feel equipped and supported to engage in response work.*

4. Increase the transparency of the evidence pathway from scientists to decision-makers making it easier for those across the scientific community to contribute as well as making the evidence base for decisions clearer to the general public.

> *Government bodies standardise rapid open access to the minutes of scientific advisory meetings and encourage input from a wider range of sources.*

5. Implement best practices for a sustainable work environment.

> *Employers promote leave-taking and respecting work hours, and clarify communication about processes and rewards across career stages, integrate support roles into research teams, and standardise the onboarding of new team members.*

Discussion

This reflective workshop brought together 27 individuals from the UK infectious disease modelling community to engage in a dialogue around the personal and professional impacts of their COVID-19 response work. Participants represented various career stages, institutions, and disciplines, enabling a diverse exploration of experiences and perspectives. We identified areas of improvement in the current approach to modelling during epidemic responses, with these including greater support for responders, line managers, and research groups. Our experiences ranged from positive to negative, with stress and exhaustion being particularly prominent. Through in-depth discussions, key themes emerged, including institutional support, mental health, career direction, and social impact. Challenges such as lack of institutional backing, insecure contracts, inadequate reward systems, and personal impacts such as mental health issues were identified. The roles of professional services staff and public health agency workers were underscored. To address these issues, we identified a variety of strategies and priority recommendations, including acknowledgement and reward of impactful response work including for professional service staff, enhanced academia-public health collaboration, minimising dependence on key individuals, increased transparency in decision-making processes, and the adoption of sustainable work practices. These findings offer valuable insights for the ongoing pandemic response and future public health emergencies.

Our approach benefitted from being embedded in the experience of the UK modelling community. The session was community-driven, adopted an informal approach, and included participants from various career stages and perspectives on the response. Prior input from the community ensured the event's relevance for attendees, while employing an external facilitator helped create a safe and structured environment for discussion. We then collectively agreed on key themes and recommendations.

However, a key limitation was participant representation. This was exacerbated by it being a one-day workshop, meaning we could only represent the views of those who were available and able to attend in person on that day. Attendees were primarily from London and Southeast England, possibly due to limited support for travel costs. Additionally, despite efforts to involve individuals who had left the infectious disease modelling field, few were able to attend. Our collective experiences are therefore likely to be missing some of the most challenging experiences and perspectives of responders, and our conclusions may be more moderate than if a wider range of participants had attended. Despite this bias, we feel this provides valuable insights into the UK COVID-19 modelling experience but should be viewed as a summary of a small group's experiences and opinions, with potential differences across jurisdictions and groups. We encourage responders in other locations to conduct similar exercises and to synthesise these findings for a broader understanding.

Ongoing efforts have begun to evaluate UK modelling work during COVID-19 both in terms of modelling results (e.g. forecasts or scenario projections[6,7]), and the systems and processes enabling the response[2,15,16]. However, so far little has been done to report the experiences of responders themselves as we have done in this work. In the context of more general crisis response, more work has been done to understand the key challenges, particularly on healthcare workers (HCW). For example, hospital disaster preparedness plans may incorporate mental and behavioural health interventions (such as resource signalling, peer support, and referrals for at-risk individuals), which have proved to be effective in reducing mental health morbidities[17,18]. Lessons from previous epidemics also emphasise the importance of effective staff support and training in preparing for future outbreaks. Perceived adequacy of training and support had a protective effect on adverse outcomes in HCW responders to the SARS epidemic[19]. These approaches, which have established use in high-stress occupations, could be adapted and applied to support modellers during epidemic response situations.

The workshop identified priority recommendations aimed at enhancing support and sustainability in pandemic response work. Our discussions underscored the importance of recognising and rewarding significant contributions to public health crises at all levels. We advocated for fostering closer ties between academia and public health agencies, building well-resourced, resilient teams, and ensuring their psychological well-being. Discussions also emphasised the need for increased transparency in the evidence-to-policy pipeline, improved work-life balance, and clear institutional communication. Further suggestions included standardising onboarding procedures and integrating support roles into teams.

Whilst we identified several themes and recommendations during our workshop, we did not explicitly separate issues specific to the pandemic response from broader academic

challenges. Some recommendations, for example, recognising non-traditional contributions or normalising annual leave, pertain to broader issues. It is important to discern whether these concerns are long-standing systemic issues that have been simply exacerbated by the pandemic, or if they have been particularly highlighted due to the unique stressors of the pandemic.

Conclusions

As a community we want to acknowledge that the pandemic has engendered widespread hardship, stress, and ill health throughout various populations. It is crucial to reflect on and address these profound impacts as we continue to tackle the crisis and prepare for future epidemic responses.

The consequences of the COVID-19 pandemic have been profound on those at the forefront of the UK modelling response. In this work, we have summarised our experiences and whilst we recognise that many of the issues we have identified impact those in our field more generally we believe that they are particularly problematic for epidemic response work. It is evident that changes are required across multiple domains, including individual work, team dynamics, and institutional structures, to enable future effective epidemic modelling responses.

Achieving these changes necessitates investment from governments, funding bodies and institutions. The solutions needed to foster a healthy and sustainable environment for future epidemic response work will not be attainable without such investment. Additionally, there is a need for teams aiming to respond to epidemics to redefine their working methods, developing response preparedness plans that emphasise wellbeing, training, and career development. It is clear that even these localised initiatives demand time investment from those leading them, and as a result, require support.

As it stands, future epidemic responses are likely to raise similar challenges to those we have identified here, including reliance on a select number of individuals, excessive workloads and the exacerbation of systemic inequalities. It is critical we act outside of response contexts; for example, by implementing the recommendations we have outlined, to mitigate these issues and respond more effectively in future.

Consent

This work is the sole product of collaboration among the named authors. All inputs used in this work were those of the authors, with no data collection from any additional participant or data source. Therefore, all participants in this work are named authors of this manuscript and have approved both the manuscript and supplement for publication.

Data availability

Underlying data

Open Science Framework: Underlying data for 'Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic', https://www.doi.org/10.17605/OSF.IO/4JNCB[13].

This project contains the following underlying data:

- Data supplement.pdf
  - Survey questions
  - Snapshot questions
  - SA pandemic timeline
  - Session questions
  - Group discussion themes
  - Themes recommendations
  - Priority recommendations

Data are available under the terms of the Creative Commons Zero "No rights reserved" data waiver (CC0 1.0 Public domain dedication).

Acknowledgements

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We also wish to thank everyone who so graciously shared their experiences as part of this project. An earlier version of this article can be found on bioRxiv (**doi:** https://doi.org/10.1101/2023.06.12.544667).

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✔ **Robert Moss**
The University of Melbourne, Melbourne, Victoria, Australia

This article reports findings arising from a one-day workshop held in London in early 2023, in which participants were asked to reflect on the personal and professional impacts of their involvement in informing strategy and policy during the COVID-19 pandemic, for the purposes of identifying key issues (and potential recommendations to mitigate these issues) and to provide guidance for future public health crises.
As the authors acknowledge in the introduction, there is a well-established body of literature concerning the impact of disasters on exposed persons, and ways to mitigate these impacts, but this literature primarily focuses on direct responders and survivors. Less attention has been given to persons who contribute indirectly to disaster response, such as the infectious disease modelling community represented in this article. I'm only aware of one paper that reflects on (professional) impacts of the COVID-19 response on modellers: "The COVID-19 response illustrates that traditional academic reward structures and metrics do not reflect crucial contributions to modern science", which was written in 2020 by three authors of this article.
The common themes and personal impacts identified here may not necessarily be surprising (many reflect broader, long-standing issues in academia) but they deserve genuine attention and reflection. Many of these issues resonate strongly with my own experiences, and those of my colleagues. I wholeheartedly agree with the authors' conclusion that it is "critical we act outside of response contexts".
The workshop was carefully planned and conducted. The supplementary materials attest to the level of engagement from the participants, and to how thoroughly the organisers have captured and reported the findings. It is unfortunate, but entirely understandable, that only a few individuals who had left the field were able to attend the event.
I only have a few minor comments regarding the article text.

1. In "Snapshot reflections", the third sentence begins "Affirmative responses were more common ...", referring to the question "Who should ensure that individuals are adequately supported and credited for their response work".
My first thought was that this is not a question with a "yes" or "no" answer. It took me a moment to recall that this was a multiple-choice question, and so participants were selecting responsible

organisational units from a predefined list.
The first half of this sentence could potentially be removed, so that it begins "Respondents considered themselves (9/56), line managers (16/56), and research groups (12/56) more responsible ...".

2. In the "Team and work dynamics" results section, the authors report that we might learn from other professions that are better adapted to emergency response work. I think this is a great suggestion, and worth highlighting as an example action for the third priority recommendation ("Ensure response teams are well-staffed, well-resourced, stable, and provided psychological support").

3. The "Non-academic contributions" results section highlights the importance of strengthening the collaboration between academics and public health agencies. This reminds me of Pan-InfORM (Pandemic Influenza Outbreak Research Modelling), a Canadian initiative that was established in 2009 for this very purpose. The authors could cite a recent review of Pan-InfORM activities (published in 2021, doi:10.3934/publichealth.2021020) as an international example.

4. Regarding the final sentence of the discussion:
"It is important to discern whether these concerns are long-standing systemic issues that have been simply exacerbated by the pandemic, or if they have been particularly highlighted due to the unique stressors of the pandemic",
I'm not sure I fully appreciate this distinction. If a concern has been "particularly highlighted" by the pandemic, I still interpret it as meaning that the concern was relevant prior to the pandemic, and I wonder if the intended meaning is that the concern was not identified or appreciated prior to the pandemic?
This sentence also led me to expect an outline of different approaches that might be used to address long-standing systemic issues versus those were specific to the pandemic response. Otherwise I don't understand why this distinction is being made.

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Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and is the work technically sound?
Yes

Are sufficient details of methods and analysis provided to allow replication by others?
Yes

If applicable, is the statistical analysis and its interpretation appropriate?
Not applicable

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Infectious disease modelling

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Reviewer Report 03 May 2024

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Srikanth Umakanthan
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The authors have touched on a critical topic that was very much neglected during the COVID-19 pandemic.
This study very well highlights the role and coordination between the employer, employee and the community using the available resources in the best possible manner.
However the following comments need to be addressed:
1. The keywords should not be similar to those in the title.
2. The workshop's type, duration, and target audience should be indicated.
3. The prior aims and objectives of the workshop and the percentage of it being achieved should be included.
4. The professional details of the external facilitator who aided in developing the discussion topics need to be mentioned.
5. The link between the results and the study methods needs to draw a clear conclusion. In the methods the authors look to "identify the Key issues such as lack of institutional support, insecure contracts, unequal credit and recognition, and mental health impacts" but the results look into the "to propose recommendations for future epidemic responses".
6. The list of specialties and the topics of the keynote speakers should be included.

Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and is the work technically sound?
Yes

Are sufficient details of methods and analysis provided to allow replication by others?
Yes

If applicable, is the statistical analysis and its interpretation appropriate?
Yes

Are all the source data underlying the results available to ensure full reproducibility?
Yes

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Pathology, infectious diseases

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Supplementary Information: Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic

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Supplementary Information

Survey questions
Snapshot questions
SA pandemic timeline
Session questions
Group discussion themes
Themes
Recommendations

Priority recommendations

Survey questions

How best would you describe your role in responding to COVID-19?

- Academic researcher
- Civil servant
- Professional services staff
- Other

How best would you describe the role you are doing now?

- Academic researcher
- Civil servant
- Professional services staff
- Other

Approximately how many years experience do you have in your role?

- 0-4 years experience
- 5-9 years experience
- 10+ years experience

If you feel comfortable doing so, please provide an overview of the work you did during the COVID-19 response.
Examples of this might include:

- facilitating meetings
- data management
- producing routine estimates for surveillance and submitting them to advisory bodies.
- developing tools and methods and support their use by other responders
- providing scenario estimates in response to policymakers requests
- writing reports
- academic paper writing
- attending meetings
- supporting researchers
- speaking to the media, ... etc.

On a scale from 0 to 10, how would you rate your professional experience in responding to COVID-19 in the UK?
0 - extremely negative
5 - neutral
10 - extremely positive

What professional costs, if any, did you experience from being involved in the response?

Think about both short- and long-term costs

What professional benefits, if any, did you experience from being involved in the response? Think about both short- and long-term benefits, for example:

1. number of first author papers
2. promotions/career progression
3. successful grants

On a scale from 0 to 10, how would you rate your personal experience in responding to COVID-19 in the UK?
0 - extremely negative
5 - neutral
10 - extremely positive

What personal costs, if any, did you experience from being involved in the response? Think about both short- and long-term costs

What personal benefits, if any, did you experience from being involved in the response? Think about both short- and long-term benefits

Has your employer or the wider community taken action to help mitigate any of the personal or professional costs you identified?

- Yes
- No
- Somewhat

Can you identify areas where action has been taken and areas where it has not?

Do you think there were barriers to doing effective and sustainable COVID-19 outbreak response work?

- Yes
- No
- Maybe

What were some of the barriers to doing effective and sustainable outbreak response work? For example, thinking about:

1. funders
2. employer organisations
3. supervisors
4. peers
5. computing resources
6. human resources

What can be done to reduce barriers and better support those involved in outbreak response work in the future?

What were some of the things that helped assist you to do effective research during the outbreak response?
For example, thinking about:

1. funders
2. employer organisations
3. supervisors
4. peers
5. computing resources
6. human resources

Do you think sufficient action is currently being taken to improve future outbreak responses to the standard you think is acceptable?

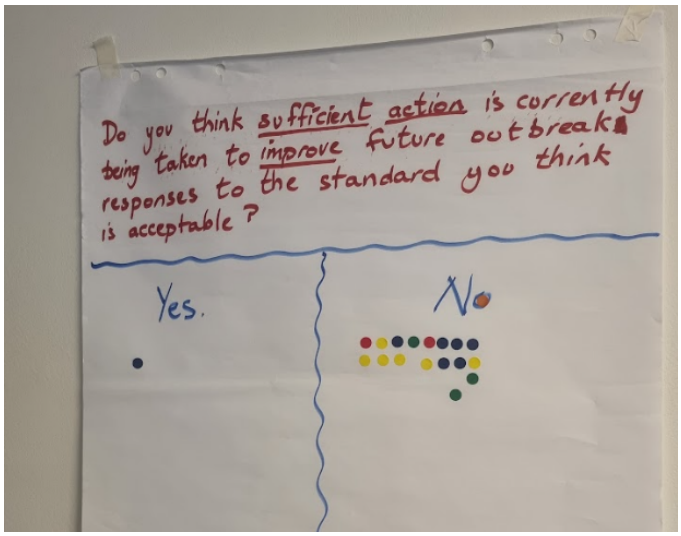
- Yes
- No
- Unsure

Is there anything else you'd like to add?

Snapshot questions

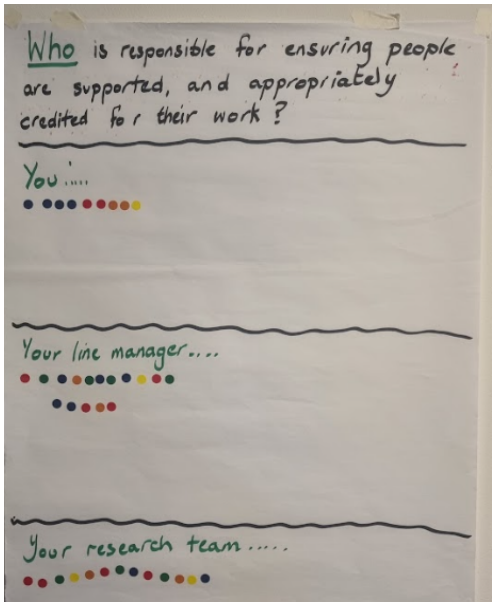
Whiteboards with the following questions were available for the first hour of the session for participants to add to with either post it notes or sticker dots. Numbers in brackets [#] represent the number of sticker dots placed by participants.

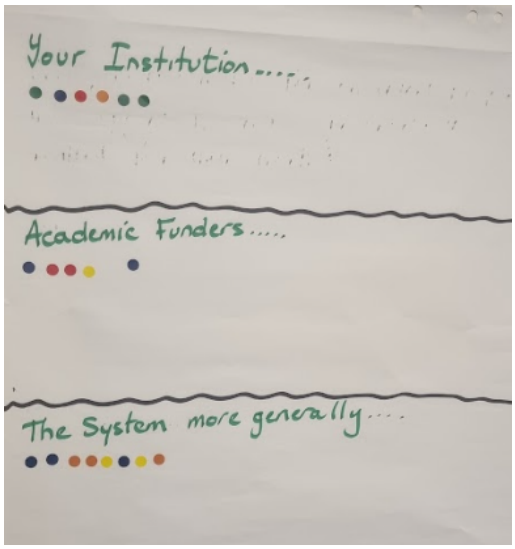
1. Do you think sufficient action is currently being taken to improve future outbreak responses to the standard you think is acceptable?



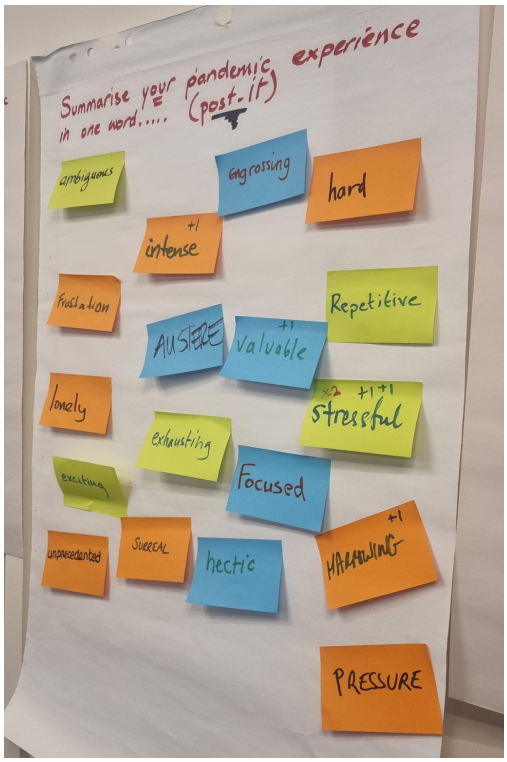
   - Yes [1]
   - No [18]

2. Who is responsible for ensuring people are supported, and appropriately credited for their work?
   - You [9]
   - Your line manager [16]
   - Your research team [13]
   - Your institution [6]
   - Academic funders [5]
   - The system more generally [8]
   - Other? [0]





3. Summarise your pandemic experience in one word.



- Ambiguous
- Engrossing
- Hard
- Intense (+1)
- Frustration
- Austere
- Valuable (+1)
- Repetitive
- Lonely
- Exhausting
- Focused
- Stressful (x2) (+1) (+1)
- Exciting
- Unprecedented
- Surreal
- Hectic
- Harrowing (+1)

Session questions

1. What was your pandemic timeline? What were the highs and lows?
2. What was your experience of pandemic work like?
   Below are some suggested directions to take this:
   - What did you expect to experience when you started working on the pandemic response, and how did this differ from what you actually experienced?
   - Did you get any benefits?
   - Were there any negatives?
   - How did your professional and personal experiences differ?
   - Do the professional gains outweigh the personal costs or vice versa?
   - Do you think privilege played a role in your experience?
3. What were some of the things that helped assist you to do effective research during the outbreak response?
   Some examples could be:
   - Management practices
   - Contract duration
   - Professional services staff
   - Funders
   - Employer organisations
   - Supervisors
   - Peers
   - Computing resources
   - HR policies
4. Do you think team science was appropriately supported over the pandemic? Some potential directions:
   - What elements of team science were most important to your work?
   - Has anything changed since prior to the pandemic in terms of support for team science?
   - What role did professional services staff, research administrators and managers, press and media teams - have on your ability to do effective response work?
5. Has your employer or the wider community taken action to help mitigate any of the personal or professional costs/challenges you identified? What more can be done?
6. Do you think there were barriers to doing effective and sustainable COVID-19 outbreak response work? If so, what were they?
7. What has been done and what more can be done to reduce any barriers to effective outbreak response work in the future

Group discussion themes

Participants formed six groups and were asked to summarise their pandemic experience using post it notes, organised into self-identified themes on a whiteboard (one per group). All text is reproduced below, source data photos available on request.

Group 1

- Life outside
    - Partner in field or not
    - Personal circumstances
    - Comradeship - shared experience
    - Importance of camaraderie in fields like military response - conflicts w/ academic reward structure
- Rewards
    - Rewards not attributed equally
    - Institutions got awards, not individuals (not all key players)
    - Don’t have good mechanisms for rewarding teams (as opposed to individuals)
- Work pressure
    - Work life balance
    - Overwork
    - Lack of strong management
    - Lack of leadership
    - Line managing
    - Mismatch org - task
- Bad stuff
    - Authorship issues
    - Professional conflicts
    - Professional character assassination
- Access & privileges
    - Different layers of privilege
    - Institutional size (bigger uns = easier)
    - Country of origin
    - Well connected researcher = better access to data

Group 2

- Career direction
    - Responsibility at expense of research breadth
    - Need to get back to long term planning
    - Some career opportunities (but added stress)
    - Missing out on non-publication experiences
    - Big picture vs detail
- Work process
    - Loose in a tornado
    - Lots of competing demands

        - Difficult to stop rapid work
        - Outsourcing prioritisation
        - Hybrid work
- (Negativity about) positives
    - Shame associated with “doing well”
    - Positive ? networking + connections
    - Career advancement
    - Acknowledge previous bad work habits
- Work “feeling” / support
    - Not recognising MH impacts when you’re in it
    - Lack of understanding from non C19 colleagues
    - Missing “in person” cues in interaction
    - Need for more support - burnout
    - Lack of support / less guidance
    - Missing informal “check-ins” working remotely
    - Pride in overwork
- Personal
    - Better boundaries / priorities
    - Paused personal commitments
    - “Is it still March 2020?”

Group 3

- A
    - Team spirit
        - New vs existing
        - Lack of support
        - Individual shared
        - Competitive
        - Collaboration
    - Institutional
        - Leadership
        - Protection
        - Recognition
        - Junior members
- B
    - Mental health
        - Lack of desire
        - Reintegration
        - Uncertainty
        - Work-life balance
        - Children, partners / home, family
        - Travel
        - Guilt
        - Feeling stuck in time-space-role

        - Variable speed of time
        - Dealing with losses
    - Societal impact
        - Responsibility
        - Value of work
        - Representation of work
        - Sustainability
        - Guilt

Group 4

- COVID modeller specific
    - Everyone busy = harder to get support
    - Talking about & acknowledge the relative difficulties of each others experiences
    - Surreal level of public + media interest (good or bad)
    - Differences in levels of success & contribution resulting in inequality in various forms, i.e. lots of success based partly on luck
    - Hard to onboard and shift responsibility (sorry new people!)
    - Difficult to balance routine modelling and interesting innovative work
    - Being lumbered with “shit” jobs
    - Worked too much because too much work to do
- More general experiences
    - Worked too much because nothing else to do
    - Work + the world were one and the same - neither was an escape from the other
    - Pressure came in waves “not again…”
    - And coincided with Xmas
    - Feels like time didn’t exist i.e. no memories for a lot of it
    - Really hating the policies for young people

Group 5

- Work
    - Lack of operational data
    - What is “good enough”?
    - Not being able to say no
    - What was the point of some of the work
    - Other work pressures e.g. teaching
    - Waves of work
    - Working overnight
    - Clashes over motivation/priorities
- Emotions
    - Powerlessness
    - Sadness
    - Frustration + anger
    - Uncertainty
    - Desensitised

    - Pride
- Life
    - Skewed sense of time
    - Life events
    - Child care (x2)
    - False memories
- What to do next?

Group 6

- Structures / work
    - Shift from egalitarian structure to pre existing hierarchies
    - Collaborative work excitement
    - Team work challenges
    - Gambling / return on investment
    - All a blur
    - Survivor bias
    - Responsibility of managers to look after - who is responsible?
    - Trust
- Structures / life
    - Burn-out creeps up on you - how to avoid
    - Life events (juggling)
    - Self-care
    - Coping mechanisms
    - Personal pandemic preparedness
    - Caring for others
    - Impact of team members' personal challenges on everyone
- Emotions / work
    - Reconciling different perspectives on hierarchy & team working
    - Who gets credit
    - Space for communicating
    - Guilt
    - Comparing self with others
    - Frustration
    - Feeling useful
    - Feeling of lack of legitimacy
    - Pride
    - Self perception vs others' perception
    -
- Emotions / life
    - Trading off career vs health / everything "just 1 more"
    - Personal responsibility (feelings of)
    - Maternity leave
    - "Fresh blood" (return from leave)
    - Fear of missing out

Themes

Participants were tasked with grouping top-level themes from the group discussions into broad categories as a spider-map. This created the following:

- Work life
    - Life outside
    - Life
    - Personal
    - More general experiences
- Work - covid modeller specific
    - Work
        - Work process
        - Work pressure
    - Social impact
    - Team spirit
        - Work “feeling support”
        - Mental health
        - Emotions
- Rewards
    - Structures
    - Institutional
    - Access & priveleges
- Career direction
    - Bad stuff
    - (Negativity about) positives

Recommendations

We reproduce text from the poster boards. We include the implementor suggested by contributors at the end of each statement where this was done.

Social impact

- Educate public / policy maker of modelling knowledge for better communication
- Develop primers/training and build on existing links [*research team*]
- Ensure impactful COVID work understood at funder/institution level (esp. If less obvious/visible)
- Support for dissemination of work/case studies [*team/institution/funder*]
- Funder buy-in: creating opportunities to support consolidation of developed methods and tools
- See credit [and fund] all outputs - incl. Software and communication, media, public engagement etc. [*funders/managers/institution*]
- Formalise connections between teams/disciplines/functions - make sure we retain “what worked”
- Provide support (e.g. MH) to reduce burnout + increase retention of institutional memory
- Do[...?]ting connections [*line managers*]

Mental health

- Culture shift
    - Reframe excellence
- Raising awareness (literacy) - also of team role in crisis response
- Preparedness; peace time team building / leadership training [*wellbeing manager, centre manager*]
- Institutional safeguarding
- “We think we do this well, but we don’t” needs embedding over a sustained time
- Departmental leadership buy in; lead by example (processes - not emailing at night etc) [*funders, institution, people*]
- Resilience training (peers support) [*institution*]
- Psychological first aid *[institution*]
- Individual + management training
- Adequate resourcing of central services + research teams to mitigate against burnout = funding. [*funders, institution*]

Career direction

- Now
    - Possibly greater acceptance of portfolio PHDs
    - Leeway in examiners for PhDs during covid/emergencies
    - More recognition of non-traditional outputs
- Future
    - Improve ease of movement in/out of public health agencies (eg UKHSA), eg dual positions
    - Shifting routine data analysis to public health bodies ASAP

        - Stop/end routine activities as soon as they're not useful
        - Find a way to credit confidential work
- How/who
    - Institutions & senior academics to write letters of support for PHD students & staff with 'non-traditional' outputs [*managers, institution*]
    - Joint appointments at public health agencies [*funders*]
    - Credit all outputs: academic, tool development, communications, public engagements, confidential reporting
    - Clear expectations for promotion [*managers, funders, institutions*]
    - Distinguish "academic" vs "emergency" response
    - Disaster roster + exercise

Work process

- Capacity for cycling between response & research
- Reduce structural reliance on 1-2 people [in] a team performing specific task
- [O]n call system?
- [A]nnual leave
- Prioritise capacity
- Project manager incorporated in team [*department, funder*]
- More transparency from government committees so groups without people on them didn't get as left behind as it seemed (to me at least)
- Transparent preparedness plan *[UKHSA, govt]*
    - Who sits on gov committees
    - What their roles/responsibilities
    - White paper [*dedicated working group incl funders*]
- Clarify roles for pandemic response: software engineering, policy-related roles
- Broader reward system, e.g. [*funder*]
    - Code/software
    - reports/briefing notes
    - "Middle" authors
    - Data collection
- Automation / routine vs. one-off - value/impact?
- Regular re-assessment of cost-effectiveness of tasks
- Ask yourself if you <u>really</u> need to do this so often/ so quickly
- Weekly discussions: priorities [*team leads*]

Life/personal

- Mechanism for feeding back & instituting good working practices
- "GWP" [good working practices] reps & a committee that reports to univ./institution executive board
    - Identify people to take on the role of rep
    - Regular surveys to gauge feelings & elicit suggestions
- Time off in lieu (mandatory)
- Paid overtime (capped)

- Establish working group to make recommendations at a national level [*institution/funders*]
- Realistic funding <> deliverables
- Hobby
- Normalise taking annual leave
- Respect (enforce) working hours [*individuals & institutions*]
- Role models
- Don't try very hard (Relax)
    - Active role
- Guidelines for line managers = cultural change
- Delay emails on weekends and 7pm-7am (allow control)
    - Auto-send emails to remind people to take break from work
- PDR [performance and development review] for work/life balance (part of PDR but checked by welfare manager)
    - Palm trees on slack
    - Normalise people taking holidays > holiday snaps
- "Covid impact statement" but for more general issues
- Have paid wellbeing officer(s)/manager to check in on personal issues & balance, give professional advice, etc…

Institutional

- Credit/rewards
    - Clear comms on processes
    - Meetings non-science (coping, emotion, credits etc)
        - > mandatory incorporation in protocols; normalising [*institution*]
- Starter pack
    - Processes
    - Support & social
    - Technical
    - Mentorship/buddies
    - Available training
    - survey input regular
    - share experience between institutions [*department groups*]
- Resilience
    - Firefighter mentality
    - Back-up / replaceable
        - learn from other professions
        - Structure ahead of time & practice [*dedicated working group*]
- Responsibility
    - Hierarchy
    - Map roles
    - Back-up/replaceable
    - Function of/process for support roles (software, admin/comms)
        - Starter pack

        - available guidelines
        - structured updating [*institution, department*]
- Clear guidance
    - Communication
    - Expectations
    - Reference resource
        - Project manager
        - career path for support roles with growth
        - comms / software / admin project management [*funder, institution*]
- Manager training
    - Niche to normal
    - Timely, frequent
    - across seniority
        - Mandatory training [*institution*]
- Lessons learned
    - Exit interview feed into updating processes
    - Left since start of pandemic
    - All staff, not only academic
        - Survey / exit interview [*manager/department*]

Additional

- Pay rise
- Now now v just now prioritisation

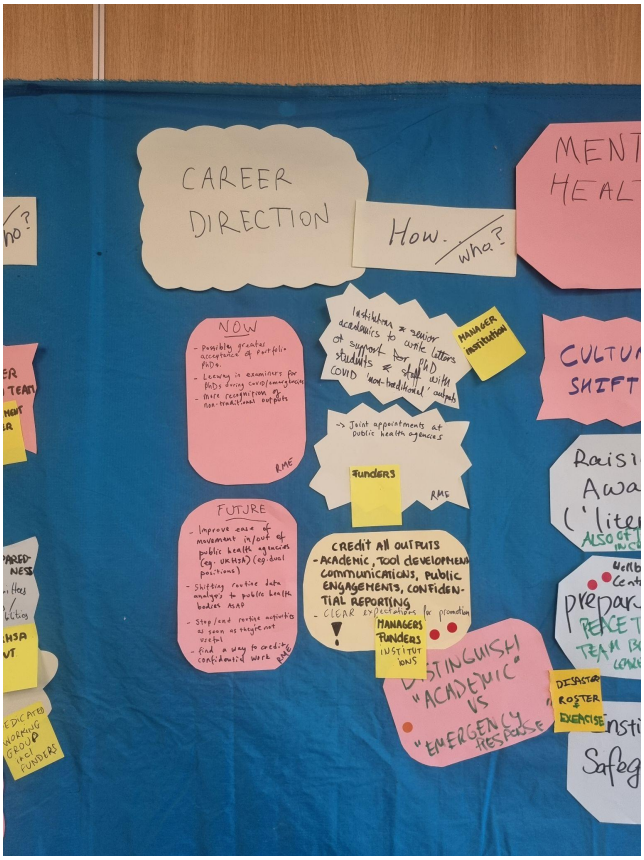


Figure 1. Example of board showing recommendations

Priority recommendations

These recommendations were highlighted by at least one participant during a group discussion. Recommendations which were highlighted by two or more participants are shown in bold.

1. Research teams should ensure that impactful COVID work is understood by funders and institutions, especially if less obvious/visible.
2. **Research teams should initiate sustainable team building and training programs during non-response periods.**
3. Employers should ensure sustainable funding for academic and professional services roles to reduce burnout risks.
4. **Employers should develop processes and guidelines for career growth support for professional services staff.**
5. **Employers need to create or expand Wellbeing Officer roles to monitor work-life balance and provide guidance.**
6. Incorporation of work-life balance components in annual performance and development reviews is essential for employers.
7. Employers should implement capped paid overtime and formal Time Off In Lieu policies and routinely analyse and act upon staff survey and exit interview data.
8. Funding bodies should adjust eligibility criteria to adequately compensate non-academic staff for activities.
9. **Funding bodies should refine impact measures to acknowledge all outputs, including academic, tool development, communications, public engagement, and confidential reporting.**